Tunisian Republic
Ministry of Higher Education and Scientific Research
University of Tunis El Manar
Faculty of Sciences of Tunis
Department of Computer Science

FINAL YEAR PROJECT REPORT

Presented for the attainment of
the Bachelor's Degree in Computer Science
“Computer Science“ : Software Engineering and System

By

Mariem Zaiane
&
Mohamed Yassine Chalouati

Titre de Projet

Host Organization : BeeCoders

Supervised by : Saloua Zammali

Overseen by : ....

GLSI3 - S01008

Academic Year: 2023-2024

Table of Contents

List of Figures

List of Tables

List of acronyms

MVC Model-View-Controller
UML Unified Modeling Language

General Inroduction

E-learning has become a transformational force, offering aspirant learners globally with previously uncommon benefits. Acquiring proficiency in a foreign language is not just advantageous but frequently indispensable for both career and individual growth. Nonetheless, there can be major challenges associated with the traditional barriers to language learning. Our e-learning platform uses technology to provide an engaging and easily accessible learning environment, which directly addresses these issues. Through our platform, learners can seamlessly connect with experienced tutors from around the world, overcoming geographical limitations and obtaining customized education. Whether motivated by travel, professional development, or personal enrichment, our platform offers comprehensive courses and resources to meet diverse needs.

In the context of our end-of-studies project, we are conducting an internship at BeeCoders, a prominent digital services company specializing in web and mobile development, web design, IT consulting, and remote training. BeeCoders excels in crafting innovative software solutions and enhancing application performance for its clientele.

During our internship, we will be primarily focused on the design and implementation of an e-learning platform that includes an AI feature. Our objective is to provide a solution which enables individuals all over the world to access learning materials and tutoring services from highly licensed teachers.

The purpose of this report is to outline the problem we are confronting and the goals we have set for ourselves. We'll dig into the different stages of the project, from front end to back end development. Lastly, we'll discuss the results obtained and the prospects for improving our E-learning web application.

Thus, this report will provide a comprehensive outline of our internship at BeeCoders, highlighting our contributions within the field of web development. Working within this renowned company offers us a valuable opportunity to gain high-level professional experience and develop our skills in designing and implementing proficient applications. We are excited about this enhancing experience and look forward to tackling the challenges that await us in the field of application development.

In the first chapter, we are going to introduce the project's context by presenting the host organization, BeeCoders, as well as the problem to be addressed. We'll also outline the project and look at the current situation in order to propose a solution. At last, we will describe the working methodology and project planning.

The second chapter will be dedicated to the preliminary research. We will examine the requirements by identifying the actors, as well as the functional and nonfunctional requirements. We'll present the overall use case diagram, class diagram, architectural style, and product backlog.

The third chapter will focus on the project's first sprint, which is dedicated to Account Management. We will discuss all of the work completed during this sprint, including the implementation of an AI feature, as well as the results obtained.

The forth chapter will...

Chapter I

General framework of the project

Plan

1 Introduction

This chapter unveils the initial framework of our project. Initially, we will present BeeCoders, where we carried out our end-of-studies project internship, as well as the context in which our project is situated. Subsequently, we will address the existing situation and the proposed solution. Finally, we will discuss the methodology followed during the development of our platform.

2 Hosting Organization

In this section, we will present the host organization, its various values, objectives, and its organizational structure.

2.1 Overview of BeeCoders

BeeCoders, founded in 2020, is a digital services company specializing in web and mobile development (Android and iOS), web design, IT consulting, and remote training.

Table I.1 refers to the datasheet of BeeCoders

Name of the establishment	BeeCoders
Address	**El Ghazela Technology Park, Ariana 2088**
Phone	**(00 216)58 84 00 64**
Email	**admin@beecoders.tn**

Table I.1: Datasheet of BeeCoders

2.2 The values of BeeCoders

Beecoders is guided by three fundamental values that shape their approach to every project:

- **Quality**: Bee Coders is committed to developing your digital product while adhering to your guidelines and deploying the most modern and suitable technologies for your project. Bee Coders also offers high-quality training modules that align with the new requirements of the IT job market.
- **Flexibility**: Flexibility serves as the cornerstone of their agency. They rely on this attribute to ensure optimal development conditions for each project. Their professionals possess a robust capacity for adaptation and integration. Additionally, Beecoders offers high-quality training modules aligned with the evolving demands of the IT job market.

- **Support**: The realm of information technology presents a thrilling journey for every business to navigate to avoid being surpassed by technological advancements. However, supporting employees through effective professional training and engaging the services of a qualified IT consultant are crucial prerequisites for success on this journey.

2.3 Organizational structure of Beecoders

Ahmed Neffati leads Beecoders as its CEO, steering the company's strategic vision and ensuring its overall success. Supporting him are two pivotal directors: Jihed Ben Gharbia, the Web Director, who spearheads the management of web-related initiatives, and Ahmed Yahmed, the Design Director, entrusted with guiding design projects and maintaining visual consistency across all platforms. Additionally, Hbiba Mezghi, in the role of Responsable Digital Marketing, reports to Jihed Ben Gharbia, and is instrumental in crafting and executing the company's digital marketing strategies aimed at bolstering its online presence and fostering business growth.

2.4 Introduction to the reception service

BeeCoders, the web and mobile development agency, harnesses the expertise of its professionals to create digital solutions that meet clients' needs, including websites, mobile applications, software, and more. The company also offers consulting services to enhance employees' IT performance.

3 Project Overview

In this section, we present the general framework of our subject, the problem statement, the objectives outlined, an analysis of the current state, and the proposed solution.

3.1 Project Framework

This end-of-studies project was conducted as part of the requirements for obtaining a Bachelor's degree in Software Engineering and Information Systems at the Faculty of Sciences in Tunis. The project spanned a duration of 4 months, starting from February 1st, during the academic year 2023/2024.

3.2 Context and Problem Statement

In today's globalized world, the ability to communicate effectively in multiple languages is increasingly valuable. Language learning is not just a skill but a necessity for many individuals, whether for academic, professional, or personal reasons. However, traditional language learning methods often lack the dynamism and real-world interaction necessary for effective learning. The problem lies in the gap between the theoretical knowledge gained through traditional language learning methods and the practical application of that knowledge in real-life communication scenarios. Learners often struggle to confidently use their language skills in conversation due to limited opportunities for immersive, interactive practice.

3.3 Project Objectives

The objective definition phase is the most important stage as it allows us to better understand our project, enabling us to deduce the solution. For this reason, we must accurately determine and specify the objectives of our project.

Our project aims to develop a platform whose primary mission is to facilitate language learning through real-time communication between learners and tutors. This platform must ensure:

- Engaging Learning Experience
- Expert Connections
- Cultural Exchange
- Technological Enhancement
- Progress Tracking

3.4 Current State Analysis

The current state analysis is an essential phase to thoroughly understand the current system and define its objectives. This phase allows us, through an examination of the current state, to leverage its advantages while minimizing its drawbacks as much as possible.

Existing language learning platforms and services offer a range of resources and tools for learners, including mobile apps, online courses, and language exchange websites. However, many of these solutions focus on self-paced learning through pre-recorded lessons or gamified exercises, rather than real-time interaction with instructors or native speakers. While language exchange platforms exist, they often rely on user-generated content and lack structured guidance from experienced instructors. Additionally, finding compatible language partners can be hit-or-miss, leading to inconsistent practice opportunities. Furthermore, traditional language schools and tutoring services may offer face-to-face instruction but can be costly and inconvenient for learners with busy schedules or limited access to local resources.

3.5 Proposed Solution

After conducting an analysis of the current landscape, we have identified several limitations. Therefore, we propose the conception and development of an e-learning platform/application that serves as a dynamic alternative to language learning services. This platform is specifically designed to connect individuals eager to learn different languages in real-time with experienced, professional/native speakers, creating an engaging and effective learning experience.

4 Development Methodology: AGILE-SCRUM

Scrum is a development framework in which cross-functional teams create products iteratively and incrementally. Its objective is to enhance team productivity by replacing heavier methodologies, and to remain adaptable to project adjustments throughout its progression.

4.1 Principles and Different Phases

The principle of the Agile Scrum method is to develop software incrementally while maintaining a completely transparent list of enhancement or correction requests to implement.
The figure I.1 represents the Scrum process.

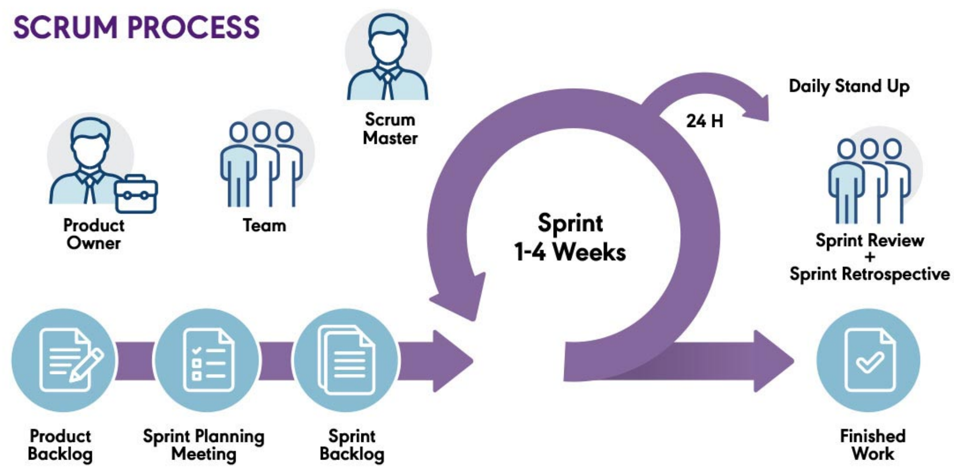

Figure I.1: The overall diagram of the SCRUM methodology. [10]

In practice, this method requires 4 types of meetings:

- **Daily meetings**: Every day, the entire team gathers, typically standing, for about 15 minutes to answer the following questions: What did I do yesterday? What will I do today? Are there any obstacles hindering progress today?
- **Planning meetings**: The entire team gathers to decide on the features that will compose the next Sprint and update the overall list.
- **Work review meetings**: During this meeting, we present what we have accomplished during the Sprint duration. A demonstration of new features or architecture presentations is organized. This is an informal meeting lasting about 2 hours, attended by the entire team.
- **Retrospective meetings**: At the end of each Sprint, the team reflects on what worked well and what didn't. During this meeting, which lasts 15 to 30 minutes and where everyone speaks for themselves, a vote of confidence is organized to decide on improvements to be made.

To better understand the Scrum methodology, it is necessary to define some terms presented in Table I.2.

Keyword	Definition
Product Backlog	Definition of functional requirements in the form of "User Story".
Sprint Backlog	List of User Stories to be implemented during a Sprint.
User Story	Literal (non-technical) description of a given functionality.
Sprint	Short-term iteration (typically 2 to 4 weeks). Each iteration defines an objective to achieve, associating it with a list of items from the Product Backlog to be implemented.
Burndown Chart	Graphical representation of the "remaining work" re-estimation.
Product Owner	Person responsible for managing the product backlog.
Scrum Master	Person responsible for involving the development team and ensuring a good understanding of the methodology and its application.
Development Team	The development team responsible for delivering an increment of the product at the end of each Sprint.

Table I.2: Key Terms in Scrum Methodology

4.2 Organization

The Scrum methodology involves three main roles:

- **Product Owner**: Responsible for maximizing return on investment by identifying product features, transferring them into a prioritized list, selecting those to be at the top of the list for the next Sprint, and continuously reprioritizing and refining this list.
- **Scrum Master**: Acts as a true project facilitator, ensuring that everyone can work to their maximum capacity by eliminating obstacles and protecting the team from external disturbances.
- **Team**: Constructs the product defined by the Product Owner. It is "cross-functional": including all the expertise necessary to deliver a potentially shippable version of the product at the end of each Sprint. It is also "self-organized", with a high degree of autonomy and responsibility.

We have presented the Scrum roles in a general manner, and for our project, we have only had the following roles:

- Product Owner: Ahmed Neffati.
- Scrum Master: Ahmed Neffati.

- Team: Mohamed Yassine Chalouati and Zaiane Mariem.

4.3 Choice of Methodology

Agile methods are increasingly emerging as a credible alternative to traditional software development models such as the V-cycle or the Unified Process, which face significant implementation challenges. For the development of this project, we have chosen to work with the agile Scrum methodology because this approach offers a range of advantages, including:

- Fully developed and tested for short iterations.
- Increased Productivity.
- Personal Organization.
- Improved Communication.

5 Conclusion

In this first chapter, we have introduced the hosting organization and outlined the general framework of our project.We provided a comprehensive overview of the organization, its mission, and how our project aligns with its goals. Subsequently, we conducted a detailed analysis of the current situation, identifying key challenges and opportunities within the existing system. Based on this analysis, we derived the proposed solution and highlighting its potential benefits. Additionally, we specified the working methodology that will guide our project, including the Agile principles and practices we will employ to ensure iterative progress and continuous improvement.

In the next chapter, we will examine the "pre-game" phase. This phase will be dedicated to specifying the functional and non-functional requirements of our application. By thoroughly defining these requirements, we aim to ensure that our development efforts are well-directed and aligned with the project's objectives, paving the way for a successful implementation.

Chapter II

Pre-game Phase

Plan

1 Introduction

In this chapter, we address "the pre-game phase," which is the first phase of the Scrum method. This phase is of great importance because we will identify the product backlog. To do this, we begin by capturing the user stories necessary for the project's realization. Next, we identify the infrastructure and tools/software for the development of this platform. Then, we detail the architecture adopted for the implementation of our platform. Finally, we present the conceptual phase.

2 Product backlog

One of the most important artifacts is the Product Backlog, which is a list of all the expected features of a product called user stories. Each user story has a priority order defined by the Product Owner. We use the MOSCOW technique to prioritize needs:

- M for Must Have: Must be done. The requirement is essential. If it is not done, the project fails. It can also be considered high priority.
- S for Should Have: This is an essential requirement that should be done if possible. However, if it is not done, it can be worked around and delivered later.
- C for Could Have: This is a desirable requirement. It could be done if it does not impact other tasks.
- W for Won't Have: This is a « luxury » requirement. An interesting task but not mandatory.

The MOSCOW technique has several advantages. First, it allows clear prioritization of project requirements and features, focusing on essential elements. This helps manage stakeholder expectations better and deliver key features quickly. Additionally, MOSCOW offers flexibility in managing priorities, allowing the team to adapt to changes and make informed decisions.

Table **??** represents all the user stories to be completed during the project.

Module	User stories	Sprint	Priority
Account Management	As a user, I want to sign up by providing essential information such as email and password.	Sprint 1	M
	As a user, I want to edit my account information, including profile picture, email address, and personal details, to keep it up to date.	Sprint 1	M
	As a user, I want to log in to access my account.	Sprint 1	M
	As a user, I want the option to recover my password in case I forgot it.	Sprint 1	M
	As a user, I want to see my account's information.	Sprint 1	M
	As a user, I want to be able to delete my account.	Sprint 1	M
	As a user, I want to receive a confirmation email to verify my account after signing up.	Sprint 1	M
	As a tutor, I want to be able to upload introductive videos.	Sprint 1	M
	As a user, I want to have a profile.	Sprint 1	M
Teaching and Learning Activities	As a tutor, I want to see the first upcoming lessons for the current week on my profile.	Sprint 2	M
	As a user, I want to have a Calendar.	Sprint 2	M
	As a user, I want to consult upcoming lessons in Calendar.	Sprint 2	M
	As a learner, I want to book lessons through my Calendar with Tutors based on the lesson's parameters.	Sprint 2	M
	As a learner, I want to consult tutors' profiles.	Sprint 2	S
	As a learner, I want to schedule lessons with tutors through their profiles and get notified when they respond to my request.	Sprint 2	S
	As a learner scheduling from the tutor's profile, I want to easily identify available times that overlap with both my schedule and the tutor's.	Sprint 2	M
	As a learner, I prefer to view only my available time slots when scheduling lessons.	Sprint 2	M
	As a user, I want to be able to mark All my notifications as read.	Sprint 2	C
	As a user, I want to filter my lessons notifications into categories such as accepted, rejected, and pending.	Sprint 2	S

Module	User stories	Sprint	Priority
Communication and Interactions	As a learner, I want to search for tutors with filtering options and pagination feature.	Sprint 2	M
	As a tutor, I want to get notified when a learner requests to schedule a lesson with me and efficiently provide a response.	Sprint 2	M
	As a learner, I want to filter tutors based on criteria, such as their availability, names, languages spoken, and topics they teach.	Sprint 2	S
	as a learner, I want to study 1 on 1 with a tutor	Sprint 3	M
	As a learner or tutor, I want to talk to a built-in chatbot for assistance.	Sprint 3	C
	As a learner, I want to like my favorite tutors.	Sprint 3	C
	As a learner, I want to have a list of my favorite tutors.	Sprint 3	C
	As a tutor, I want to chat with learners who seek language tutoring.	Sprint 3	M
	As a learner, I want to chat with tutors.	Sprint 3	M
	As a learner, I want to see tutors through a recommendation system.	Sprint 3	C
	As a learner, I want to receive messages' notifications.	Sprint 3	W
	As a learner, I want to make a video call with tutors.	Sprint 3	M
Subscription Management	As a learner, I want to subscribe to access premium content.	Sprint 4	3
	As a learner, I want to pay for my subscription online.	Sprint 4	3
Courses Management	As a learner, I want to browse my available language courses.	Sprint 4	2
	As a learner, I want to categorize language courses by language and proficiency level.	Sprint 4	2
	As a learner, I want to download courses.	Sprint 4	2
	As a tutor, I want to be able to make courses.	Sprint 4	2
	As a user, I want to differentiate between premium courses and free courses.	Sprint 4	1
	As a user, I want to view the course information.	Sprint 4	1
Community Management	As an administrator, I want to send announcements and notifications to users.	Sprint 4	S
	As an administrator, I want to have the ability to remove courses that violates community standards.	Sprint 4	S
	As an administrator, I want to delete user accounts when necessary.	Sprint 4	M

Table II.1: Product Backlog

After identifying the user stories in the product backlog, we are going to compose them into five sprints.

1. Sprint 1 : Account Management.
2. Sprint 2 : Lessons Management.
3. Sprint 3 : Communication and Interactions.
4. Sprint 4 : Content and Community Management.

3 Work environment

During this phase, we present the software architecture, tools and programming languages used to implement the project, considering the choice of the right working environment as an important step.

3.1 Hardware environment

During various stages of the project, including documentation, requirements specification, design and development, we had access to three computers with the following specifications:

- Laptop 1 :
  - Operating system : Windows 11.
  - CPU : 11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz 3.00 GHz.
  - Memory : RAM 12.0 GB.
  - Storage :
    - 237GB hard drive.
    - 1TB hard drive.
- Laptop 2 :
  - Operating system : Windows 11.
  - CPU : 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz 2.42 GHz.
  - Memory : RAM 32.0 GB.
  - Storage : 475 GB hard drive.
- Desktop :
  - Operating system : Windows 10.
  - CPU : Intel(R) Core(TM) i5-4460 CPU @ 3.20GHz 3.20 GHz.
  - Memory : RAM 8.00 GB.
  - Storage : 1TB hard drive.

3.2 Software environment

The implementation of this project requires certain software and tools, which will be presented as follows :

- **Visual studio Code**
  Visual Studio Code is an extensible, open-source, cross-platform integrated development environment chosen for the development of our application on both the backend and frontend sides. In the latest survey, Visual Studio Code was ranked as the most popular development tool. [16]

- **Database server** : **MySql**
  MySQL is a database management system. The MySQL software delivers a very fast, multi-threaded, multi-user, and robust SQL (Structured Query Language) database server. MySQL Server is intended for mission-critical, heavy-load production systems as well as for embedding into mass-deployed software. [8]

- **Postman**
  Postman is an API platform for building and using APIs. Postman simplifies each step of the API lifecycle and streamlines collaboration to create better APIs—faster. We chose Postman because it allows us to execute a series of HTTP requests one after the other and save them into a collection. [11]

- **StarUML**
  StarUML is a software modeling platform that supports UML (Unified Modeling Language). It is based on UML version 1.4 and provides eleven different types of diagram, and it accepts UML 2.0 notation. It actively supports the MDA (Model Driven Architecture) approach by supporting the UML profile concept. StarUML excels in customizability to the user's environment and has a high extensibility in its functionality. Using StarUML, one of the top leading software modeling tools, will guarantee to maximize the productivity and quality of our project. [14]

3.3 Used Technologies

In this section, we will list the programming languages and all the technologies used in the back-end and front-end.

3.3.1 Programming languages

In our project, we have used the following programming languages to support different components, ensuring flexibility, efficiency and robustness in our development approach :

- **Javascript**
  Javascript is an interpreted object-based scripting language. It is a lightweight language that uses the concept of prototypes with weak and dynamic typing to create interactive websites. It is also used in many external environments like Node Js, React Js, which is our case. [7]

- **Python**
  Python is the open source programming language most used by computer scientists. This language is at the forefront of infrastructure management, data analysis, and software development. Indeed, among its qualities, Python allows developers to focus on what to do rather than how to do it. It frees developers from the syntactic constraints that previously held them back with older languages. As a result, code development in Python is faster than in other languages. [12]

3.3.2 Back-End

The back-end is the server-side part that handles user requests, processes data, performs calculations, and communicates with the database.
We have used the following technologies in the back-end :

- **Node.js**
  Node.js is a JavaScript runtime environment that allows developers to run JavaScript code outside of a web browser. It is event-driven, non-blocking, and well-suited for building scalable network applications such as web servers and APIs. It uses the V8 JavaScript engine and provides a rich module library via npm (Node Package Manager). [9]

- **Express.js**
  Express is a minimalist and flexible Node.js web application framework that provides a powerful feature set for web and mobile applications. With countless utility methods and HTTP middleware, creating a powerful API is quick and easy. Express provides a thin layer of basic web application functionality without sacrificing the Node.js features. [3]

- **Django**
  Django is a high-level Python web framework that encourages rapid development and clean, pragmatic design.In the backend context, Django is a powerful Python web framework used to manage data processing and server-side logic in web applications. It provides tools for database interaction, URL routing, dynamic content generation, and follows the Model-View-Template architectural pattern to maintain code organization. With a focus on rapid development and scalability, Django is widely adopted for building powerful backend systems. [2]

3.3.3 Front-End

The front end is the visible, interactive part that users interact with directly in the browser.
We have used the following technologies to implement the front end of the platform :

- **HTML5**
  HTML5 (HyperText Markup Language 5) is the latest version of the major web language HTML. HTML5 specifies two syntaxes for abstract models defined in terms of the DOM: HTML5 and XHTML5. The language also includes an application layer with a large number of APIs and algorithms for processing documents with non-compliant syntax. [6]

- **CSS3**
  CSS, short for Cascading Style Sheets, is a computer language used to style web pages (HTML or XML). It consists of “cascading style sheets,” also known as CSS (.css) files, which contain coding elements for formatting. [1]

- **Tailwind CSS**
  Tailwind CSS is a utility-first CSS framework that streamlines web development by providing a set of pre-designed utility classes. These classes enable rapid styling without writing custom CSS, promoting consistency and scalability. Tailwind’s approach shifts focus from traditional CSS components to functional classes, empowering developers to efficiently build responsive and visually appealing interfaces with minimal effort. [15]

- **React JS**
  React.js, also known as React, is a free and open source JavaScript library mainly used to create user interfaces by combining sections of code (components) into complete web pages . Originally developed by Facebook, it is now maintained by Meta and the open source community. React uses JSX, a combination of JavaScript and XML, to create elements and display them on web pages. Despite the steep learning curve for junior developers, React has become one of the most popular and in-demand JavaScript libraries. It is considered a JavaScript library rather than a framework, providing flexibility and speed of development. [13]

3.3.4 Version Control and Collaboration Tools

Version Control and Collaboration Tools are essential for managing code changes, facilitating team collaboration, and ensuring project stability.
We have used the following technologies for version control :

- **Git**
  Git is a DevOps tool used for source code management. It is a free and open-source version control system used to handle small to very large projects efficiently. Git is used to tracking changes in the source code, enabling multiple developers to work together on non-linear development. [4]

- **GitHub**
  GitHub is a web-based interface allowing real-time collaboration. It encourages teams to work together in developing code, building web pages and updating content. GitHub allows you to create, store, change, merge, and collaborate on files or code. Any member of a team can access the GitHub repository and see the most recent version in real-time. Then, they can make edits or changes that the other collaborators also see. [5]

3.4 Technological Choices

Our solution is a web application that follows a 4-tier architecture based on the Express.js framework, Django, ReactJS, and stores data in a MySQL database. In the following, we explain the choice of the expressjs framework, the django framework, the Reactjs presentation framework, as well as the MySQL database.

3.4.1 Choice of the Express.js framework

We chose Express.js as the backend framework for website because it is lightweight, flexible, and perfectly suited to the needs of our project. Express.js provides a minimalist approach to building web applications, allowing us to quickly set up robust APIs and efficiently handle HTTP requests. Its middleware architecture enables seamless integration of additional functionality such as authentication and data validation, which is essential to ensure the security and reliability of the platform. Additionally, the vibrant community surrounding Express.js provides extensive resources and support, making Express.js a great choice for rapid development as our platform becomes increasingly complex and the user base grows. This makes it an ideal choice for scalability.

3.4.2 Choice of the Django framework

We chose to work with Django for many reasons, primarily for Ai integration. This framework allows us to separate Ai logic from the Express.js backend server to enhance modularity and maintainability.

Plus Django uses python as its programming language, which provides a large amount of resources and libraries specifically meant for Artificial Intelligence development, making this framework a perfect option to integrate Ai features into our platform.

3.4.3 Choice of The ReactJS framework

We chose to work with ReactJS as the front-end framework for its numerous advantages. React's modular structure allows for the development of reusable components, resulting in a seamless user experience with efficient rendering and performance. Leveraging React's extensive ecosystem and community support offers us access to tools that improve functionality, also the integration of Express.js, django and MySQL provides seamless data management and communication across layers, resulting in a scalable and user-friendly e-learning platform

3.4.4 Choice of the MySQL database

MySQL is free software. It's a database management program. We picked it since it has many benefits. For example:

- **Cross-platform :** MySQL runs on different computer operating systems like Windows, Mac, and Linux. This is called being cross-platform.
- **Multi-user capability :** Many people can use MySQL without issues. Having robust multi-user functionality means multiple users can access it easily.
- **Security :** Security safeguards are strong with MySQL. Administrators decide who can fully access the 0database. This access control system protects MySQL well.

4 Architecture of the solution

Before starting to develop the solution with the software, it is essential to draft its architecture. This draft will determine what quality, maintainability and flexibility the software will have, and it will have its say during the development. The architecture must be well thought out for a great software.

4.1 4-tier architecture

4.1.1 Definition

Our platform is separated into four layers, each of which contains a well-defined collection of components.
the 4-tier architecture, includes the presentation layer, the business layer, the AI layer, and the data layer :

- **Presentation layer :** This is a layer reserved for the client, which is referred to as "lightweight" in the sense that it doesn't undertake any processing or operating functions unlike the Client/Server or two-tier model.
- **Business layer :** Associated with the server, it includes the application server or middleware, or even an intermediate server, which in many cases is a web server with application extensions.

- **AI Service layer :** This layer is responsible for integrating AI features such as face detection. It serves as the backbone for integrating advanced AI capabilities into the system, enabling automation.

- **Data access Layer :** Related to the database server (DBMS).

This architecture allows for the separation of concerns, reducing application complexity, ensuring scalability, and facilitating maintenance and deployment.

4.1.2 Advantages of the 4-tier architecture

The primary benefit of the 4-tier architecture is the improved separation of concerns. It supports modularity and flexibility by allowing each layer to be created, implemented, and scaled separately. Changes to the presentation layer, such as user interface updates, can occur without changing the underlying business logic or data access techniques. Similarly, improvements to AI services can be deployed within the dedicated AI service layer without affecting other architecture components. Also this architecture has more benefits compared to the other architectures :

- **Enhanced security**: Since the presentation and data layers cannot communicate directly, SQL injections and other malicious exploits are prevented now, In our 4-tier architecture, the application layer represents a protective barrier between the presentation layer that is responsible for user interactions, the Data layer which manages database operations and the AI service layer which manages the AI features within the platform.

4.2 RESTful API

4.2.1 Definition

RESTful API is an interface that two computer systems use to exchange information securely over the internet. Most business applications have to communicate with other internal and third-party applications to perform various tasks. For example, to generate monthly payslips, your internal accounts system has to share data with your customer's banking system to automate invoicing and communicate with an internal timesheet application.

RESTful APIs support this information exchange because they follow secure, reliable, and efficient software communication standards. [17]

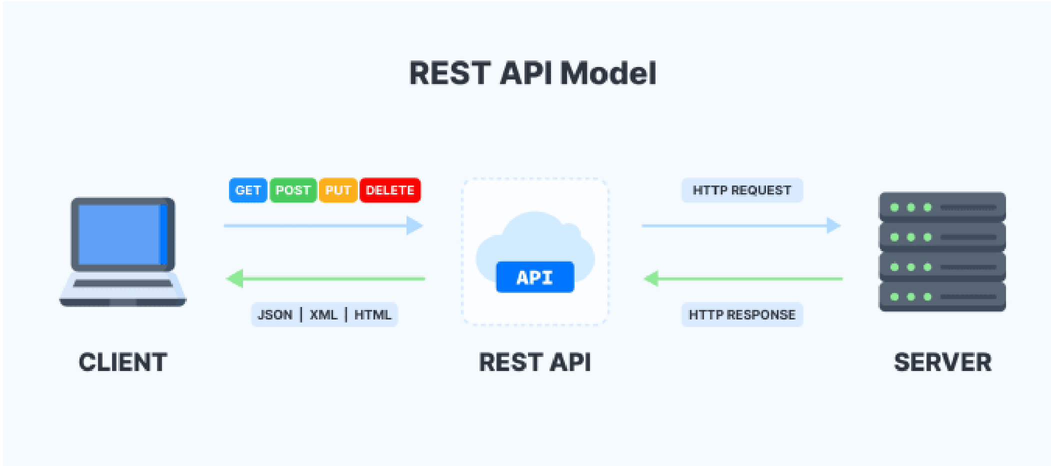

Figure II.1: RESTful API [18]

4.2.2 Advantages of RESTful APIs

RESTful APIs include the following benefits [17]:

- **Scalability**: Systems that implement REST APIs can scale efficiently because REST optimizes client-server interactions. Statelessness removes server load because the server does not have to retain past client request information. Well-managed caching partially or completely eliminates some client-server interactions. All these features support scalability without causing communication bottlenecks that reduce performance.

- **Flexibility**: RESTful web services support total client-server separation. They simplify and decouple various server components so that each part can evolve independently. Platform or technology changes at the server application do not affect the client application. The ability to layer application functions increases flexibility even further. For example, developers can make changes to the database layer without rewriting the application logic.

- **Independence**: REST APIs are independent of the technology used. You can write both client and server applications in various programming languages without affecting the API design. You can also change the underlying technology on either side without affecting the communication [17].

5 Requirements specification

In this section, we will present the functional and non-functional requirements of our application.

5.1 Actors identification

In this section, we identify the various actors who have an influence on our application. The rights attributed to them are determined according to their roles. Following the study of the different interactions of the system, we conclude that our application comprises three actors:

- **The visitor** :
  he is then either a learner or a tutor

- **The learner** :
  engages in finding tutors, enrolling in classes, learning new languages, improving skills, and communicating with tutors through chat and video calls.

- **The tutor** :
  responsible for teaching students, creating and managing classrooms, assisting learners, posting introductory videos, and maintaining a profile that showcases their expertise and background.

- **The administrator** :
  responsible for the administration of the application and the management of users.

5.2 Functional requirements specification

Functional requirements represent the client's requirements regarding the features to be included in the development of the application. Following several meetings with the Product Owner, led by Mr. Ahmed Neffati, we have listed these requirements in Table II.2 :

Actor	**Description of Functional Requirements**
Visitor	Consult landing page Sign up
Learner	Authentify Consult profile Edit profile Engage with tutors Enroll in courses Manage calendar Subscribe
Tutor	Authentify Consult profile Edit profile Connect with Learners Manage calendars Manage courses
Administrator	Authentify Manage users Create admins Delete courses Consult statics

Table II.2: Functional requirements' description

5.3 Non-functional requirements specification

To effectively meet the needs of users, the application must respect the following non-functional requirements:

- **Security** :
  our application ensures secure access through user authentication based on their access rights and incorporates factors that protect it against accidental or malicious attacks.our application ensures secure access through user authentication based on their access rights and incorporates factors that protect it against accidental or malicious attacks.

- **Fiability** :
  the system must be available and functional at all times.

- **Scalability** :
  the application must be scalable and open to any new additions.

- **Performance** :
  the system must respond quickly, regardless of the user's action.

- **The ergonomics of interfaces** :
  ease of use of the interfaces. The objective is to create interfaces that meet the following ergonomic criteria:

  – **Accessibility** :
    this application should be accessible from different browsers (Chrome, Firefox, Safari...).

  – **Simplicity** :
    It must also be intuitive so that even a beginner user does not encounter difficulties in understanding its operation.

  – **Consistency** :
    the application must be consistent, meaning it should ensure homogeneity across interfaces.

6 Conceptual phase

The project design phase is a critical step before implementation. Its goal is to organize, structure, and plan the project. The design phase is critical, as it sits between the initial design phase and the actual implementation of activities.

6.1 Global Use Case

The global use case diagram depicts the interactions between various actors and the system. It also covers the functionalities of our suggested solution to provide a comprehensive picture of its functional behavior.

Figure II.2 shows the global use case diagram of our solution.

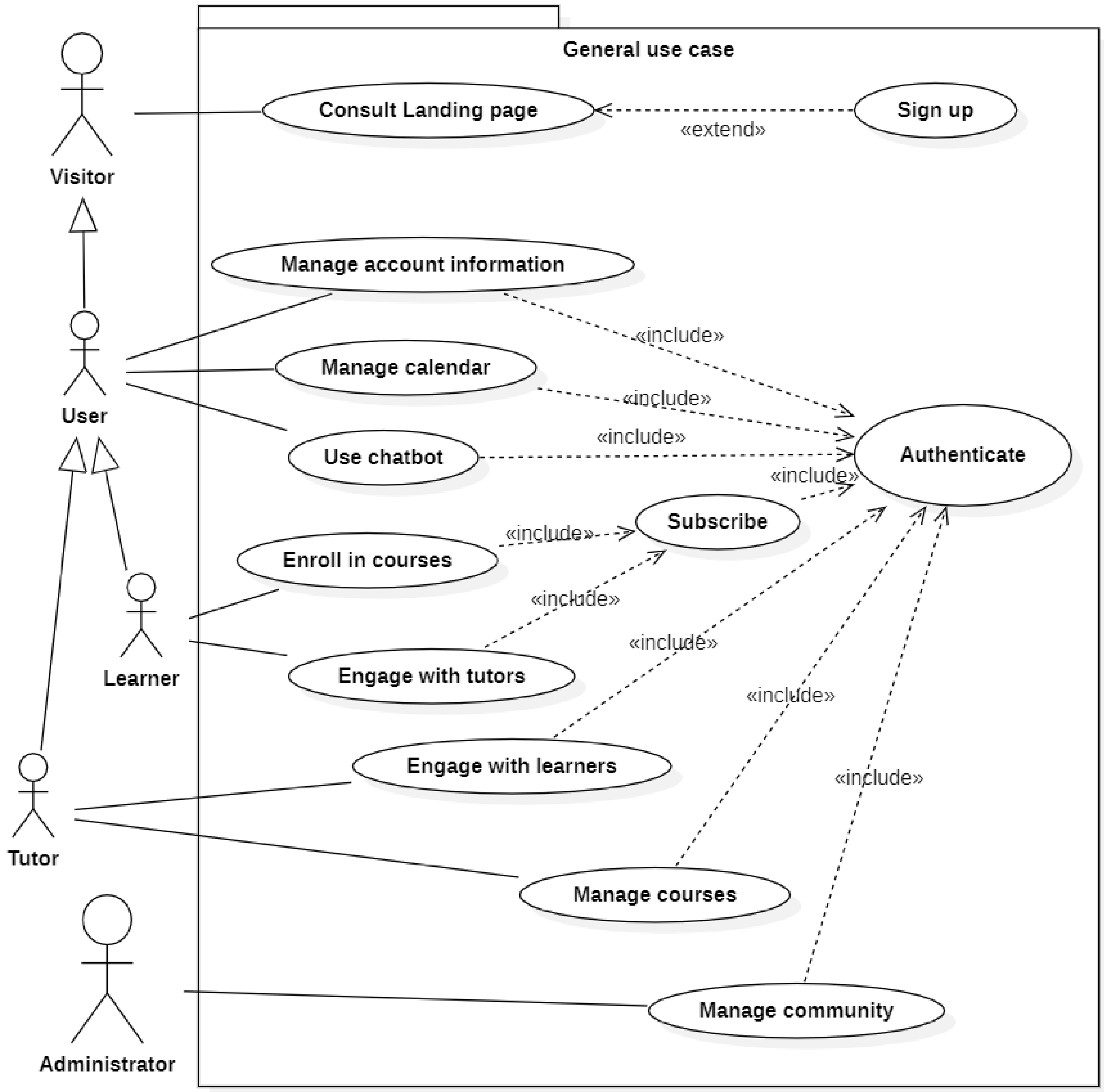

Figure II.2: Global use case diagram

6.2 Global Class Diagram

The global class diagram depicts the relationships and interactions between classes, their attributes, and methods, providing an overview of a system's static structure. It provides a core blueprint for understanding the arrangement and behavior of the system's components, allowing for more efficient system design and development.

Figure II.3 shows the Global class diagram of our solution.

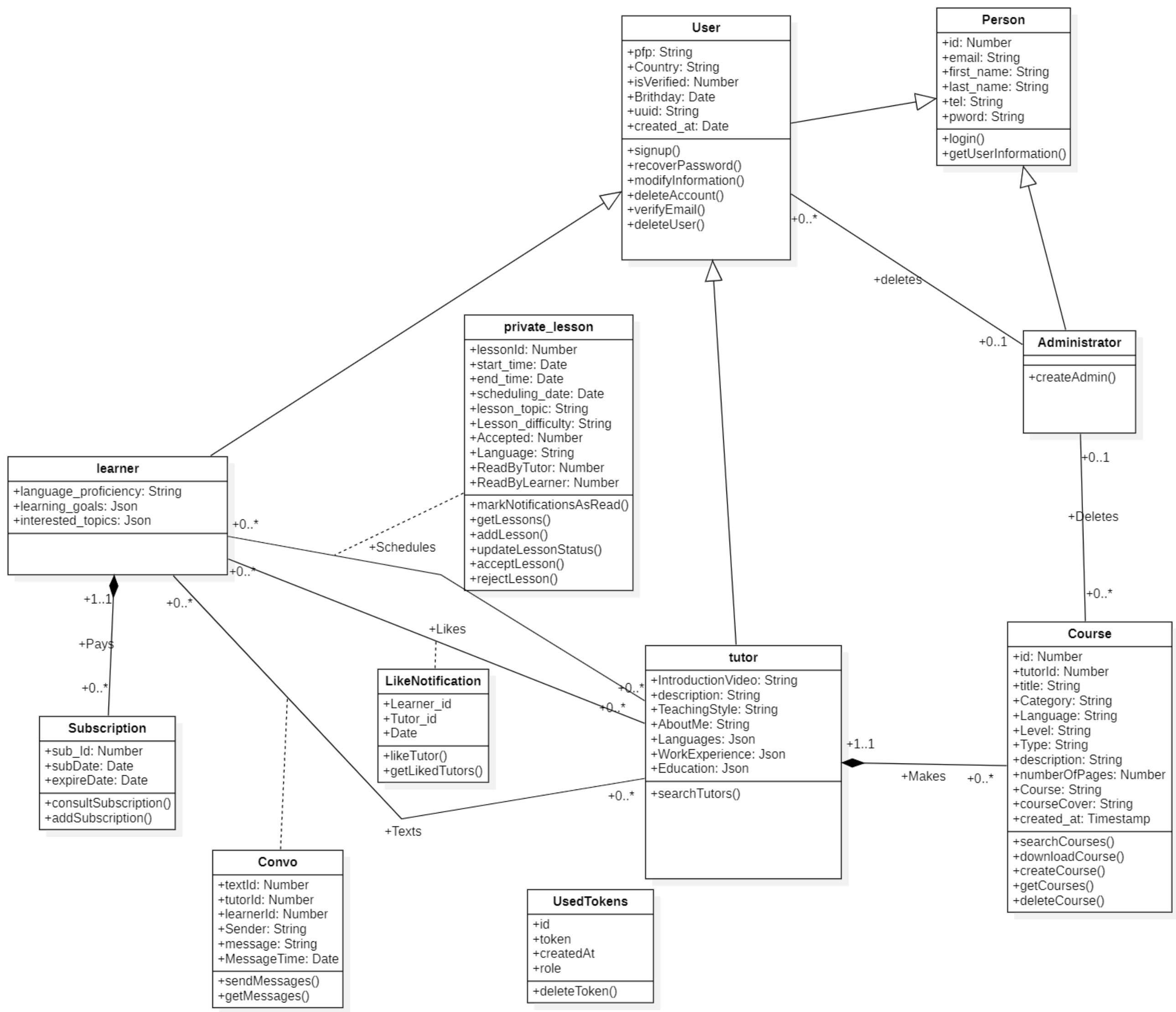

Figure II.3: Global class diagram

7 Conclusion

Finally, this chapter presented a detailed summary of the project by highlighting its different critical design stages. Functional and non-functional requirements have been properly stated, providing a solid foundation for efficient system development. Furthermore, the presentation of the global use case and class diagram provided a clear representation of the many aspects to be implemented in the system. Overall, this chapter, which culminates in the presentation of the produced backlog, represents an important milestone in the project's development by providing a coherent and clear summary of its primary components. The following chapter will go into the conceptual investigation.

Chapter III

Sprint 1: Account Management

Plan

1 Introduction

After analyzing and defining the product backlog, we held a meeting during which we agreed on the allocation of tasks and the division of the application into five sprints. We are now starting the first sprint, which focuses on part of the "Account management" of the web application.

2 Backlog of Sprint 1

As a start, we identify the list of tasks to be completed during this Sprint. The plan for this iteration includes nine user stories.

In our sprint planning, we use the T-Shirt Size Estimation system to categorize user stories based on their complexity. This system uses intuitive sizes—Small (S), Medium (M) and Large (L) to represent different levels of work. Small tasks are straightforward and quick to complete, Medium tasks require moderate effort, and Large tasks are complex and take several days. This approach simplifies estimation, fosters team collaboration, and helps ensure balanced workload distribution in our sprints.

Table III.1 represents the user stories of our sprint.

Model	User stories	Complexity	Estimation
Account Management	As a user, I want to sign up by providing essential information such as email and password.	L	3
	As a tutor, I want to be able to upload introductory videos.	S	1
	As a user, I want to edit my account information, including profile picture, email address, and personal details, to keep it up to date.	M	2
	As a user, I want to log in to access my account.	M	2
	As a user, I want the option to recover my password in case I forgot it.	M	2
	As a user, I want to see my account Information.	S	2
	As a user, I want to be able to delete my account.	S	1
	As a user, I want to receive a confirmation email to verify my account after signing up.	S	1
	As a user, I want to have a profile.	S	1

Table III.1: Sprint 1 User stories

3 First Sprint Refinement

3.1 Identification of functional needs

A use case is a collection of activity sequences that fully describes the system's functional needs. Each use case relates to a system's business function as seen by one of its actors. In the following, we present the list of the main functionalities that our platform must offer.

- **Account creation** : This feature allows users to create an account by providing basic information and answering questions to personalize their experience on the platform. During the account creation process, an email confirmation is sent to verify the user's email address.
- **Authentication** : The goal of this functionality is to allow users to log in using two methods: through their Google account or by providing their email address and password.
- **Managing account** : This functionality enables users to manage their account by editing essential information such as their phone number, first and last name, profile picture, and password. Users can also delete their account or review their activity history. To access these features, users must first authenticate themselves.

3.2 Global model of use cases for the first Sprint

3.2.1 Use Case Diagram of the Sprint

The first sprint focuses on the use case that describes the various functionalities represented in the use case diagram shown in Figure III.1

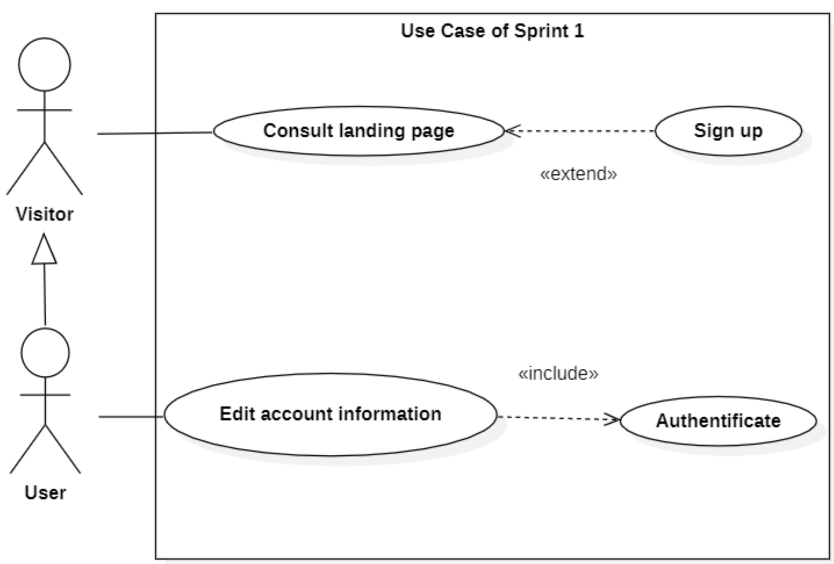

Figure III.1: Use case diagram sprint 1

3.2.2 Refinement of use cases

quad In this section, we will start with detailing the"Edit account information" use case from sprint 1 as illustrated in Figure III.2.

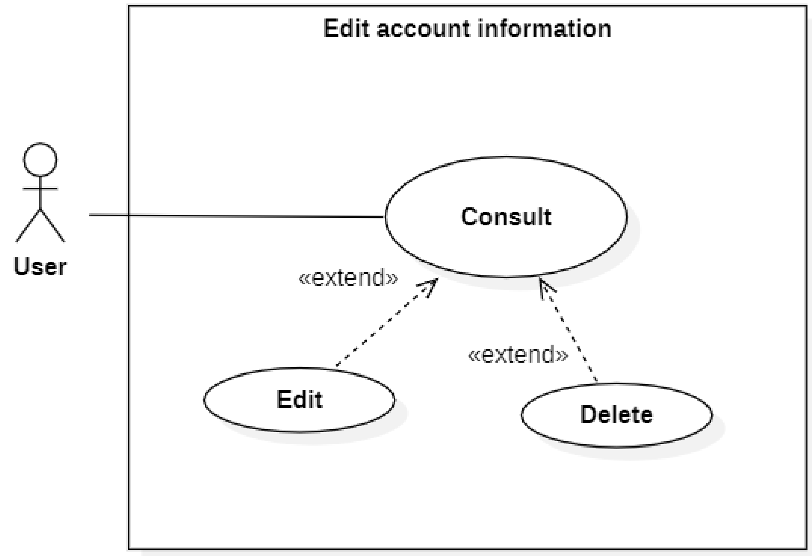

Figure III.2: Refinement of «Edit account information» use case

Table III.2 represents a description of the "Authentication" use case.

IDENTIFICATION SUMMARY	
Title	Authentication
Actor(s)	User
DESCRIPTION OF SEQUENCES	
Pre conditions	Having an account
Principal Scenario	
1.The user clicks on the button "sign in" 2.The system displays two options: continue with Google, or fill in with email and password. 3.The user selects the more convenient option. 4.The user checks/ticks the reCAPTCHA "I am not a robot" box. 5.The user clicks on the button Sign In. 6.The system displays the user's profile.	
Alternative Scenario	
- If the user enters incorrect information or an incompatible email and password, the system will display an error message. - If the user leaves the fields empty, they won't be able to click on the "Sign In" button. - If the user forgets their password, they can click on "Forgot password" and follow the provided steps to reset it.	

Table III.2: Textual description of the use case "Authentication"

Table III.3 represents a description of the "Tutor’s account creation" use case.

IDENTIFICATION SUMMARY	
Title	Tutor’s account creation
Actor(s)	Tutor
DESCRIPTION OF SEQUENCES	
Pre conditions	Not already having an account
Principal Scenario	
1.The user clicks on the button "sign up as a tutor" 2.The system displays two options: continue with Google, or fill in with email and password. 3.The user selects the more convenient option. 4.The user clicks on the button "Sign Up". 5.The system triggers an email verification process and simultaneously generates a message to inform the user of the action. 6.The user clicks on the "Verify Email" button he received in his mailbox. 7.The system redirects the user to another page upon successful verification, confirming completion of the process. 8.The user clicks on the "Go back" button. 9.The system redirects the user to the "welcome" page. 10.The user selects his country. 11.The user clicks on the "Next" button. 12.The system redirects the user to the "Intro" page. 13.The user uploads a profile picture (An AI system verifies the profile picture to ensure it contains a face) and provides personal details including an introductory video, description, languages spoken, work experience, and education. 14.The user clicks on the "Next" button. 15.The system redirects the user to the "Wifi Test" page. 16.The system starts testing the wifi. 17.The system prints out the result. 18.The user clicks on the "Next" button. 19.The system displays the user’s profile.	
Alternative Scenario	
- If the user enters incorrect information, an invalid email or unmatched passwords, the system will display an error message. - If the user leaves the fields empty, they won’t be able to click on the ’Next’ button.	

Table III.3: Textual description of the use case "Tutor’s account creation"

3.2.3 Sequence diagrams

In our platform, whether the user is a learner or a tutor, they are required to input their email and password for access, or alternatively, they can choose to proceed by using their Google account. This grants them access to various features and their account within our platform.
The different steps of the authentication process are represented in sequence diagram of Figure III.3.

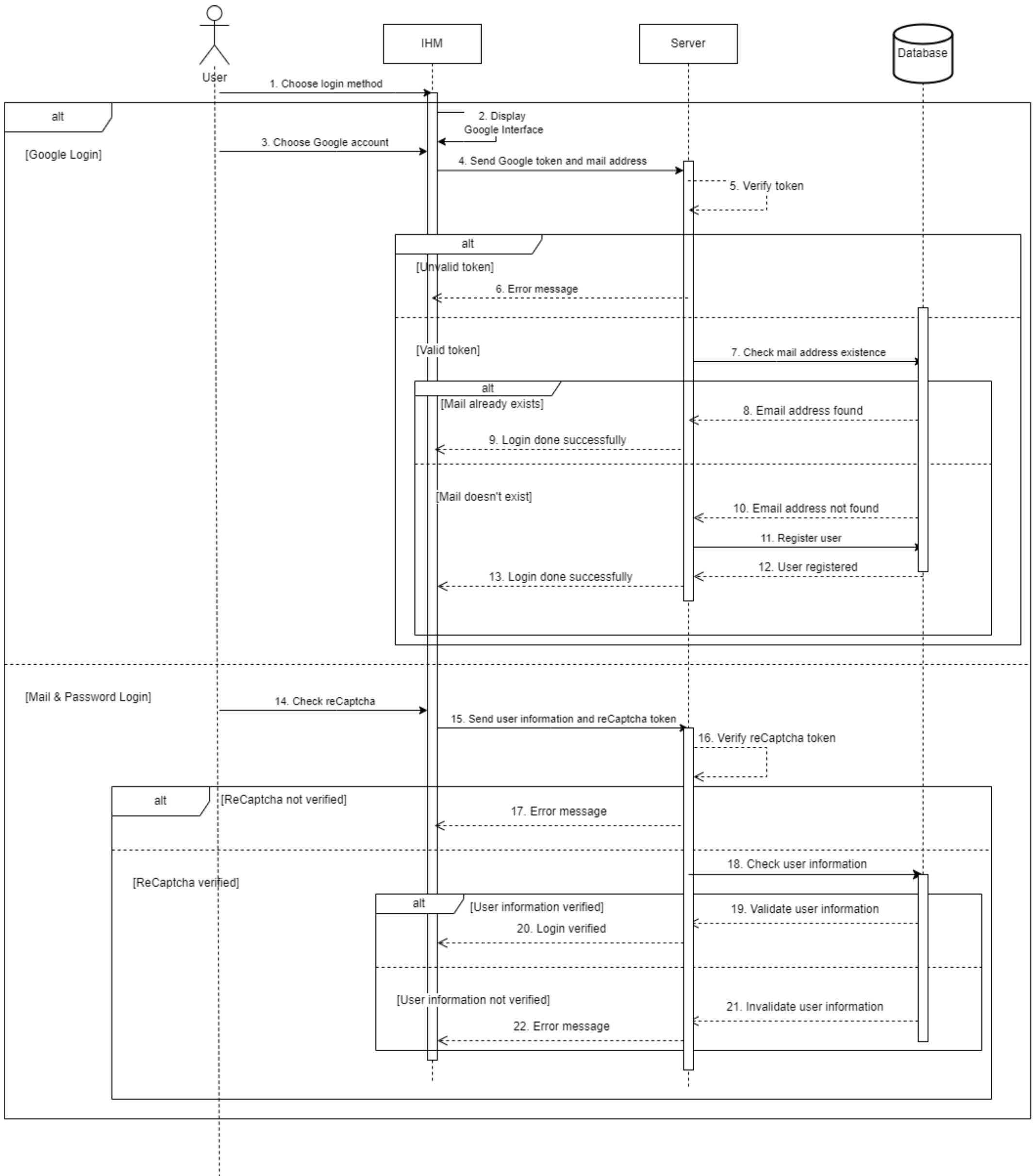

Figure III.3: Sequence diagram of "Authentication"

The figure III.4 presents the sequence diagram for Tutor's account creation.

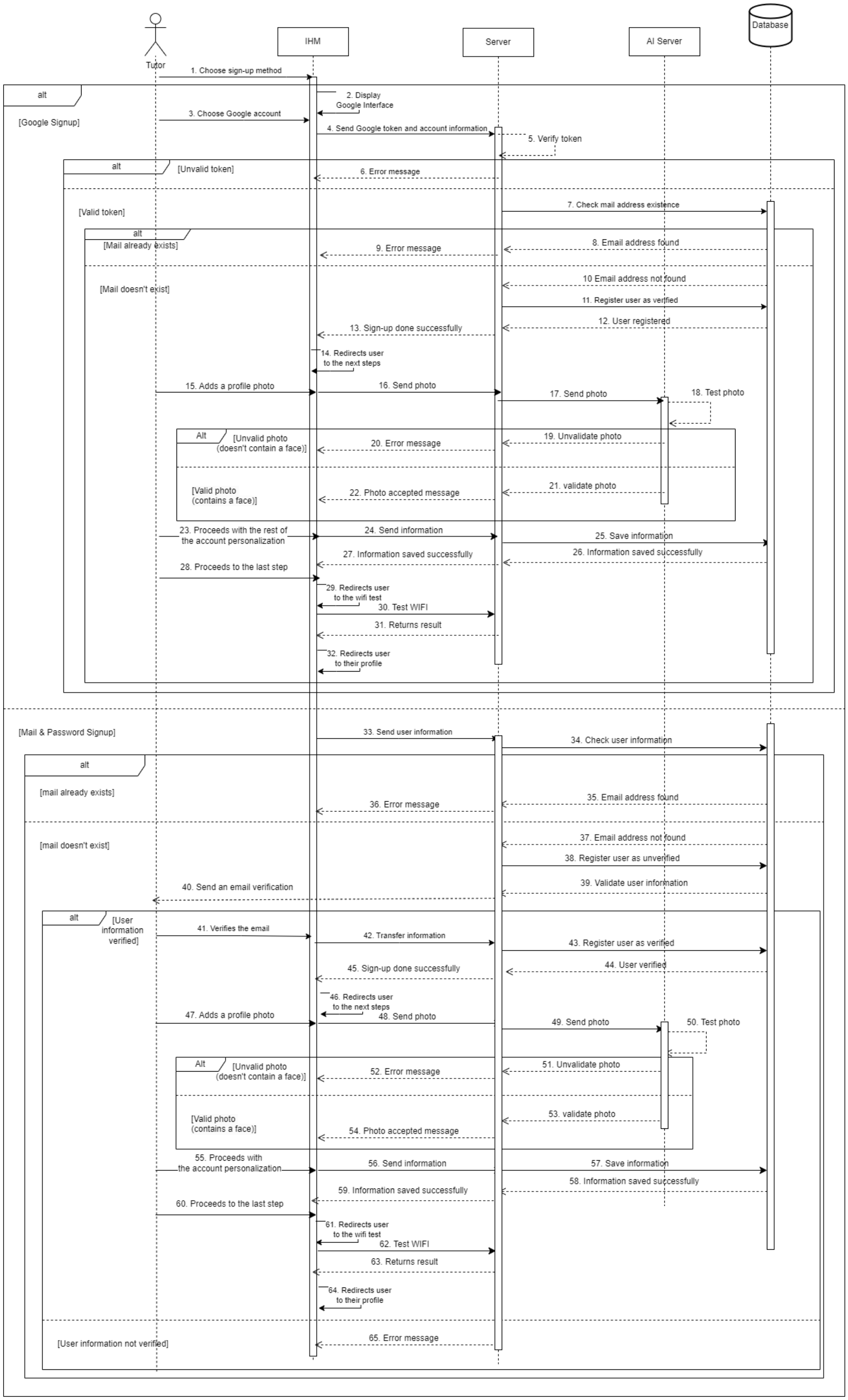

Figure III.4: Sequence diagram of "Tutor's account creation"

3.2.4 Class diagram

The use case diagram highlights the interactions between actors and the system, while the class diagram details the internal structure of the application. The latter abstractly represents the various system objects that interact to fulfill the different use cases of the application. Therefore, the class diagram serves to identify the classes and their relationships, the attributes and methods of each class, as well as the associations between classes. It is essential for understanding how the different components of the application interact with each other.

The class diagram III.5 corresponds to sprint 1.

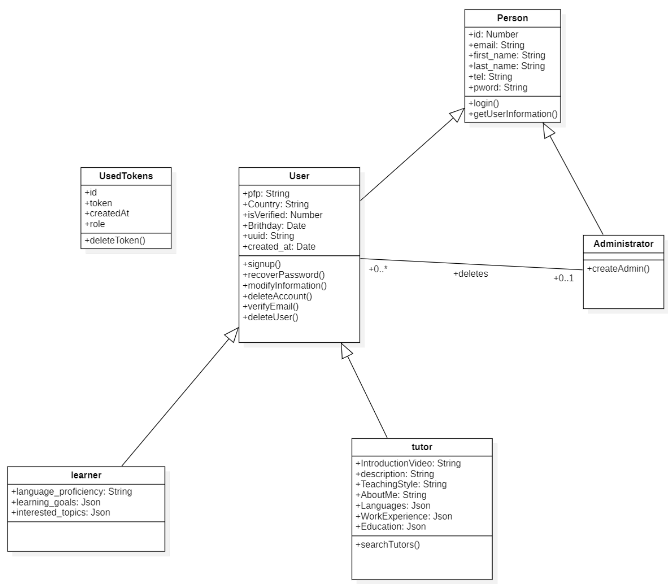

Figure III.5: Class diagram sprint 1

4 Realisation

This section will present multiple screenshots to describe the interfaces of our tool and thus provide a better understanding of its functionality.

We present the authentication process of our platform, which is divided into four main parts: Sign in as a learner, Sign in as a tutor, Sign up as a learner, and Sign up as a tutor. Each part is facilitated by an interface that serves as the initial user interface of our tool, allowing users to input their information.

The authentication interface III.6 Presents the sign-in page of both users, Learner and tutor: a simple and user-friendly interface where they simply fill in their email and password.

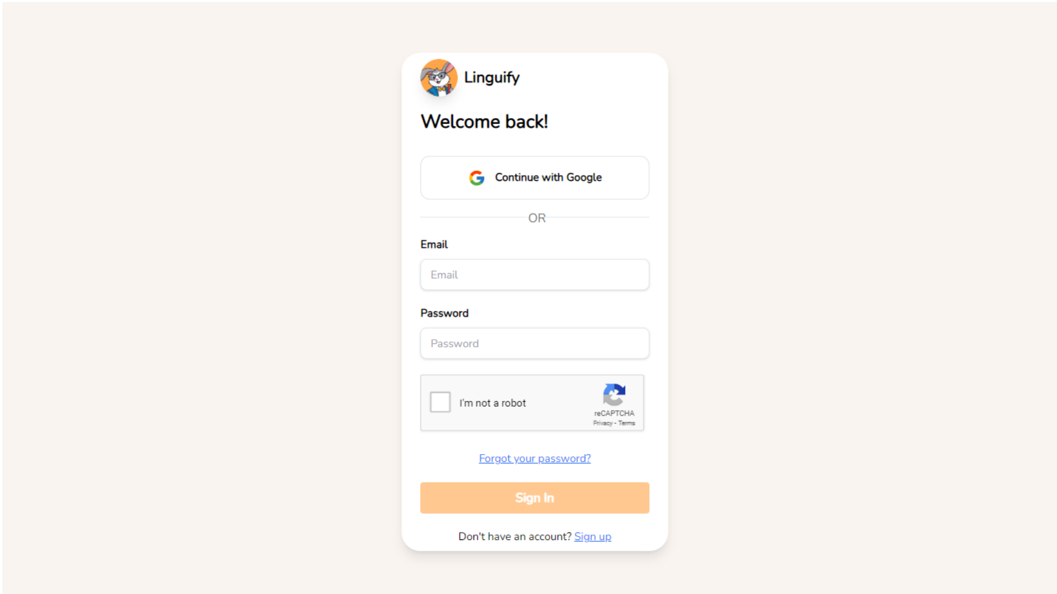

Figure III.6: The Authentication page interface

The sign up interface III.7 presents the learner's sign-up page: This process unfolds in three steps. In the first step, users choose between signing up with Google or manually entering their email, password, and confirming their password.

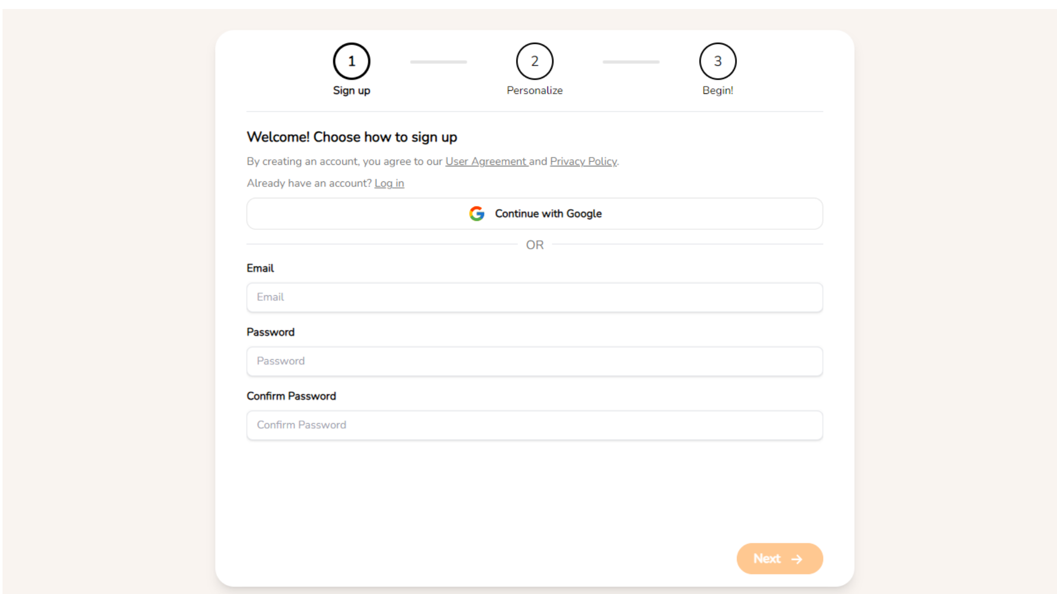

Figure III.7: Learner Sign up Interface

If they choose to manually fill in their information, an email verification will be sent to the provided email address.

(upon completion of the email verification, they will proceed to the second step), as illustrated in Figure III.8.

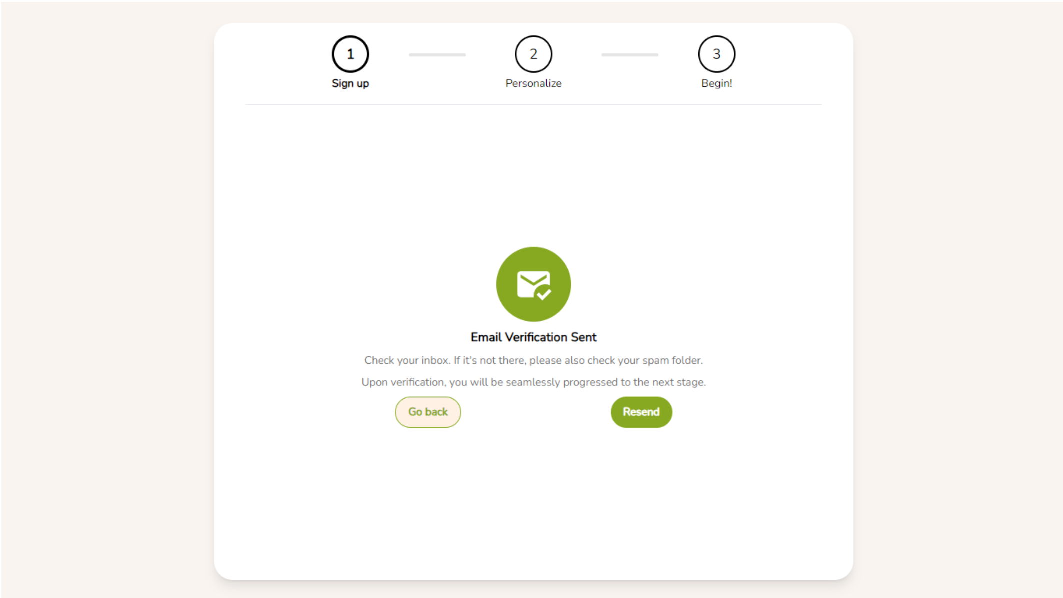

Figure III.8: Learner Email Verification Interface

If they choose to continue with Google, they will automatically proceed to the second step. In this second step, the learner selects their English proficiency level, as illustrated in Figure III.9.

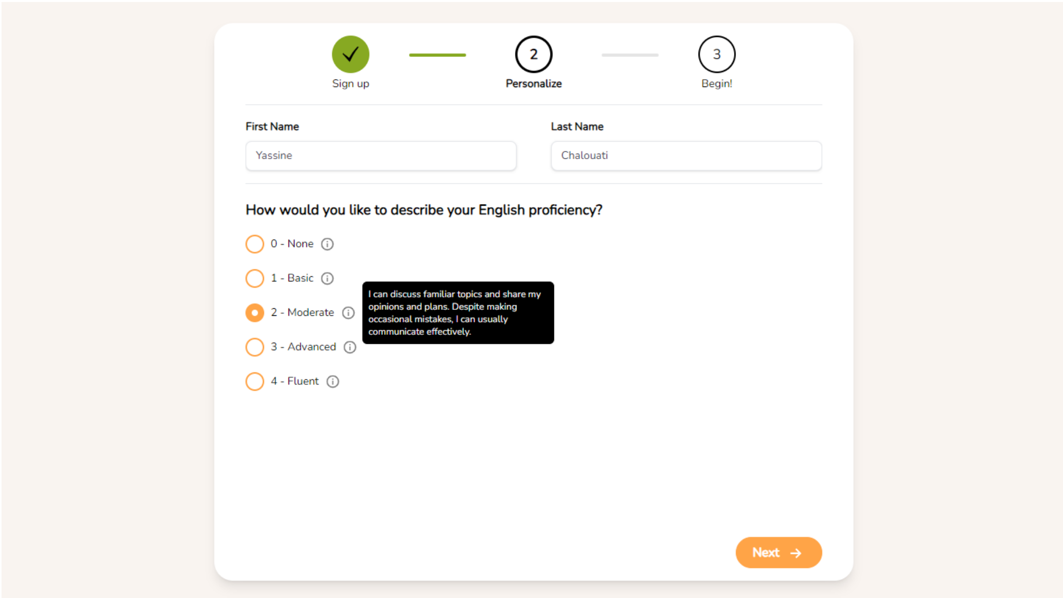

Figure III.9: Step 1: Personalizing Learner Account Interface

In the third and final step, the learner selects their learning goals and favorite topics, as illustrated in Figure III.10.

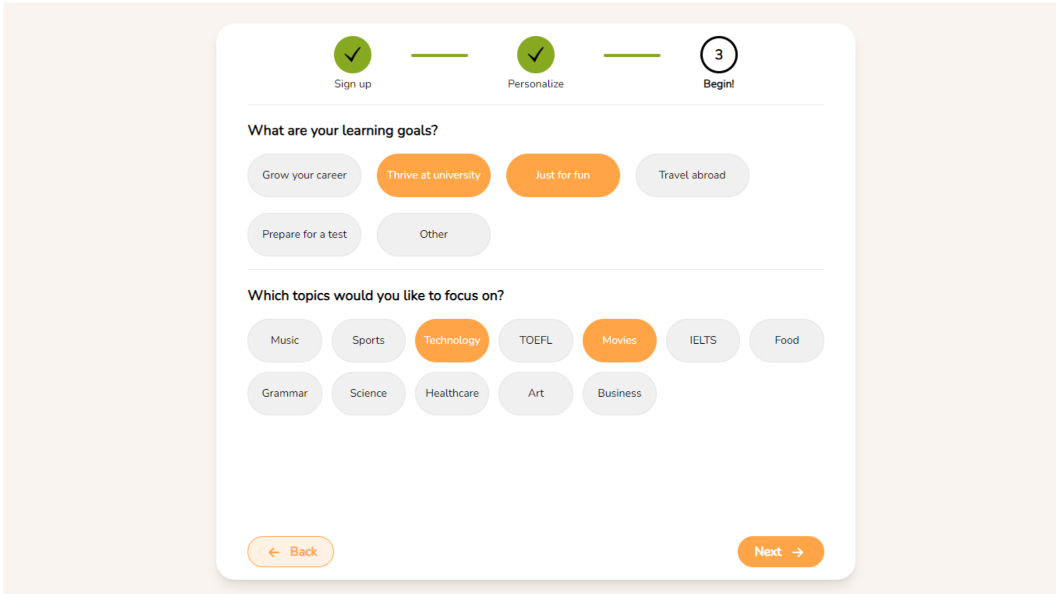

Figure III.10: Step 2: Personalizing Learner Account Interface

After completing these steps, the learner is directed to their profile page, where they can view their information. The figure III.11 displays the interface for the learner's profile.

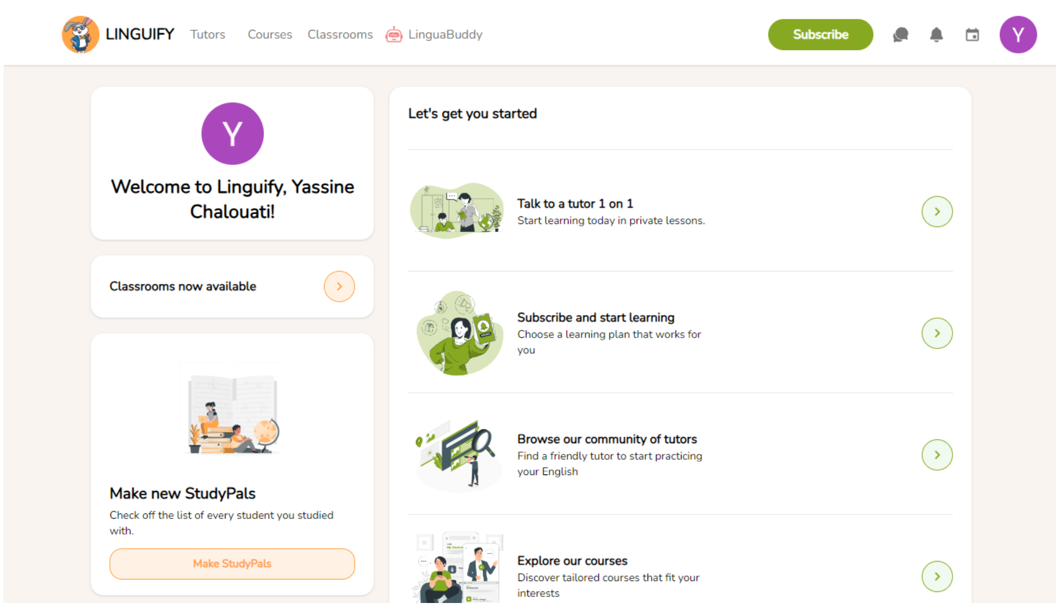

Figure III.11: Learner Profile Interface

The figure III.12 showcases the tutor's sign-up page: This process unfolds in four steps.

Figure III.12: Tutor Sign up Interface

In the first step, the user selects their country, as illustrated in Figure III.13.

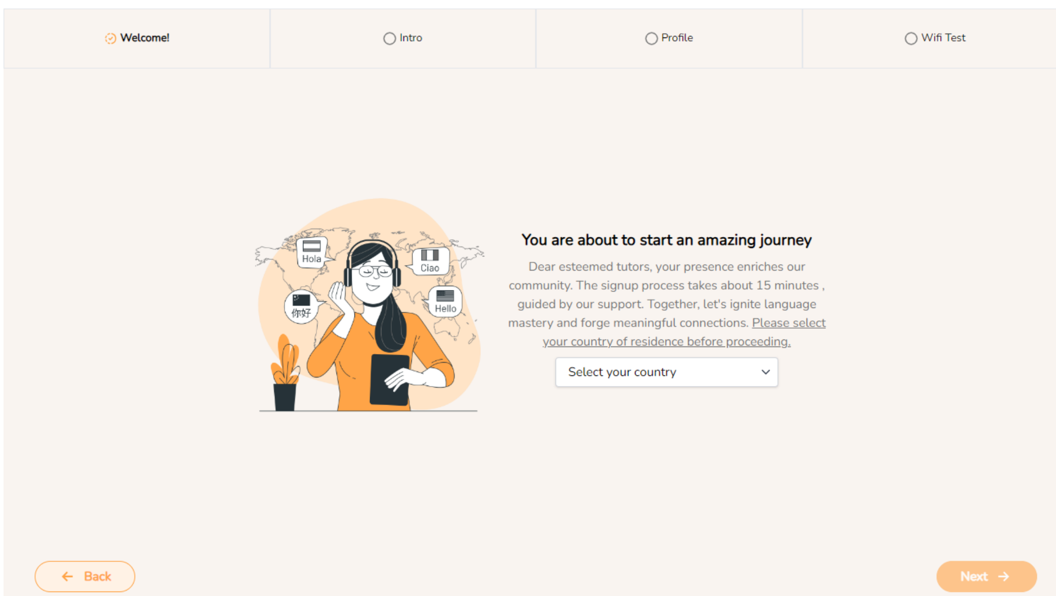

Figure III.13: Step 1: Personalizing Tutor Account - Interface

The figure III.14 shows the second step, also known as "Intro", the tutor enters their account information, which includes their photo, an introductory video, and a description paragraph. We've integrated an AI feature that verifies whether the user has uploaded a formal picture (an image containing a face). If not, the system will reject it.

Face detection AI feature: In our journey of implementing the facial detection AI feature, we pursued two distinct approaches. Initially, we utilized a pre-trained model known as ResNet18, augmenting its training with the data collected from Kaggle. Our dataset comprised two classes: 1. Faces and 2. Other objects excluding faces. However, upon manual testing, we encountered limitations in the model's performance. Subsequently, we transitioned to our second approach, which relied on object detection.Here, we employed a face recognition system capable of identifying and manipulating faces using Python. This system makes use of the power of dlib's state-of-the-art face recognition built with deep learning.

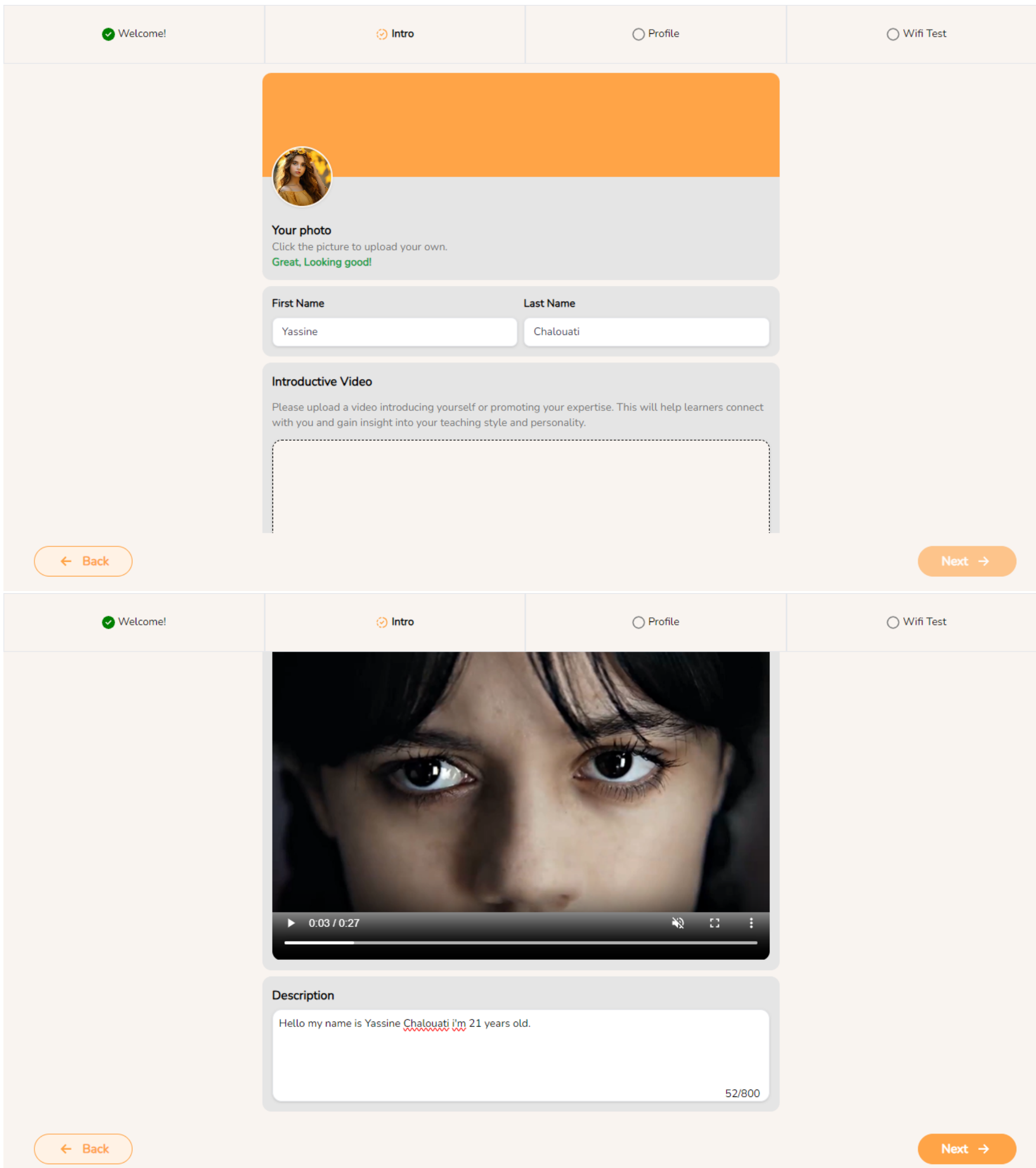

Figure III.14: Step 2: Personalizing Tutor Account - Interface

In the third step, also known as "Profile", the tutor provides more information about themselves by filling in fields such as "Teaching Style" and "About Me", selecting the languages they speak, detailing their work experience, and listing their education, as illustrated in Figure III.15.

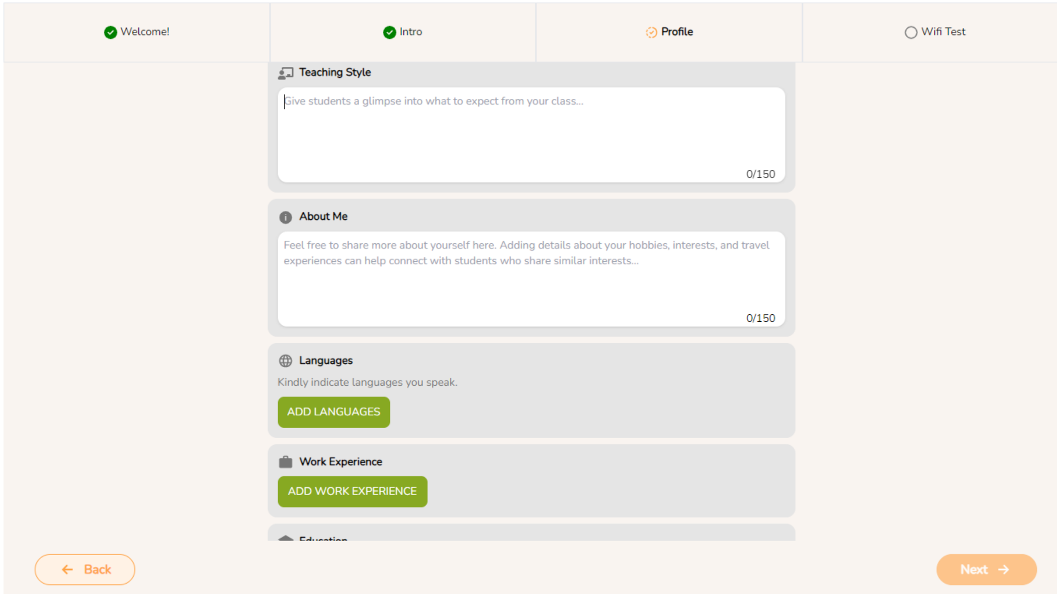

Figure III.15: Step 3: Personalizing Tutor Account - Interface

Here's an example showcased in figure III.16 of how the education section can be filled out or accessed by clicking on the education field.

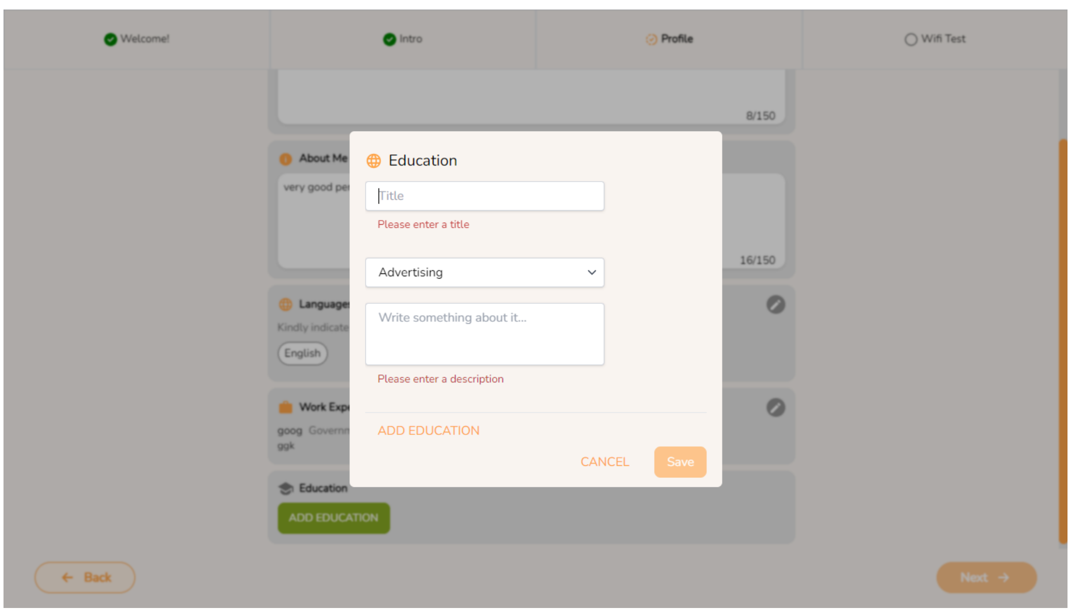

Figure III.16: Step 3: Add Education Window - Interface

In the fourth step, also referred to as "Wi-Fi Test", the user clicks on a button labeled "Start Testing", initiating the system's evaluation of the wifi quality, as illustrated in Figure III.17.

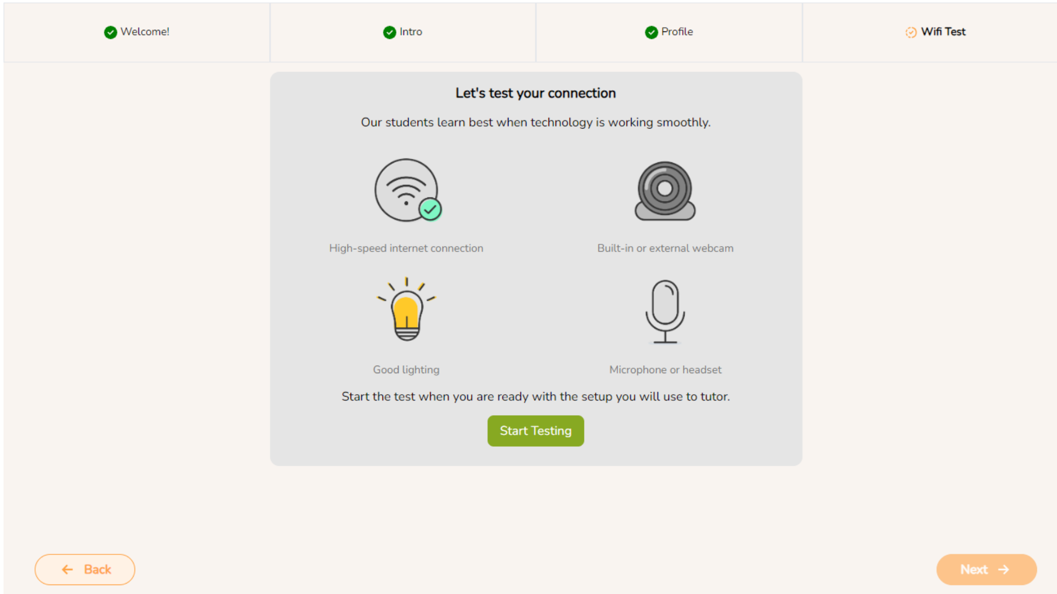

Figure III.17: Step 4: Tutor Wi-Fi Test - Interface

Following the test, the system provides feedback on the quality of the user's wifi and offers advice on potential improvements, as showcased in Figure III.18.

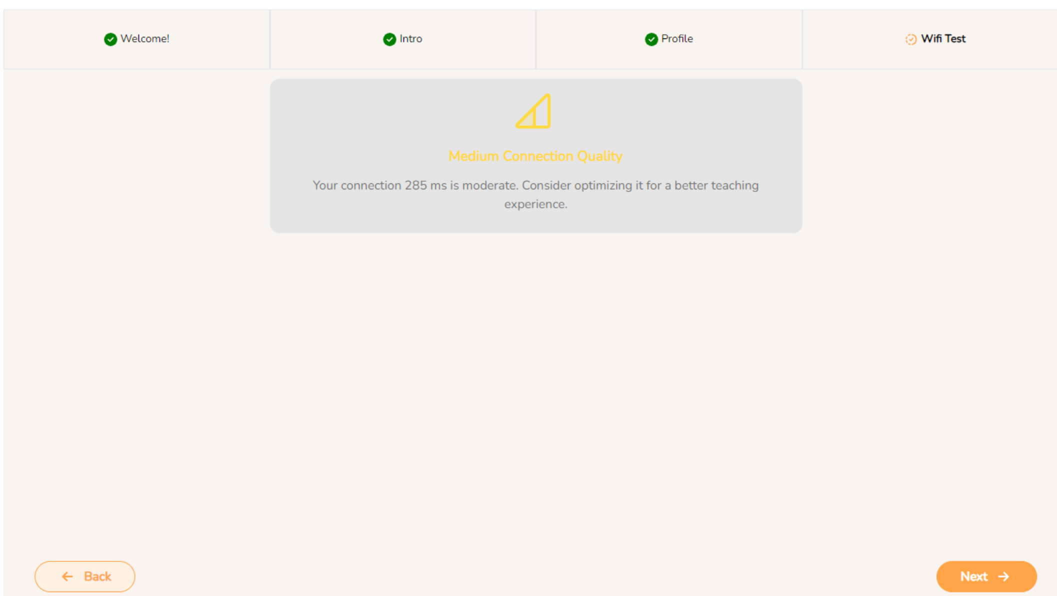

Figure III.18: Step 4: Tutor Wi-Fi Test Result - Interface

Finally, the system redirects the tutor to their profile, where they can access their information, calendar, and any feedback they've received, as illustrated in Figure III.19.

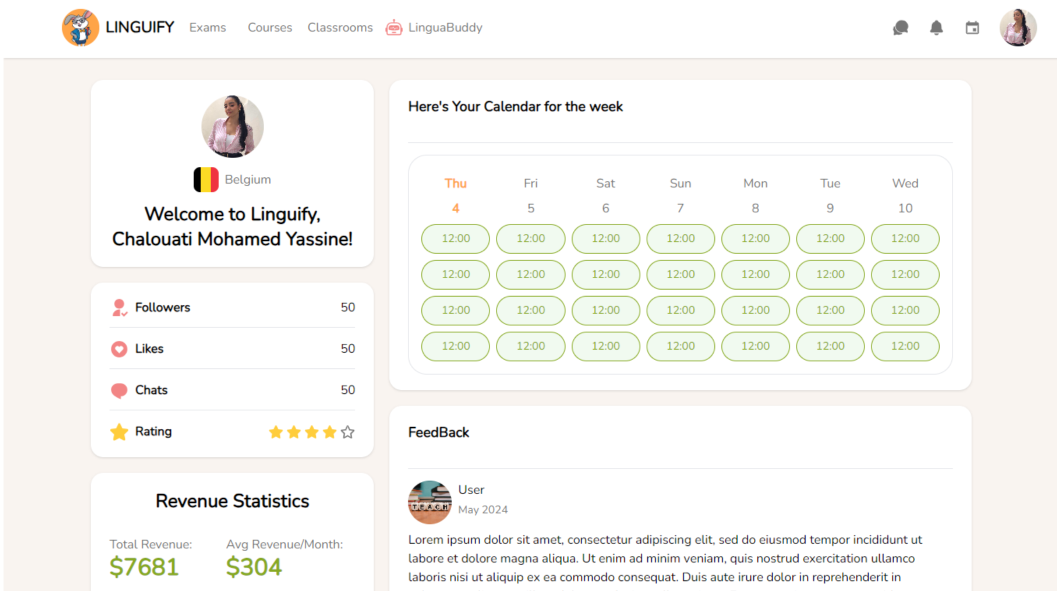

Figure III.19: Tutor Profile Interface

If a user forgets their password, they can go to the sign-in page and click on the "forgot password" link. After entering their email address, the system will send them an email to recover their password, as illustrated in Figure III.20.

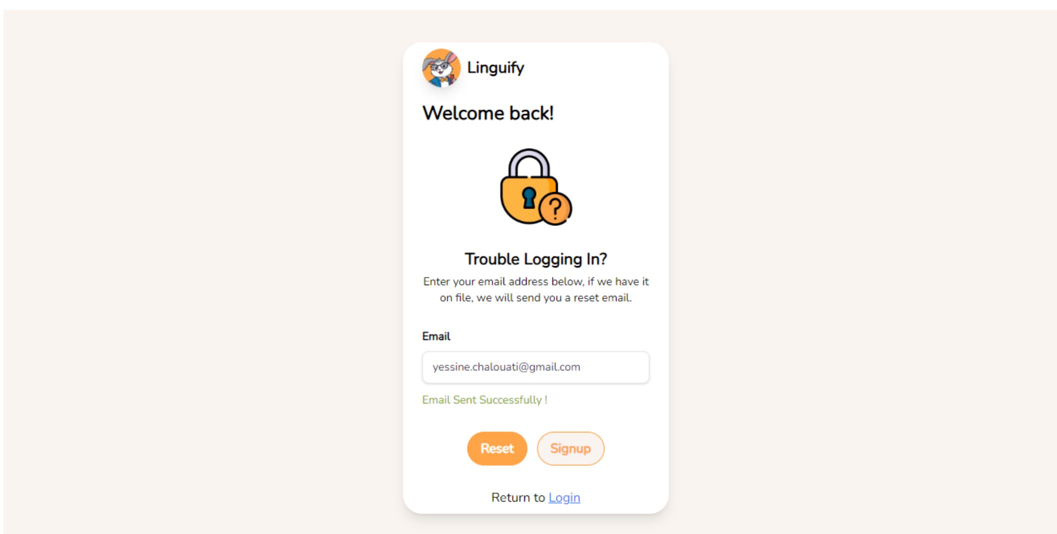

Figure III.20: Recover Password Interface

In this sprint, we also developed the edit account feature. Users can access this by clicking on "Settings," which appears when they click on their profile picture, as illustrated in Figure III.21.

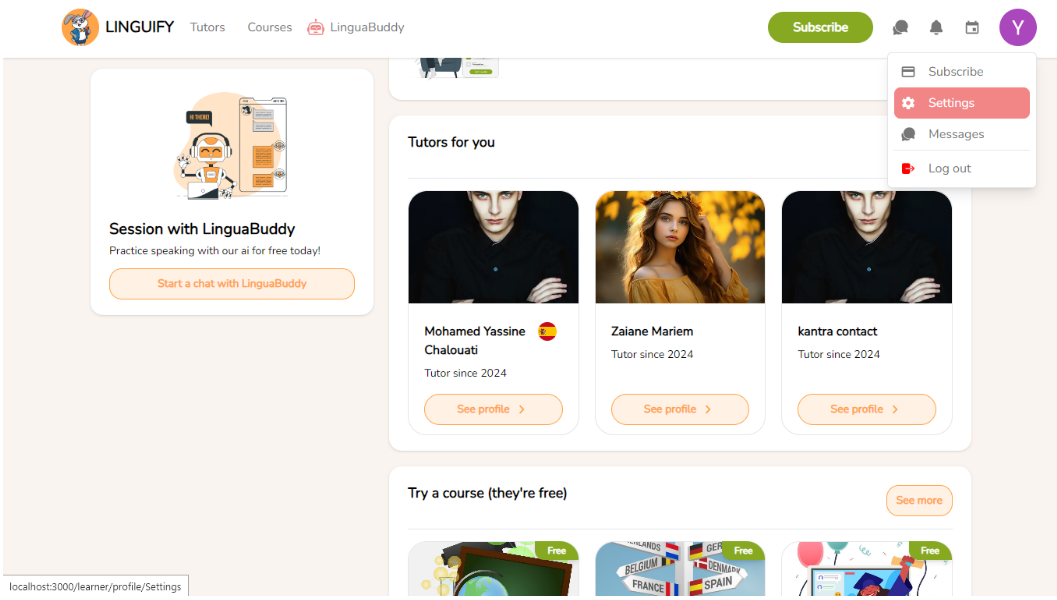

Figure III.21: Settings Interface

Upon clicking on "Settings," the user will be redirected to the settings page, where they can update their student profile, account information, subscription details, or delete account, as illustrated in Figure III.22.

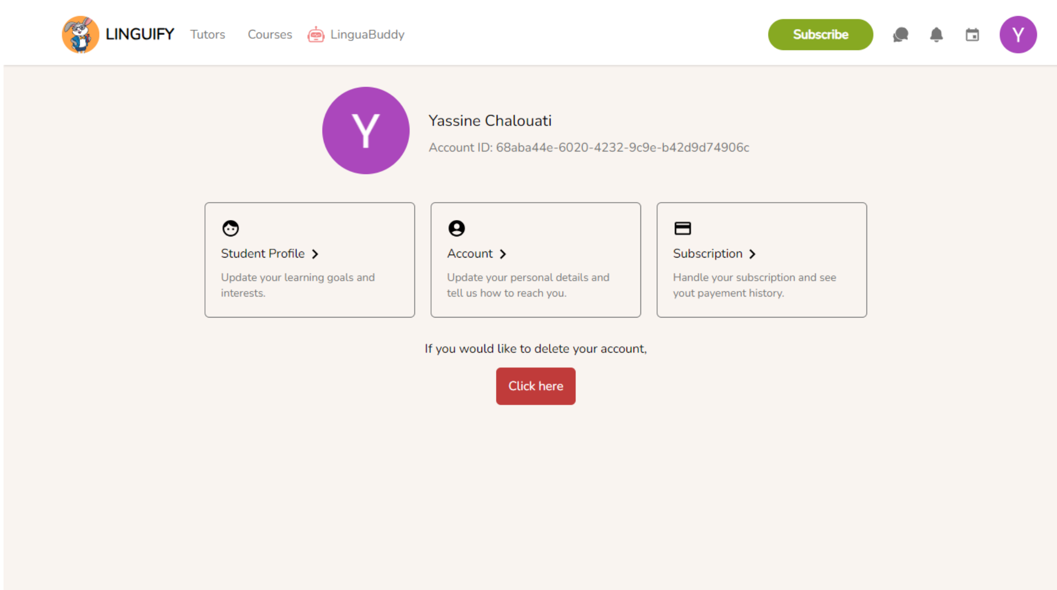

Figure III.22: Settings Page Interface

As illustrated in figure III.23, learners will be able to change their language proficiency, learning goals, and focus topics.

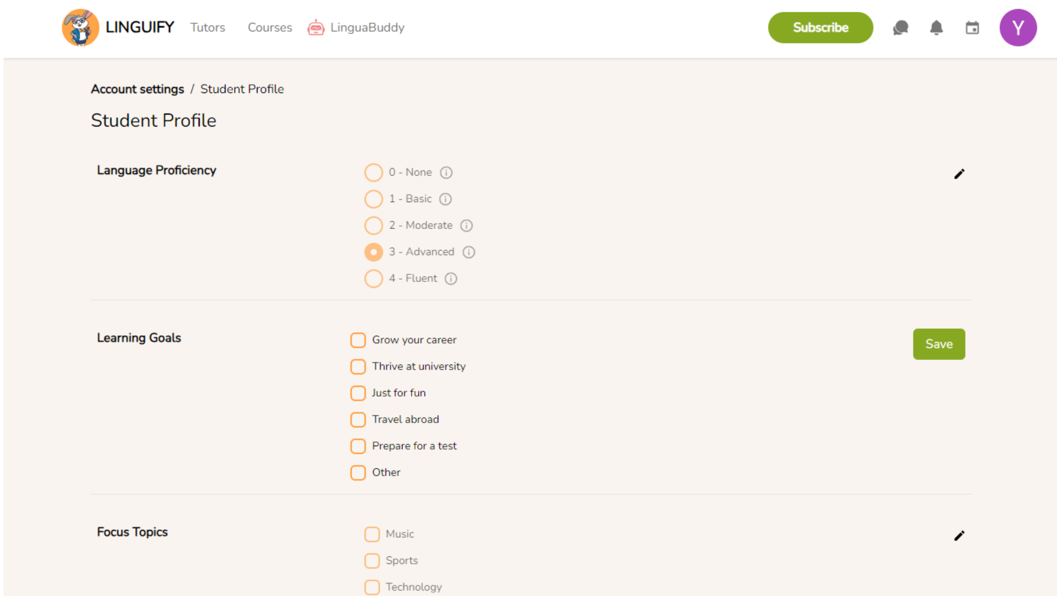

Figure III.23: Settings Page Interface

As illustrated in figure III.24, learners will be able to edit their name, password, mobile number and birthday.

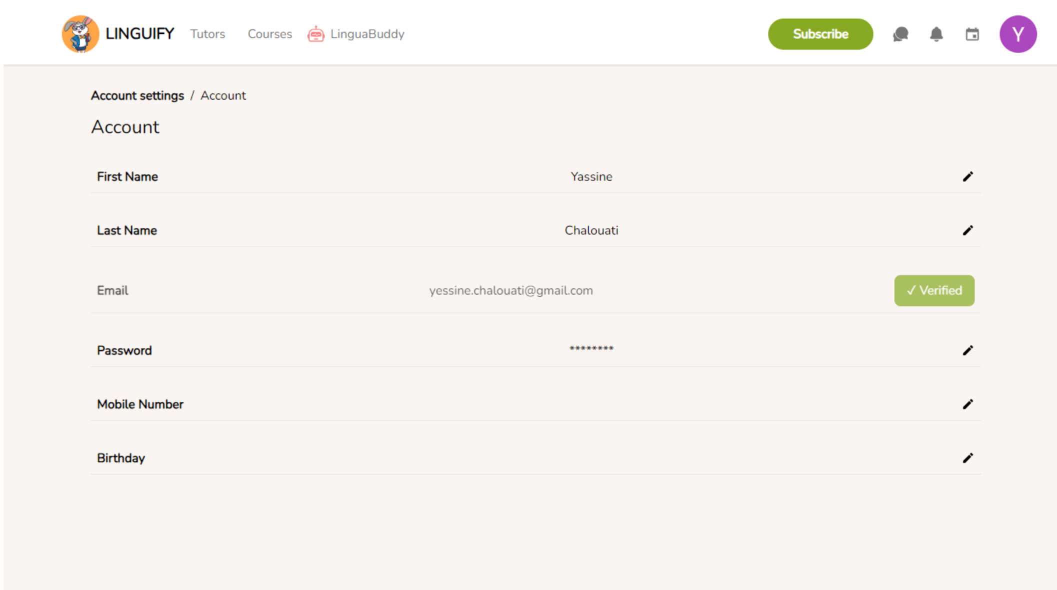

Figure III.24: Settings Page Interface

5 Sprint review

The goal of the Sprint Review is to evaluate the product increment developed throughout the Sprint and determine progress towards the Release. Following a meeting with the Scrum Master and the Product Owner, it was confirmed that the Sprint plan aligned with what was achieved throughout this iteration, validating the completion of the Sprint.

During this sprint, we developed the entire process for account creation for both tutors and learners, ensuring each had a unique flow. Additionally, we implemented authentication and features for managing account information, such as the ability to delete or edit accounts. We are pleased with the work and the features we implemented during this sprint.

6 Conclusion

In conclusion, we have addressed numerous key aspects of Agile development, including sprint backlog, functional specification, design, testing, sprint review, and retrospective. By successfully developing the entire account creation process for tutors and learners, implementing authentication, and adding features for managing account information, we have made significant progress towards our product goals. Additionally, we integrated an AI feature into the tutor's account creation process to enhance the formality of the platform and improve the tutor's image. The collaboration and alignment with the Scrum Master and Product Owner ensured that our efforts were on track and met the planned objectives. We are satisfied with the outcomes of this sprint and look forward to building on this success in future iterations.

Chapter IV

Sprint 2: Lessons Management

Plan

1 Introduction

This chapter focuses on implementing a calendar system for learners and tutors, allowing knowledge seekers to request lessons with any tutor on the platform and enabling tutors to view their upcoming lessons. Additionally, we implemented a notification system to streamline the process by notifying tutors when a learner schedules a lesson and notifying learners when their lesson is accepted. Throughout this part, we will explore the sprint's backlog, functional specification, design, sprint review, and retrospective.

2 Backlog of Sprint 2

We need to start by identifying the list of tasks to be accomplished during this sprint. Table IV.1 represents the user stories of our sprint.

Module	User stories	Complexity	Estimation
Teaching and Learning Activities	As a tutor, I want to see the first upcoming lessons for the current week on my profile.	L	2
	As a user, I want to have a Calendar.	L	2
	As a user, I want to consult upcoming lessons in Calendar.	S	1
	As a learner, I want to book lessons through my Calendar with Tutors based on the lesson's parameters.	L	3
	As a learner, I want to consult tutors' profiles.	S	1
	As a learner, I want to schedule lessons with tutors through their profiles and get notified when they respond to my request.	L	2
	As a learner scheduling from the tutor's profile, I want to easily identify available times that overlap with both my schedule and the tutor's.	M	2
	As a learner, I prefer to view only my available time slots when scheduling lessons.	L	2

Module	User stories	Complexity	Estimation
Communication and Interactions	As a user, I want to be able to mark All my notifications as read.	S	1
	As a user, I want to filter my lessons notifications into categories such as accepted, rejected, and pending.	L	2
	As a tutor, I want to get notified when a learner requests to schedule a lesson with me and efficiently provide a response.	L	2
	As a learner, I want to filter tutors based on criteria, such as their availability, names, languages spoken, and topics they teach.	M	1

Table IV.1: Sprint 2 backlog

3 Sprint Refinement

For this part, first we will be identifying the functional requirements. Then we will draw the use case and the class diagram for this sprint and finally, the textual description of each case.

3.1 Identification of the Functional Requirements

For this sprint, those are the functional requirements that we will be implementing:

- **Search For Tutors** : The goal of this functionality is to allow learners to find Tutors by their names and their specifications.
- **Manage calendar**: The goal of this functionality is to give learners and tutors the ability to have lessons together.
- **Receive notifications**: The goal of this functionality is to send notifications to learners and tutors when a learner schedules a lesson or when a tutor accepts one.

3.2 Global model of use cases for the second Sprint

3.2.1 Use Case Diagram of the Sprint

This sprint focuses on the use cases that describe the different functionalities represented in the use case diagram illustrated in Figure IV.1.

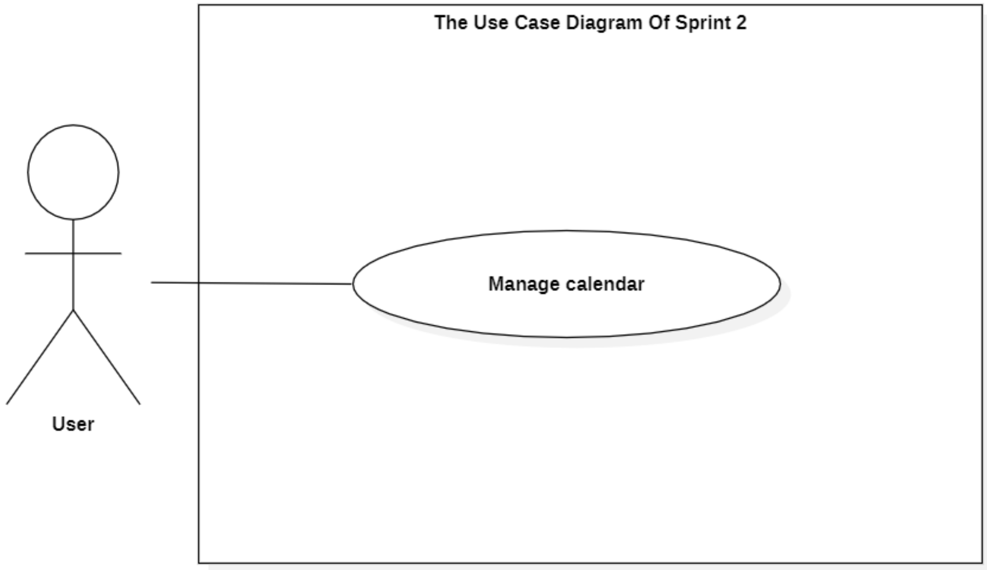

Figure IV.1: Use case diagram sprint 2

3.2.2 Refinement of Use Cases

In this section, we will start with detailing the"Manage Calendar" use case from sprint 2 as illustrated in Figure IV.2.

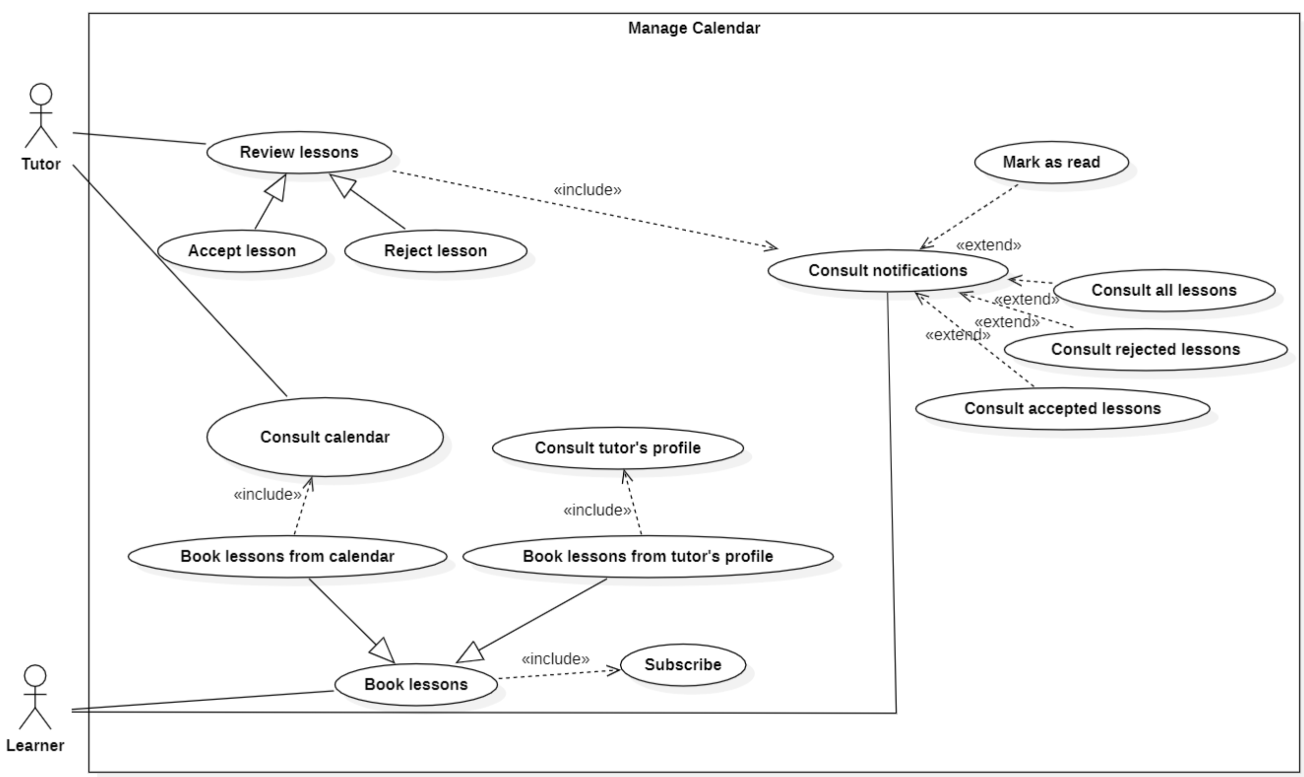

Figure IV.2: Refinement of «Manage calendar» use case

The table IV.2 represents a description for the Book lessons from calendar use case.

IDENTIFICATION SUMMARY	
Title	Book lessons from calendar
Actor(s)	Learner
DESCRIPTION OF SEQUENCES	
Pre conditions	User needs to be authenticated in order to access the system.
Principal Scenario	
1.The user clicks on the button with the Calendar icon situated in the navigation bar. 2.The system displays the Calendar with first lessons for the next days being displayed. 3.The user presses the "Add" button to schedule a lesson. 4.The system displays the first step modal, showcasing only the available free time for the current day, and the duration option. 5.The user selects the convenient duration and time for his potential lesson and presses the "Select preferences" button. 6.The systems displays the second step modal, showcasing the topic, language and the lesson's difficulty options. 7.The user fills the information of the lesson and presses the "Select tutor" button. 8.The system presents the third modal, revealing the available tutors for the selected time, matching the learner's preferences. 9.The user selects the convenient tutor for his case. 10.the system adds the lesson to the calendar and notifications of the learner as a pending lesson and sends the lesson to the selected tutor.	
Alternative Scenario	
- If the user selects lesson options for which there are no available tutors, the system will display an indicator in the third step, and the user won't be able to schedule a lesson. - If the user doesn't select the options, they won't be able to click on the button that takes them to the next step.	

Table IV.2: Textual description of the use case Book lessons from calendar

Table IV.3 represents a description for the "Manage reviewing lessons" use case.

IDENTIFICATION SUMMARY	
Title	Manage reviewing lessons
Actor(s)	Tutor
DESCRIPTION OF SEQUENCES	
Pre conditions	User needs to be authenticated in order to access the system.
Principal Scenario	
1.The user clicks on the button with the notifications icon situated in the navigation bar. 2.The system displays the Notifications. 3.The user presses the "Pending" button to see all the lessons that has been requested. 4.The system displays all the lessons that has been requested and not responded to yet. 5.The user Accepts or declines these lessons by clicking on one of the icons, either the tick or the cross, within the notification. 6.The system shifts the notification from the "Pending" section to either the "Accepted" or "Rejected" section based on the user's response, relocating it to "Accepted" if approved, and to "Rejected" if declined.	

Table IV.3: Textual description of the use case the Manage reviewing lessons

3.2.3 Sequence diagram

The sequence diagram in Figure IV.3 represents the steps to be followed by a learner in order to book a lesson.

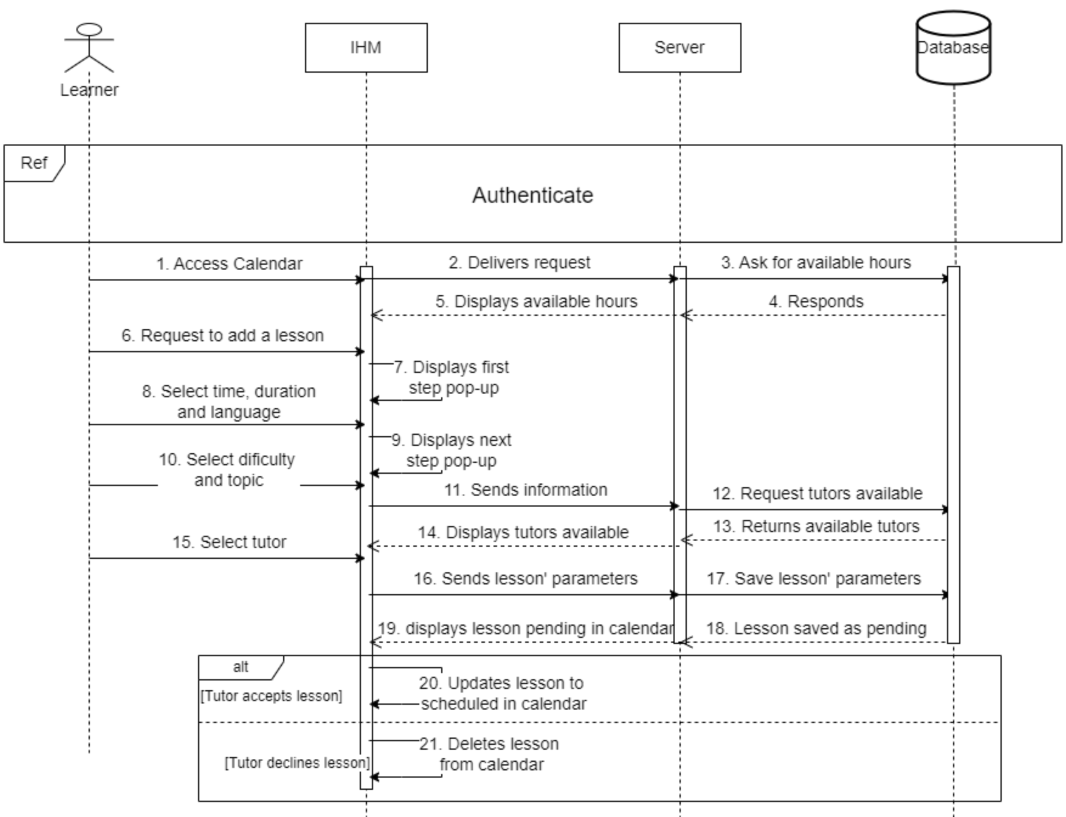

Figure IV.3: Sequence diagram of "Book lesson"

The figure IV.4 presents the sequence diagram for Consult lesson.

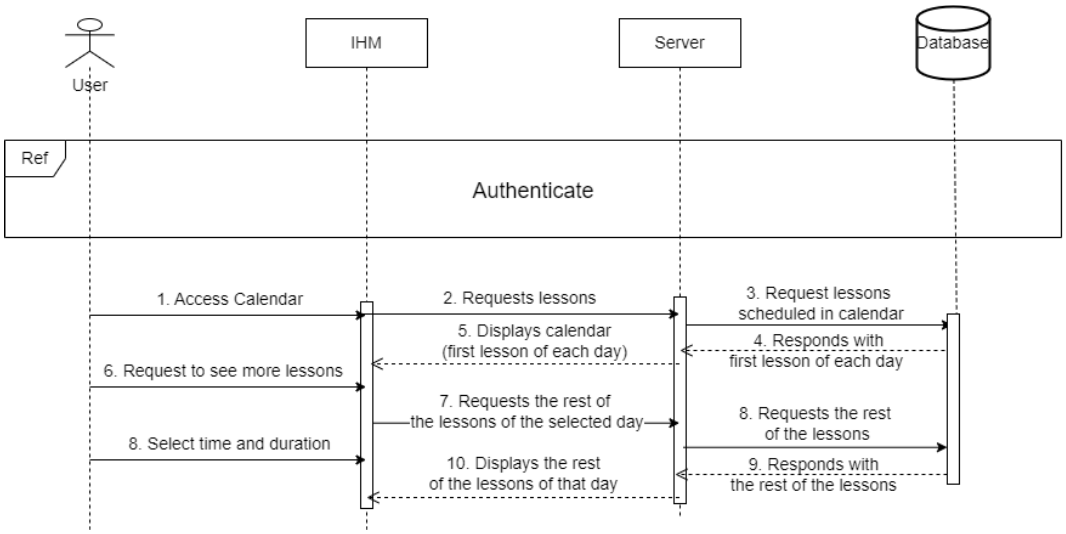

Figure IV.4: Sequence diagram of "Consult lesson"

3.2.4 Class diagrams

The figure IV.5 shows the class diagram of sprint 2.

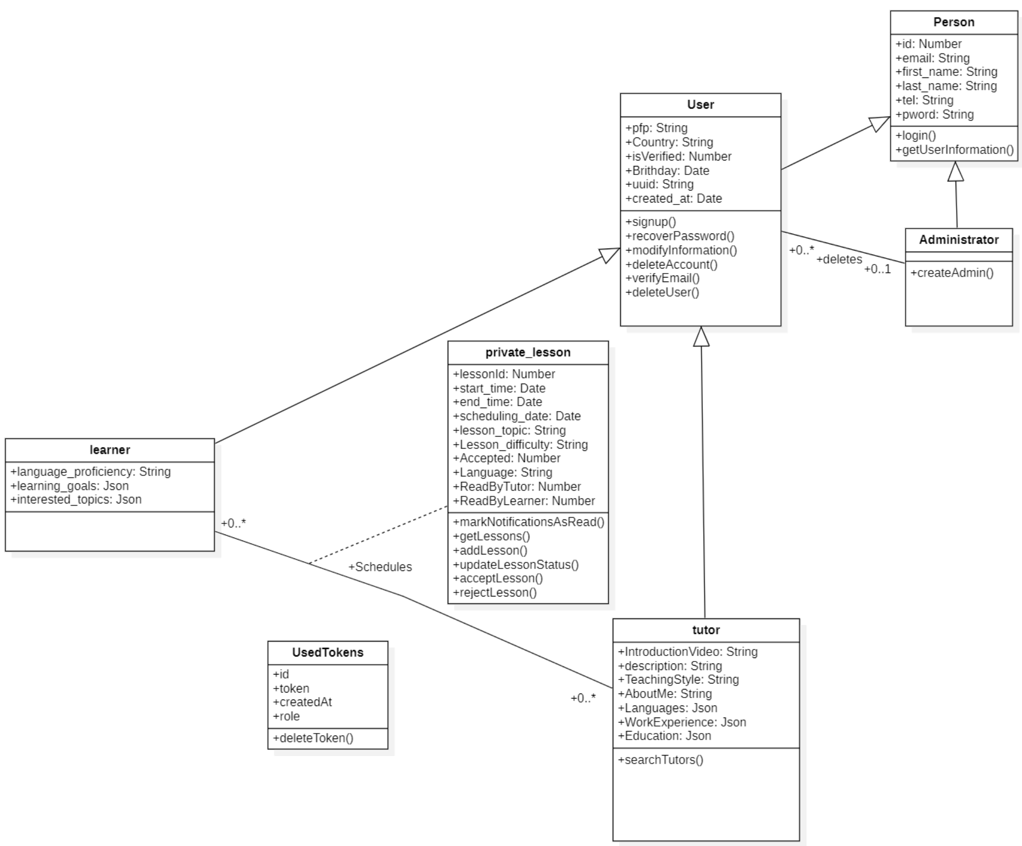

Figure IV.5: Class diagrams for sprint 2

4 Realisation

This section will present multiple screenshots to describe the interfaces of our tool and thus provide a better understanding of its functionality.

Upon clicking on the Calendar icon situated in the navigation bar, our users will have access to their calendars, as illustrated in the figure IV.6.

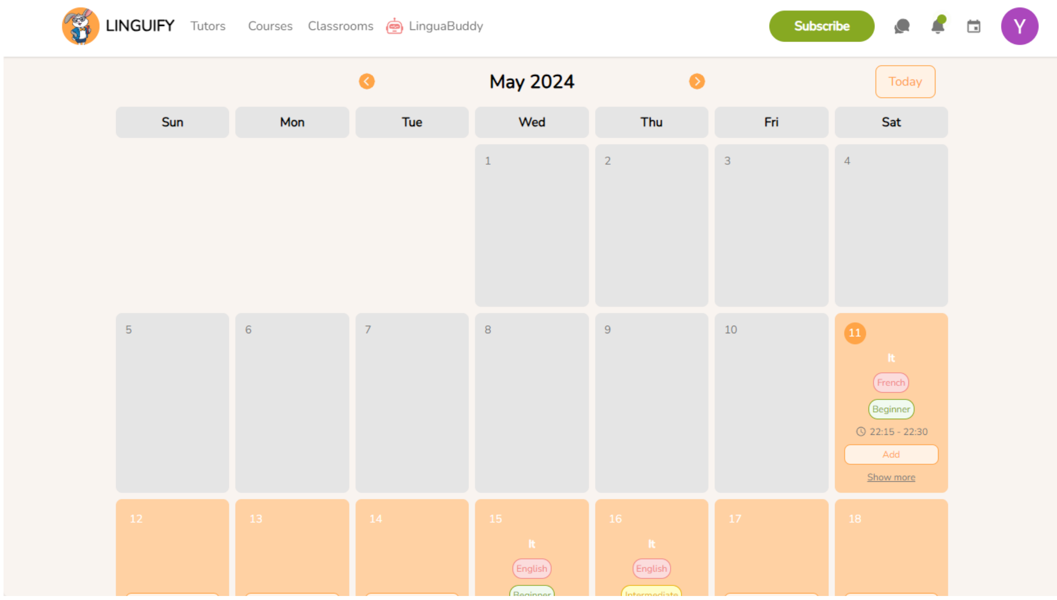

Figure IV.6: Interface of the "Calendar" page

Upon clicking on the "Add" button, the learner will be presented with the first step to schedule a lesson, as illustrated in the figure IV.7.

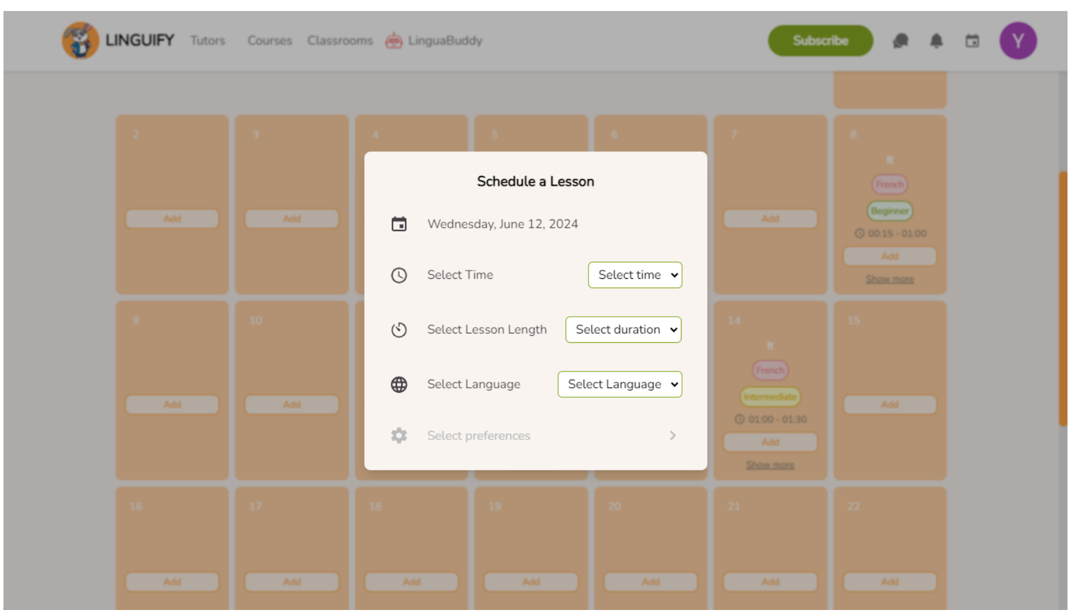

Figure IV.7: Step 1: Selecting lesson time, duration and language - Interface

After selecting the time, the duration and the language, the learner can press the "Select preferences", as illustrated in the figure IV.8.

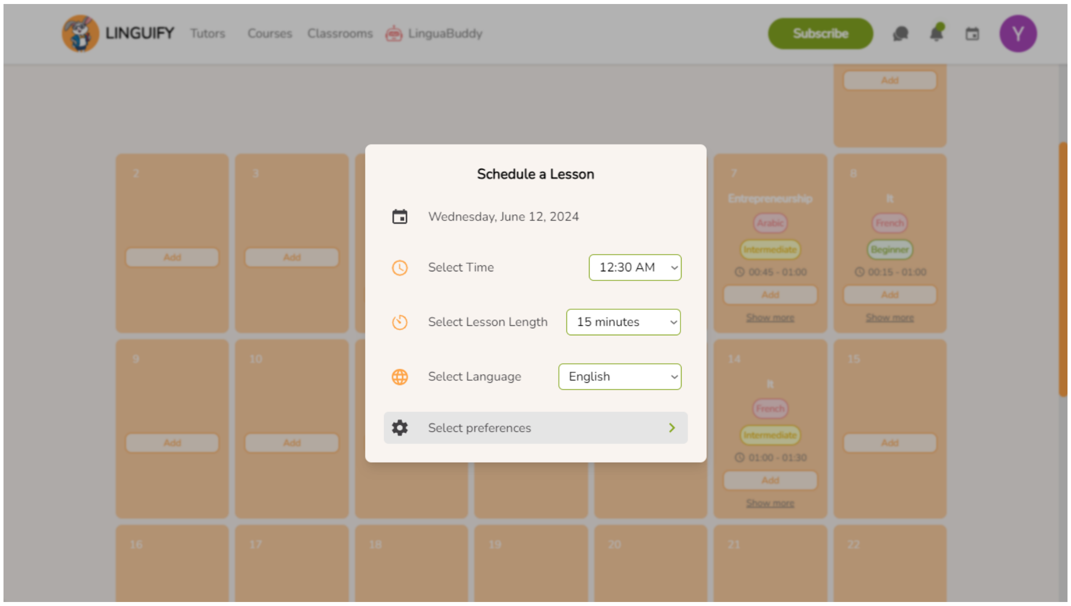

Figure IV.8: Step 1: Successful Selection - Interface

After clicking the "Select preferences" button, the learner will be presented with the second step of the schedule lesson process, as illustrated in the figure IV.9.

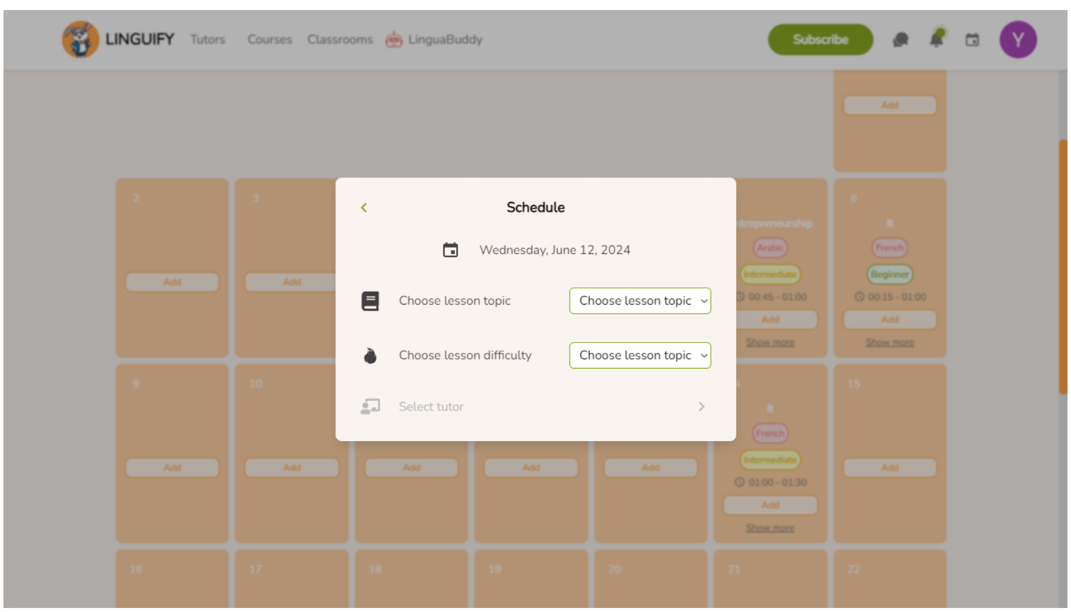

Figure IV.9: Step 2: Selecting lesson topic and difficulty - Interface

After selecting the lesson topic and difficulty the learner will be taken to the last step of the process, as shown in the figure IV.10.

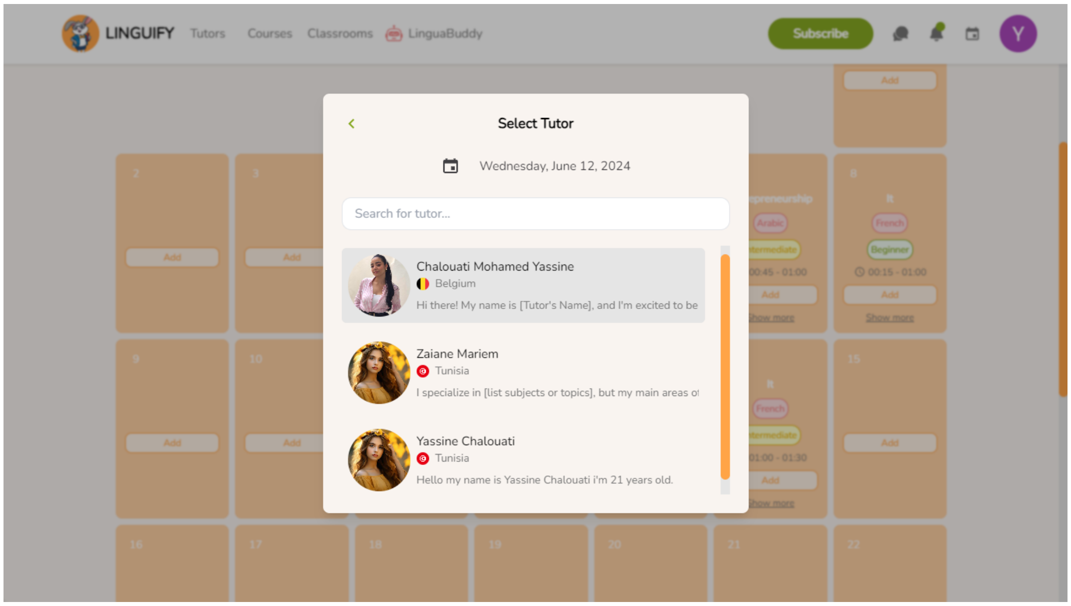

Figure IV.10: Step 2: Selecting Tutor - Interface

After finalizing the scheduling process, the learner will be notified with the scheduled lesson, as illustrated in the figure IV.11.

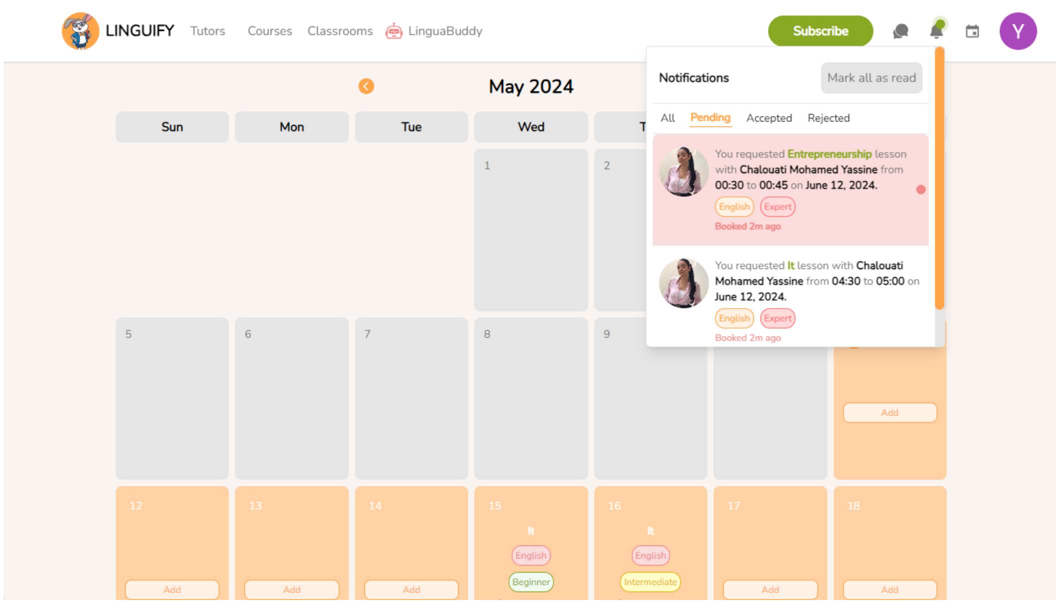

Figure IV.11: Pending learner notification Interface

Also the tutor gets notified with the scheduled lesson giving him the ability to either accept or reject it, as illustrated in the figure IV.12

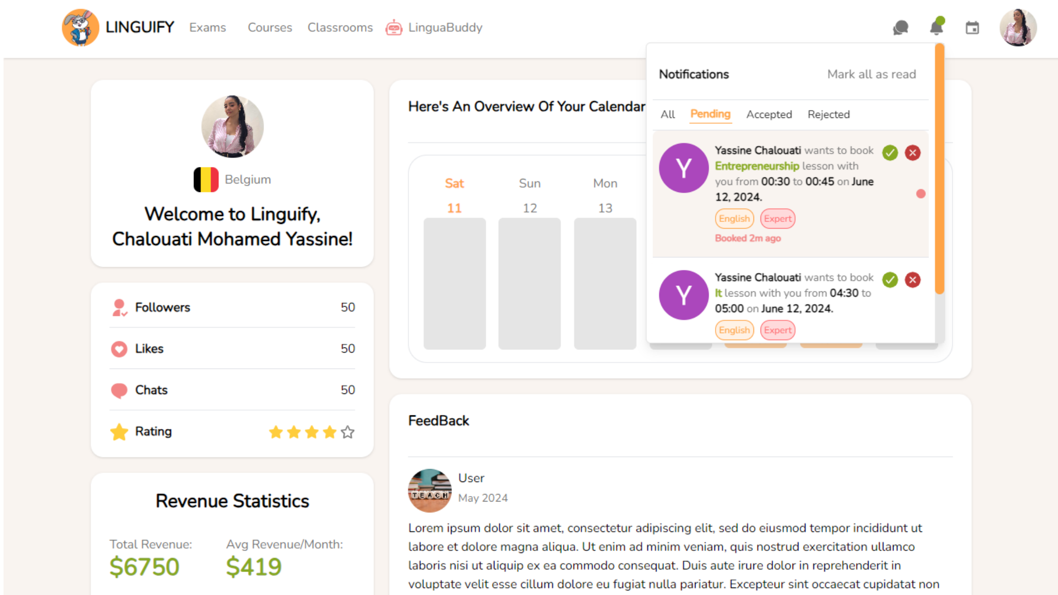

Figure IV.12: Pending tutor notification Interface

In case the tutor accepts the lesson, the lesson gets added to the tutor calendar, as illustrated in the figure IV.13.

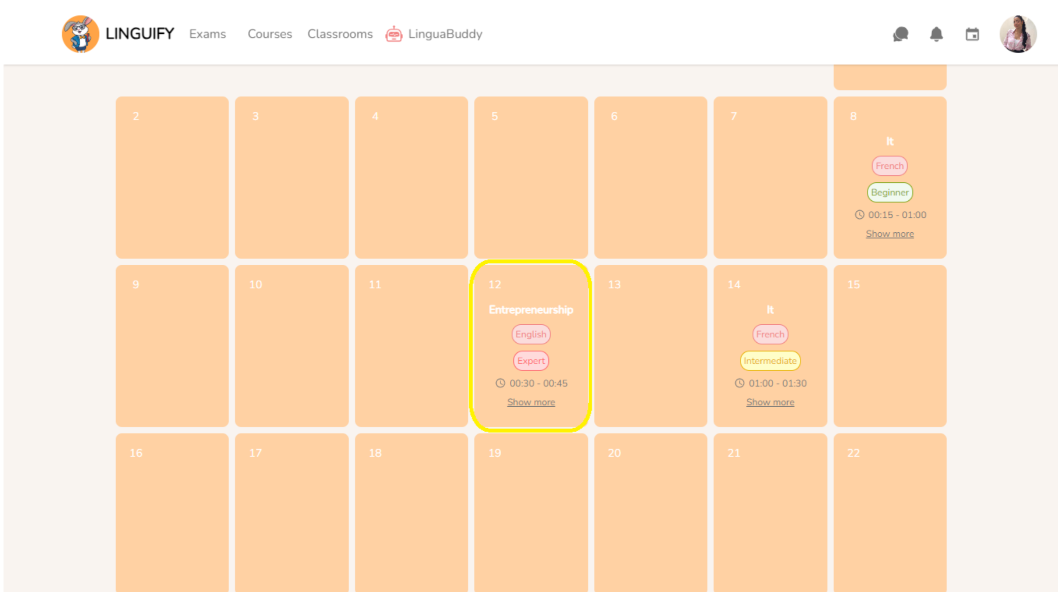

Figure IV.13: Tutor Scheduled Lesson Calendar Interface

If the lesson accepted is within the current week, it gets added in the learner's profile, as illustrated in the figure IV.14.

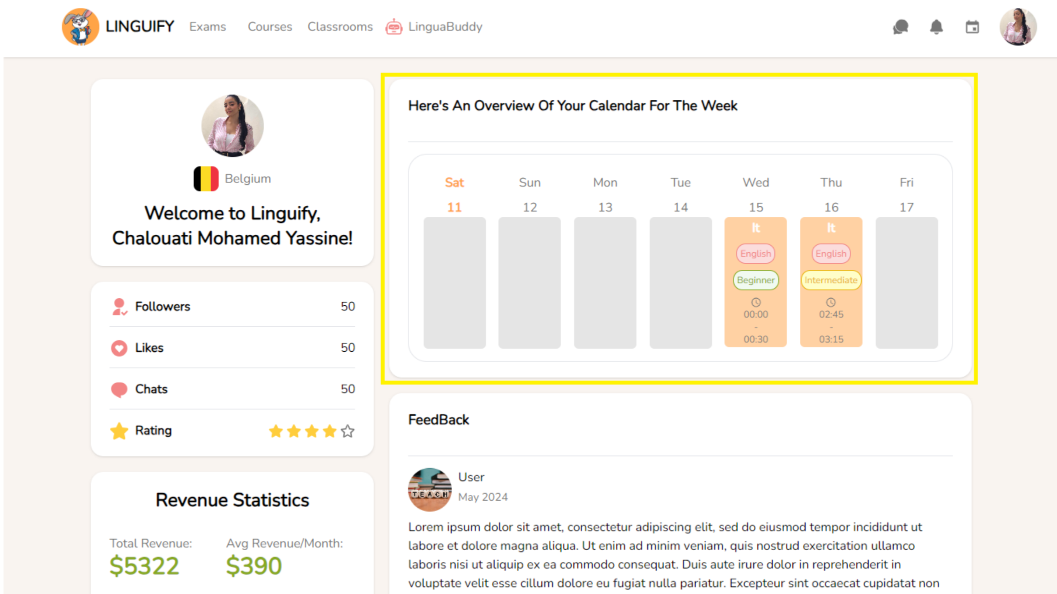

Figure IV.14: Profile lessons Interface

Upon accepting The lesson, the concerned learner will get notified, as illustrated in the figure IV.15.

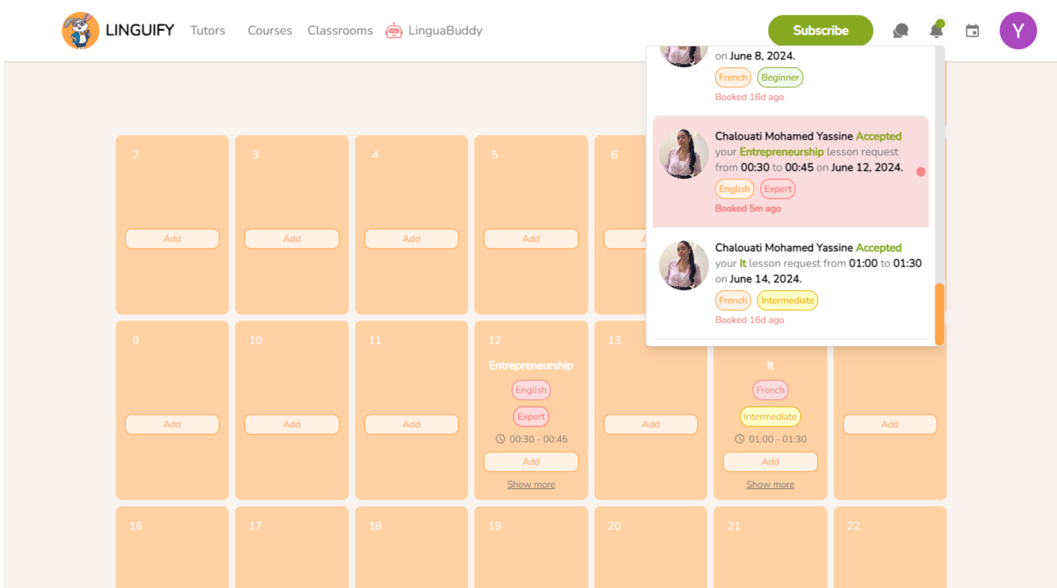

Figure IV.15: Accepted learner lesson notification Interface

Upon rejecting a lesson, the concerned learner will get notified, as illustrated in the figure IV.16.

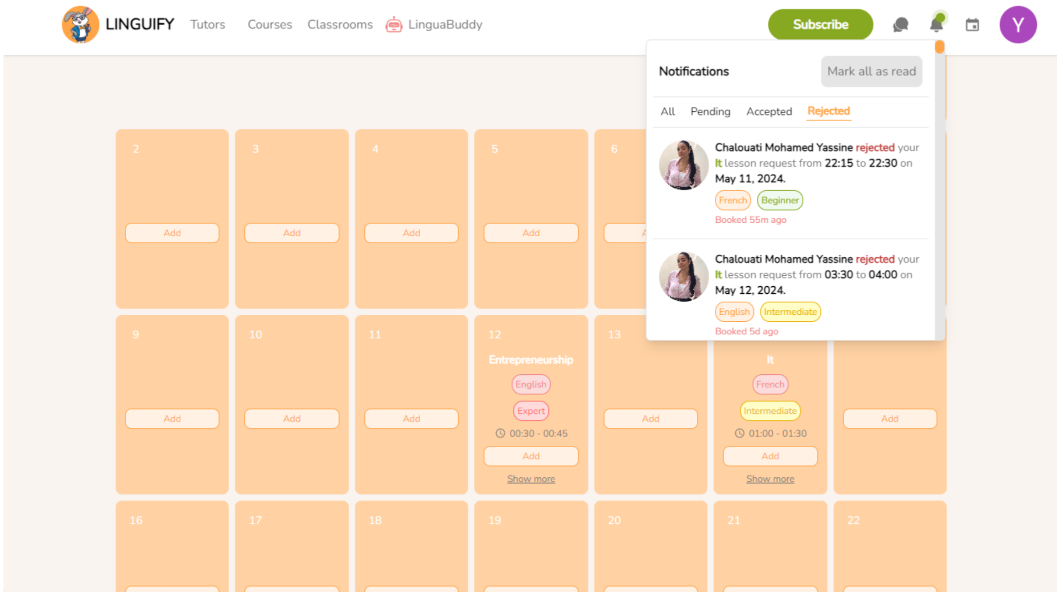

Figure IV.16: Rejected learner lesson notification Interface

After pressing the 'Mark all as read' button in the notifications section, all lessons associated with the user will be marked as read, and the notification bell will be removed, as shown in the Figure IV.17.

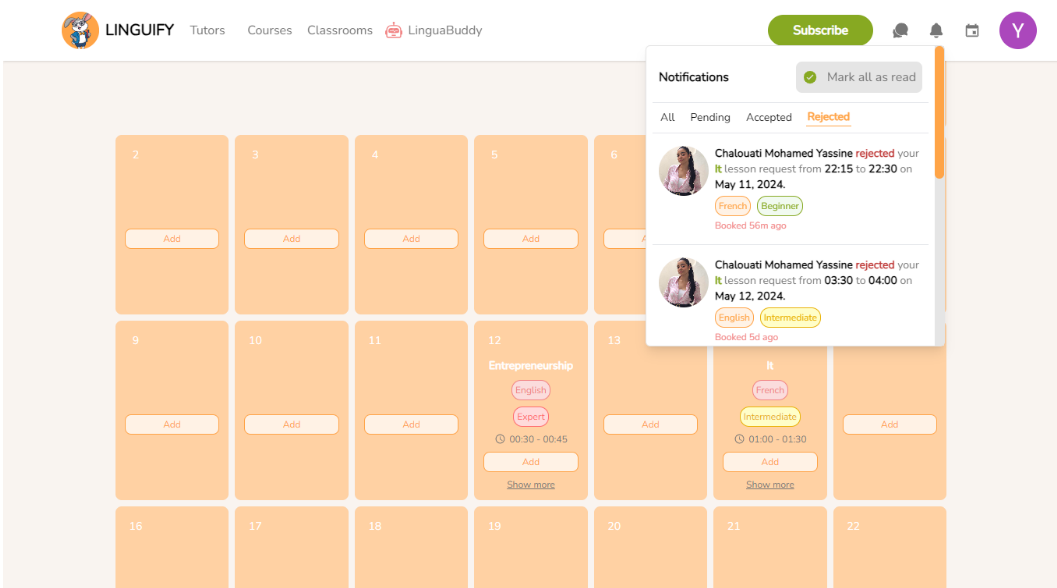

Figure IV.17: Mark all as read Interface

After pressing the 'Show more' button located within one of the calendar's days, the user gains access to every scheduled lesson for that specific day, as showncased in the figure. IV.18.

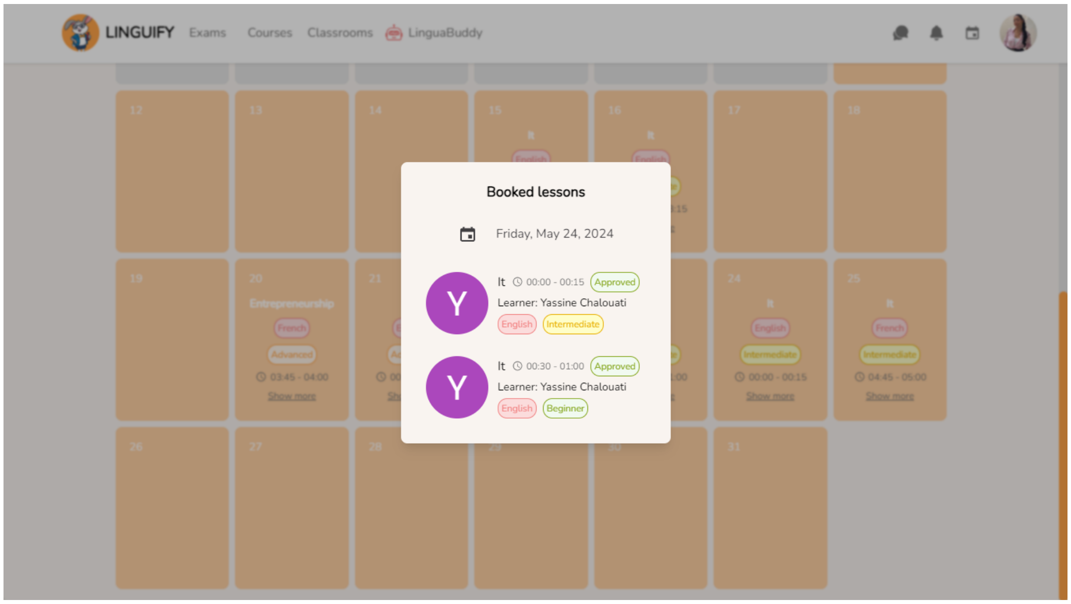

Figure IV.18: Show more lessons Interface

The learner can search for tutors after pressing the Tutors section from his navigation bar, as illustrated in the figure IV.19.

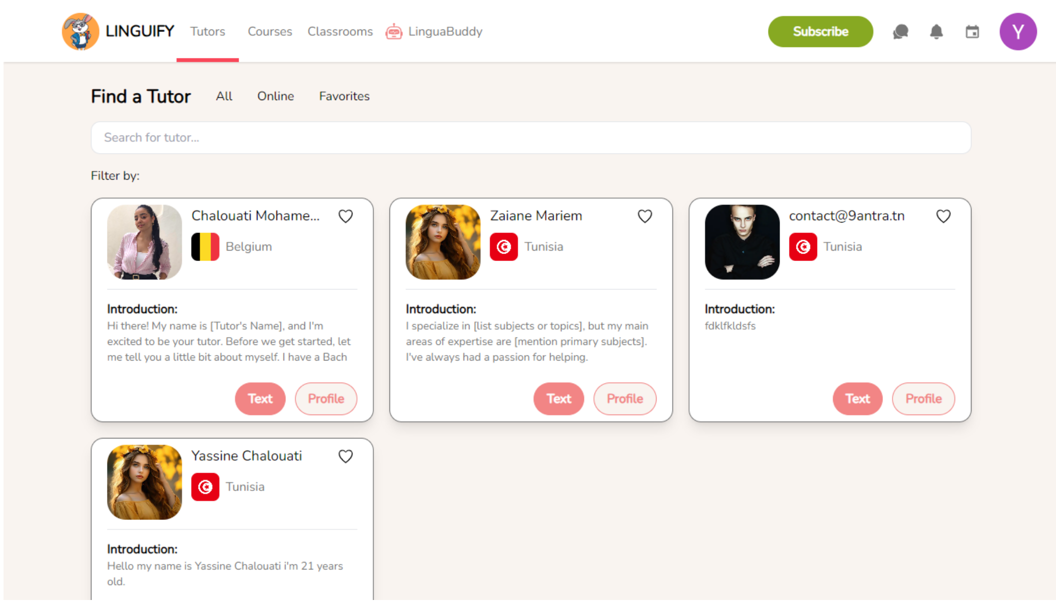

Figure IV.19: Search tutor profile for learner Interface

5 Sprint Review

Overall, the sprint was a success, and we achieved all our goals. We developed a calendar system for both tutors and learners, from which learners can schedule lessons with any tutor they want, or through the tutor's calendar after consulting his profile. Also, we ensured that lesson scheduling is ideal by implementing collision detection, preventing any overlapping lessons. We are satisfied with the results achieved during this sprint.

6 Conclusion

The sprint aimed at creating and implementing a calendar system. It has been successfully completed. We managed to develop a powerful calendar system that allows learners to schedule lessons effortlessly with tutors, also setting up a real time notification system using socket io for scheduling, we made learners able to find tutors through a searching system, consult their profiles, and schedule lessons through their profiles. We are excited about the significant impact this calendar system will have on our platform and look forward to further development.

Chapter V

Sprint 3: Communication and Interactions

Plan

1 Introduction

This chapter focuses on enhancing user interaction within the platform. Learners can schedule 1-on-1 study sessions with tutors and access a built-in chatbot for assistance. They can also favorite tutors, rate them, and receive notifications for messages and new course creations. Tutors can engage directly with learners through chat. Additionally, learners benefit from a recommendation system for discovering tutors and can initiate video calls for study sessions. In this chapter, we'll delve into the sprint's backlog, functional specifications, design, sprint review, and retrospective.

2 Backlog of Sprint 3

We need to start by identifying the list of tasks to be accomplished during this sprint. Table V.1 represents the user stories of our sprint.

Module	User stories	Complexity	Estimation
Communication and Interactions	as a learner, I want to study 1 on 1 with a tutor	L	3
	As a learner or tutor, I want to talk to a built-in chatbot for assistance.	L	2
	As a learner, I want to like my favorite tutors.	S	1
	As a learner, I want to have a list of my favorite tutors.	S	1
	As a tutor, I want to chat with learners who seek language tutoring.	L	3
	As a learner, I want to chat with tutors.	L	3
	As a learner, I want to see tutors through a recommendation system.	S	1
	As a learner, I want to receive messages' notifications.	L	2

Table V.1: Sprint 3 backlog

3 Sprint Refinement

For this part, first we will be identifying the functional requirements. Then we will draw the use case and the class diagram for this sprint and finally, the textual description of each case.

3.1 Identification of the Functional Requirements

For this sprint, those are the functional requirements that we will be implementing:

- **Tutor-Learner interactions** : The goal of this functionality is to enable interaction between learners and tutors through various means such as chatting, calling, and liking.

- **Video calls** : The goal of this functionality is to enable learners and tutors to engage in video calls with access to all associated features while having a lesson together.
- **Building a chatbot** : The goal of this functionality is to provide users with access to a built-in chatbot.

3.2 Global model of use cases for the third Sprint

3.2.1 Use Case Diagram of the Sprint

This sprint focuses on the use cases that describe the different functionalities represented in the use case diagram illustrated in Figure V.1.

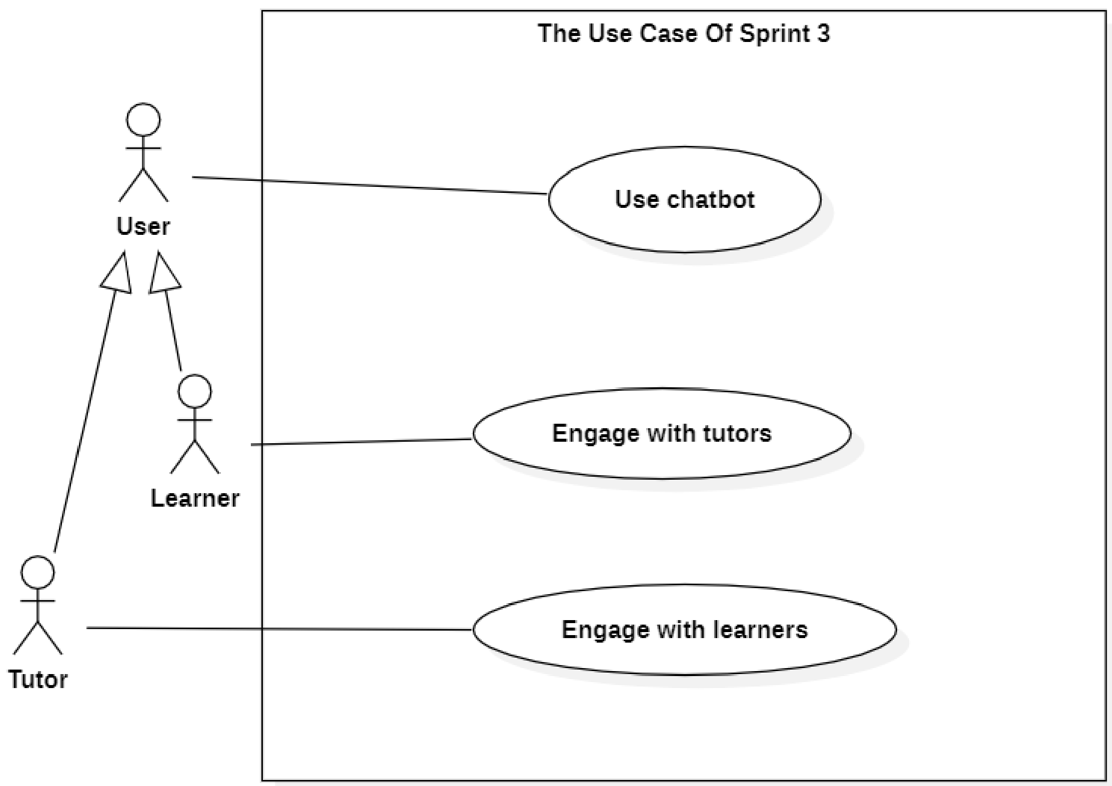

Figure V.1: Use case diagram sprint 3

3.2.2 Refinement of Use Cases

In this section, we will start with detailing the "Communication and Interactions" use case from sprint 3 as illustrated in Figure V.2.

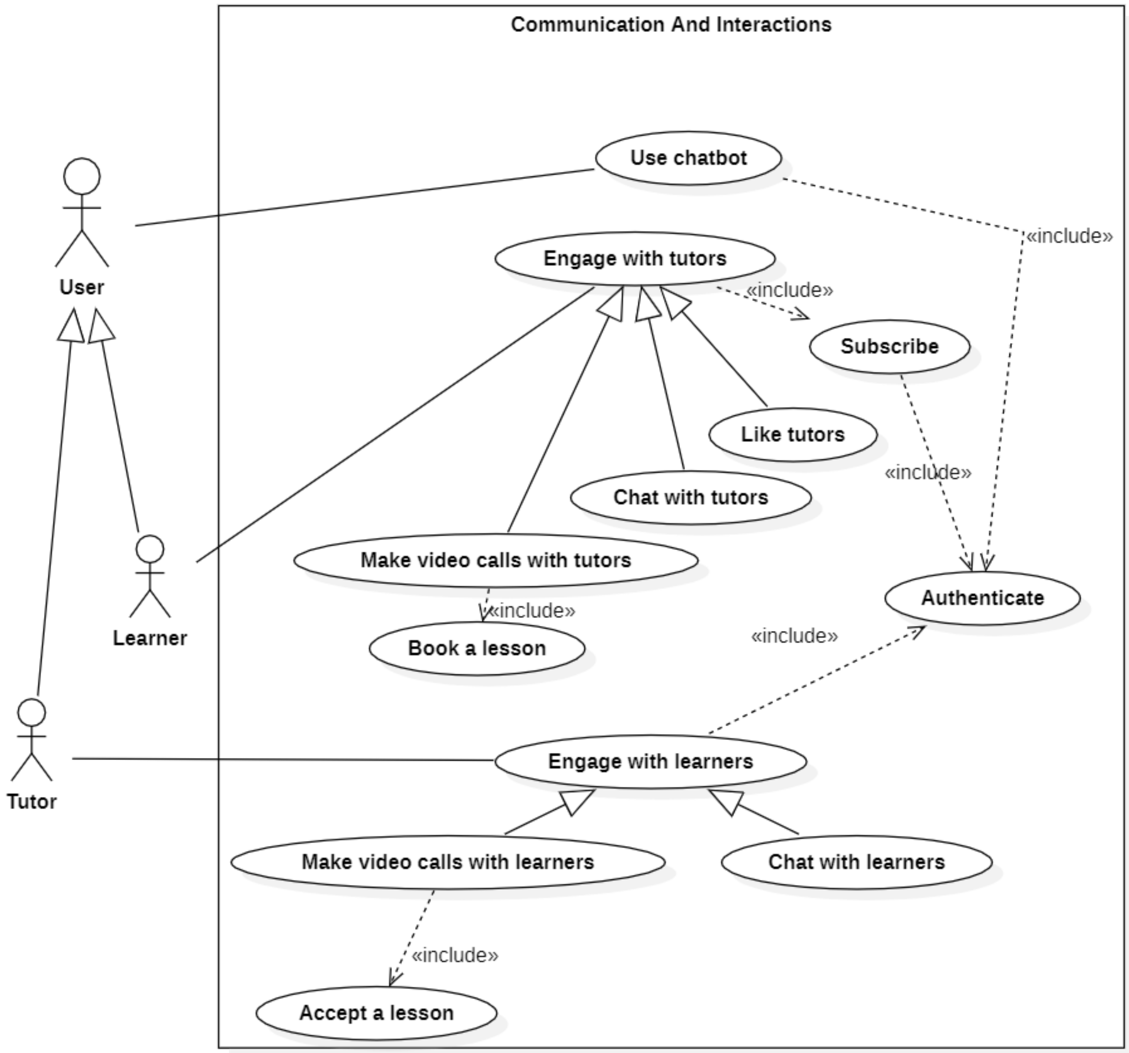

Figure V.2: Refinement of "Communication and Interactions" use case

The table V.2 represents a description for the Make video call use case.

IDENTIFICATION SUMMARY	
Title	Make a video call
Actor(s)	Tutor, Learner
DESCRIPTION OF SEQUENCES	
Pre conditions	User needs to have a lesson booked in advance.
Principal Scenario	
1.The system sends a reminder notification 15 minutes prior to the scheduled video call. 2.The system displays a pop-up with a button for both users to join the meeting. 3.The user clicks the "Join Call" button. 4.The system redirects the user to the video call interface. 5.The user adjusts the video call settings, such as entering their name, configuring the camera and microphone, and then clicks the "Join" button to enter the call. 6.The system connects the two users in the video call, allowing them to see and hear each other, mute and unmute, turn the camera on or off, share their screens, and chat. 7.When they are finished, they simply hang up by clicking the red button.	
Alternative Scenario	
- If the user doesn't receive the pop-up to join or misses it, they can enter the video call using the link sent to their email earlier.	

Table V.2: Textual description of the use case video call

The table V.3 represents a description for the "Chat with learners" use case.

IDENTIFICATION SUMMARY	
Title	Chat with learners
Actor(s)	Tutor
DESCRIPTION OF SEQUENCES	
Pre conditions	Users need to be authenticated and subscribed.
Principal Scenario	
1.The user accesses the profile of the learner they want to talk to. 2.The user clicks on the "Chat" button. 3.The system displays the chat interface. 4.The user types their message and then clicks the "Send" button. 5.The user receives a response message from the other user. 6.And the exchange of messages between them continues.	

Table V.3: Textual description of the use case Chat

3.2.3 Sequence diagram

On our platform, tutors and learners have the ability to video call each other for lessons, chat for assistance, and even seek support from a chatbot.
The sequence diagram in Figure V.3 represents the steps to be followed by a learner in order to make a video call.

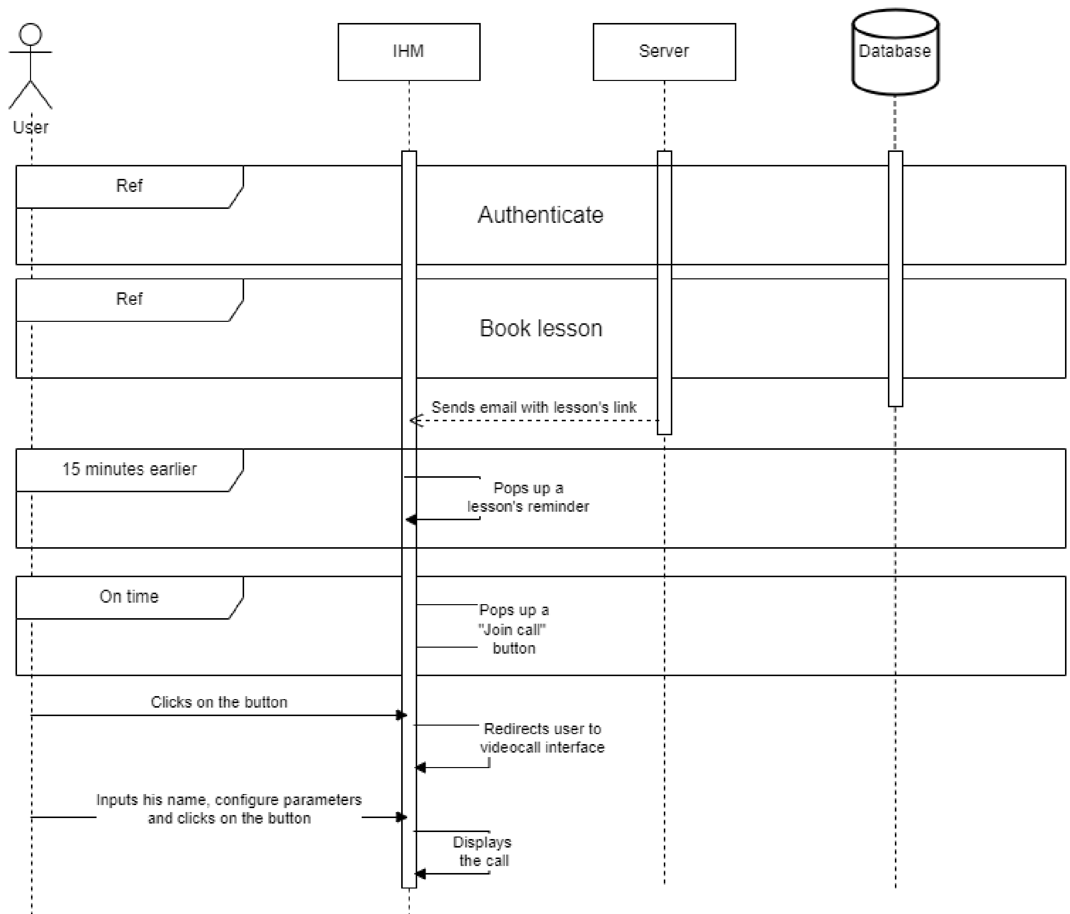

Figure V.3: Sequence diagram of "Make a videocall"

3.2.4 Class diagrams

The figure V.4 shows the class diagram of sprint 3.

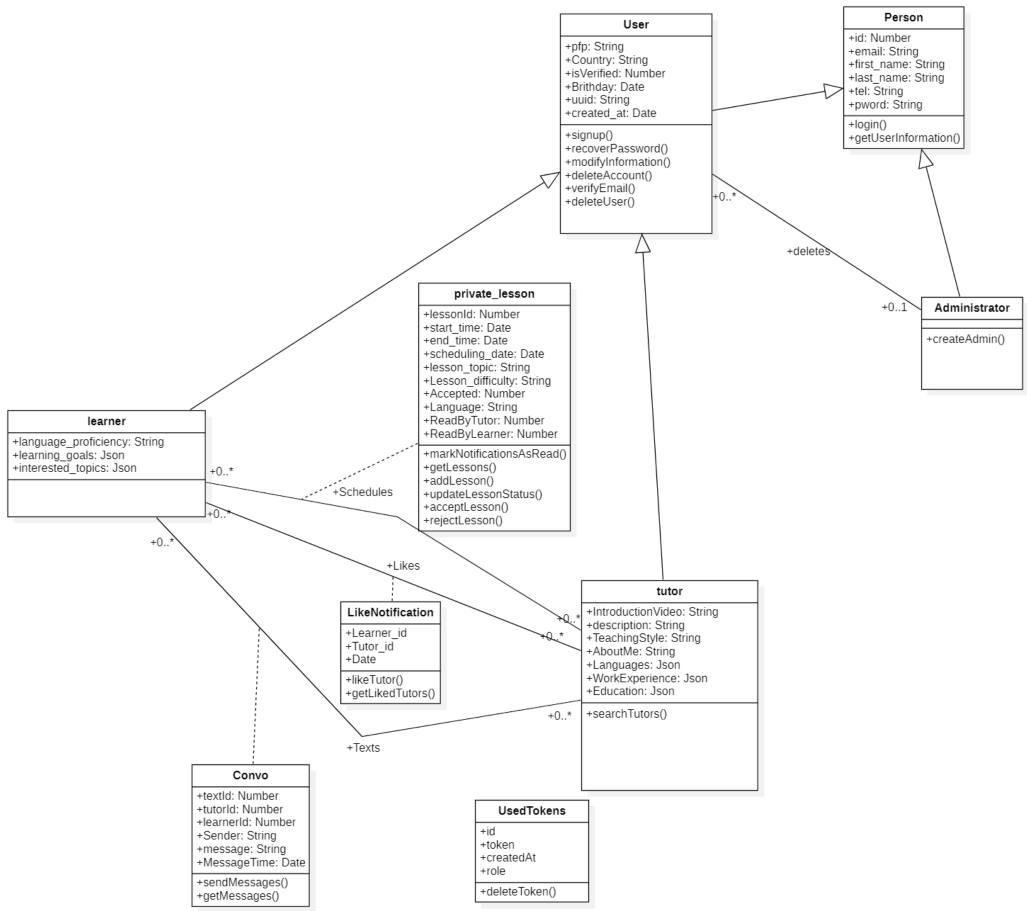

Figure V.4: Class diagrams for sprint 3

4 Realisation

This section will present multiple screenshots to describe the interfaces of our tool and thus provide a better understanding of its functionality.

To ensure both tutors and learners are well-prepared, a reminder will pop up for both parties 15 minutes before their scheduled lesson, as illustrated in the figure V.5.

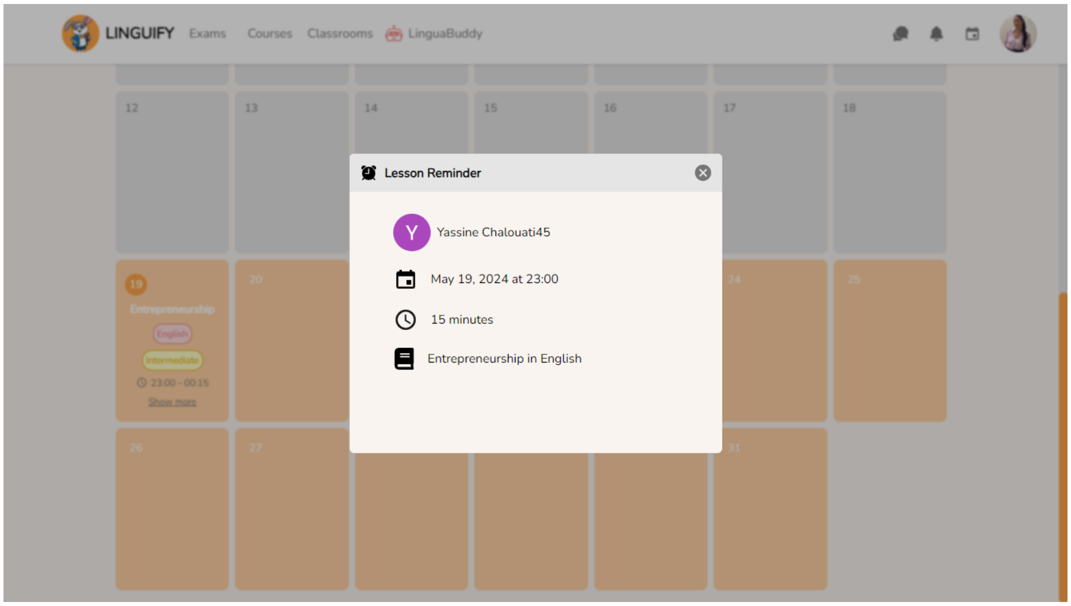

Figure V.5: Interface of the "Reminder" page

At the exact scheduled time, another notification will appear with a button to start the call, as illustrated in the figure V.6.

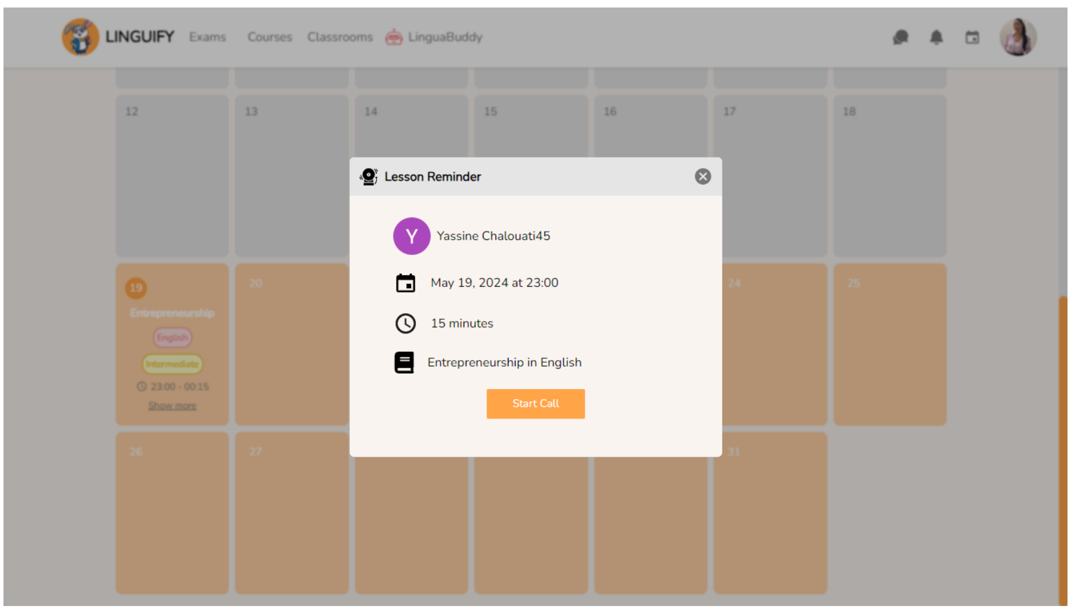

Figure V.6: Join call Interface

After clicking the button, the video call interface will be displayed, allowing users to input their names and configure essential parameters such as microphone and camera settings, as illustrated in the figure V.7.

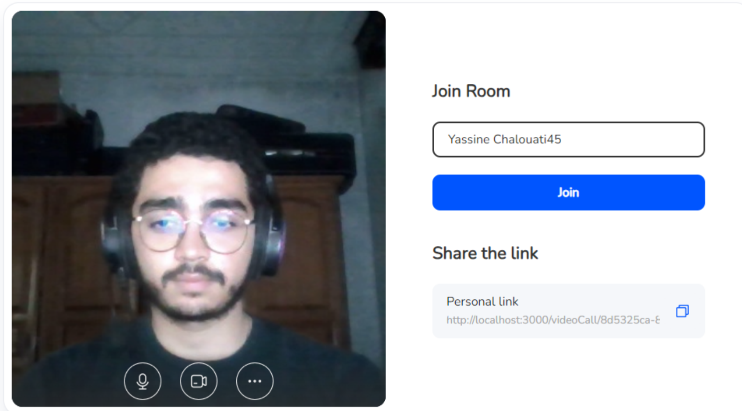

Figure V.7: Call configuration Interface

Finally, both the tutor and learner will be in their call together, where they can share screens, chat, and control their cameras and microphones, as illustrated in the figure V.8.

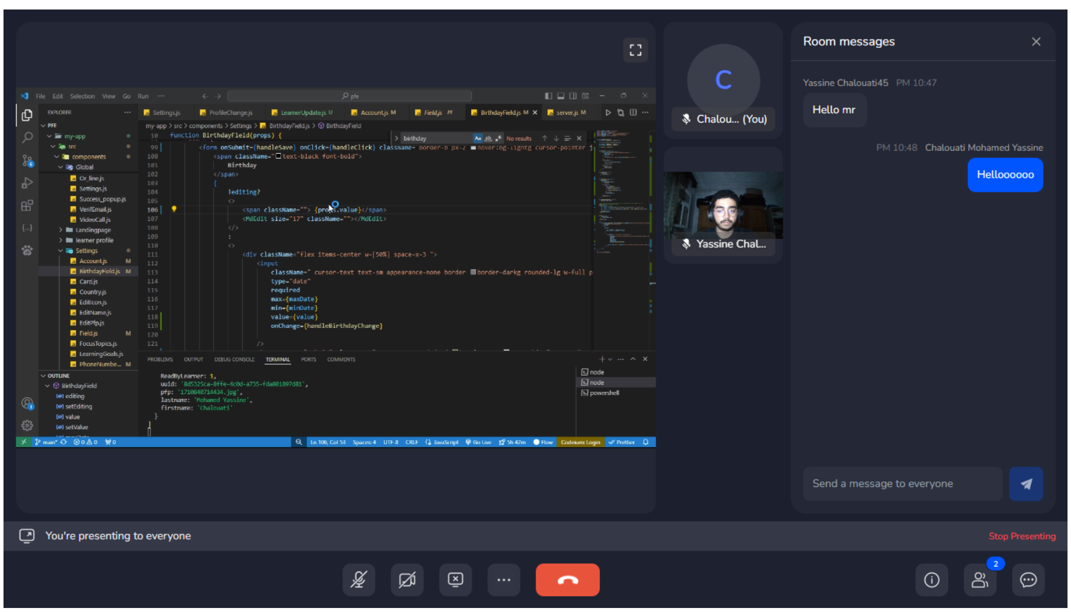

Figure V.8: Videocall Interface

If learners want to like a tutor, they can simply click the "Like" button on the tutor's profile, as illustrated in the figure V.9.

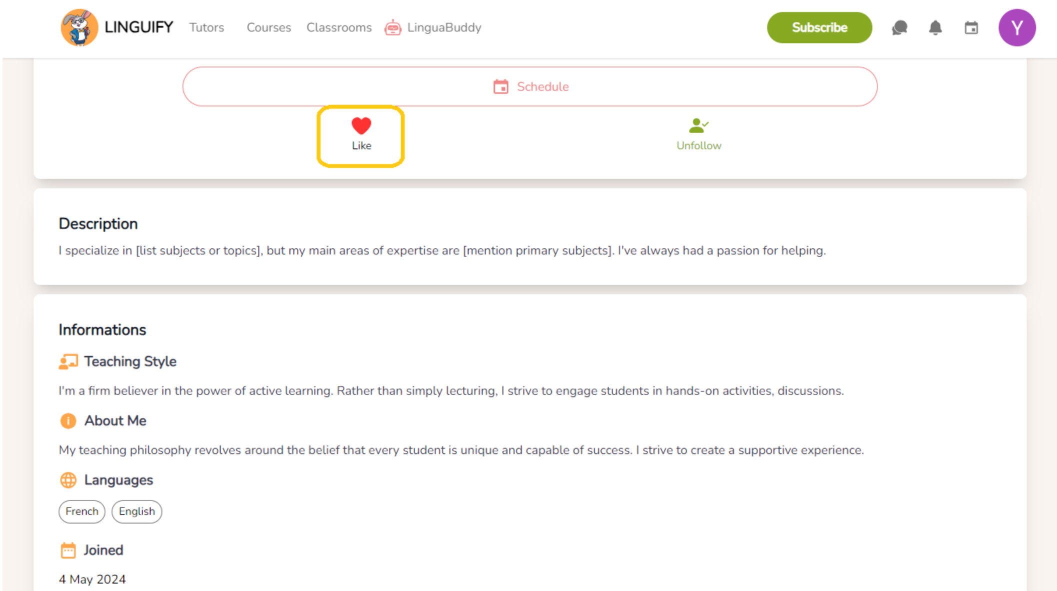

Figure V.9: Like tutor Interface

To view the messages the user received, they should click on the bubble icon located in the navigation bar, as illustrated in the figure V.10.

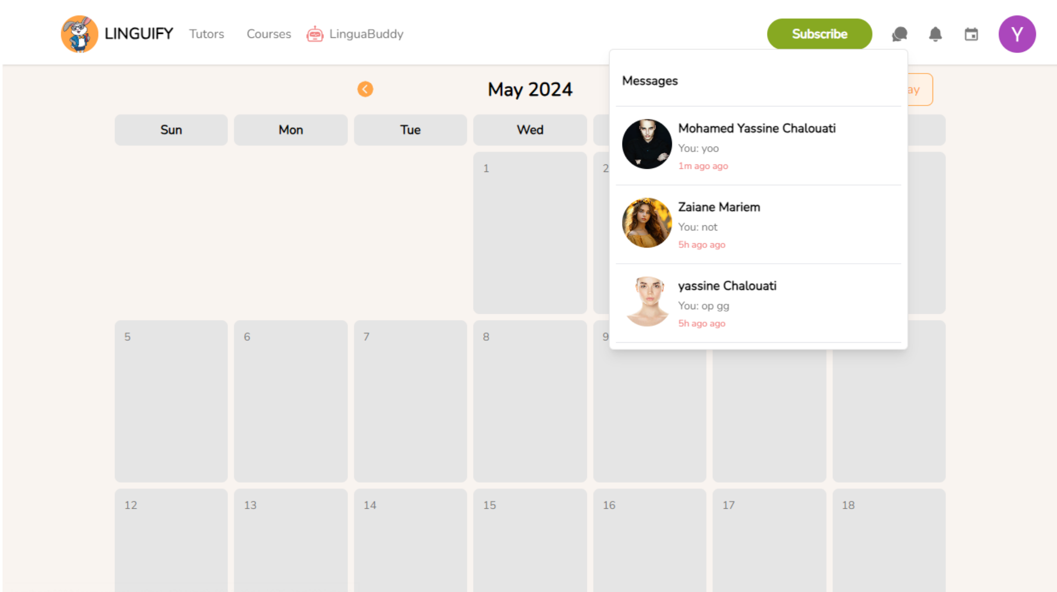

Figure V.10: Check messages - Interface

When the learner clicks on one of the tutors listed, they are redirected to the chat interface where they can communicate with the tutor, as illustrated in the figure V.11.

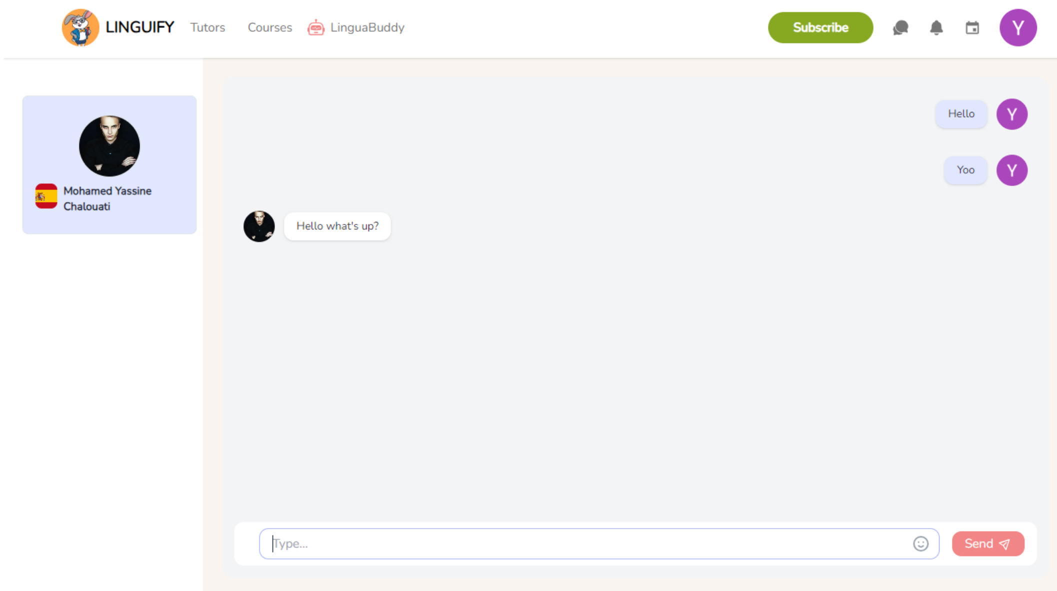

Figure V.11: Tutor-Learner chat - Interface

On the tutor's side, they will receive the message immediately, as illustrated in the figure V.12.

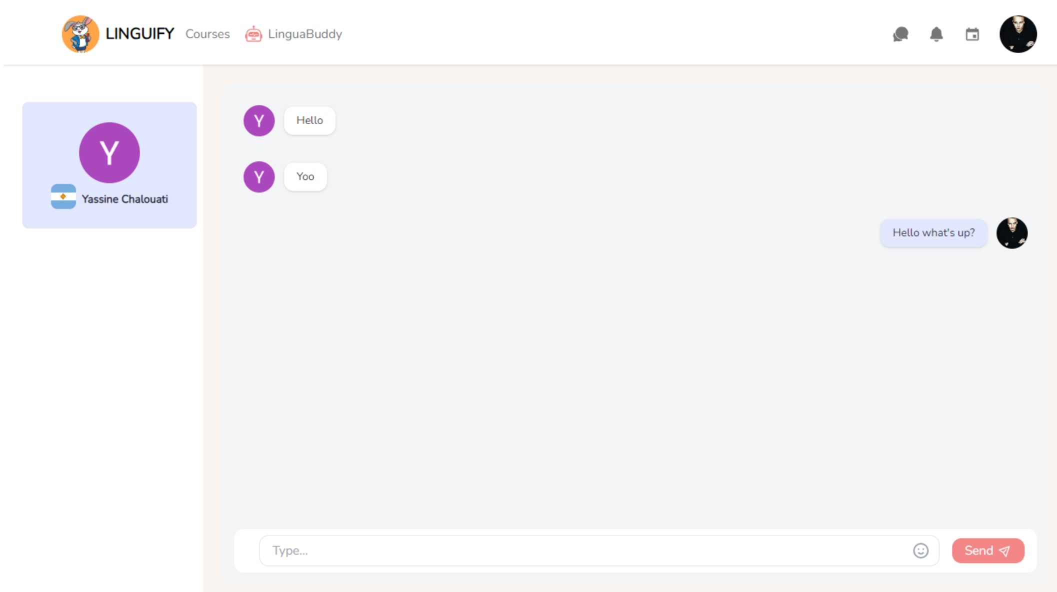

Figure V.12: Check messages - Interface

5 Sprint Review

In summary, the sprint was highly successful, and we effectively accomplished all of our objectives. We implemented various interactions between tutors and learners, including video calls with essential features such as screen sharing, camera and microphone configuration, and in-call chat functionality. Additionally, we provided users with the capability to like their favorite tutors, follow them, and communicate via text. Furthermore, we developed a chatbot that both users can engage with to seek assistance and support.

6 Conclusion

The sprint focused on developing and implementing all possible interactions between learners and tutors, covering aspects such as video calls, chat functionalities, liking, and following. We prioritized making these actions user-friendly and seamlessly integrated into our platform. We are enthusiastic about the positive impact this enhancement will bring, and we eagerly anticipate what lies ahead for our platform.

Chapter VI

Sprint 4: Content and Community Management

Plan

1 Introduction

This chapter focuses on implementing key functionalities to enhance the learning experience and strengthen the roles of administrators. Users will be able to subscribe to premium content, browse language courses by language and proficiency, enroll seamlessly, and download supplementary files. Tutors will have the ability to create engaging courses. Additionally, we prioritize empowering administrators with enhanced tools for effective content management, user interactions, and community standards enforcement.
Throughout this sprint, we will explore the sprint's backlog, functional specification, design, sprint review, and retrospective.

2 Backlog of Sprint 4

We need to start by identifying the list of tasks to be accomplished during this sprint. Table VI.1 represents the user stories of our sprint.

Module	User stories	Complexity	Estimation
Subscription Management	As a learner, I want to subscribe to access premium content.	L	3
	As a learner, I want to pay for my subscription online.	L	3
Courses Management	As a learner, I want to browse my available language courses.	S	2
	As a learner, I want to categorize language courses by language and proficiency level.	M	2
	As a learner, I want to download courses.	S	2
	As a tutor, I want to be able to make courses.	M	2
	As a user, I want to differentiate between premium courses and free courses.	S	1
	As a user, I want to view the course information.	S	1
Community Management	As an administrator, I want to send announcements and notifications to users.	Sprint 5	S
	As an administrator, I want to have the ability to remove courses that violates community standards.	Sprint 5	S
	As an administrator, I want to delete user accounts when necessary.	Sprint 5	S

Table VI.1: Sprint 4 backlog

3 Sprint Refinement

For this part, first we will be identifying the functional requirements. Then we will draw the use case and the class diagram for this sprint and finally, the textual description of each case.

3.1 Identification of the Functional Requirements

For this sprint, those are the functional requirements that we will be implementing:

- **Learners' subscriptions** : The goal of this functionality is to enable learners to subscribe to our application's premium content seamlessly, utilizing a safe and user-friendly payment method.
- **Courses' management** : The goal of this functionality is to provide learners with the ability to easily enroll in courses of their choice and to equip tutors with the capability to upload courses.
- **Community management** : The goal of this functionality is to give administrators control over community activities, such as deleting user accounts or removing courses that do not comply with our standards.

3.2 Global model of use cases for the forth Sprint

3.2.1 Use Case Diagram of the Sprint

This sprint focuses on the use cases that describe the different functionalities represented in the use case diagram illustrated in Figure VI.1.

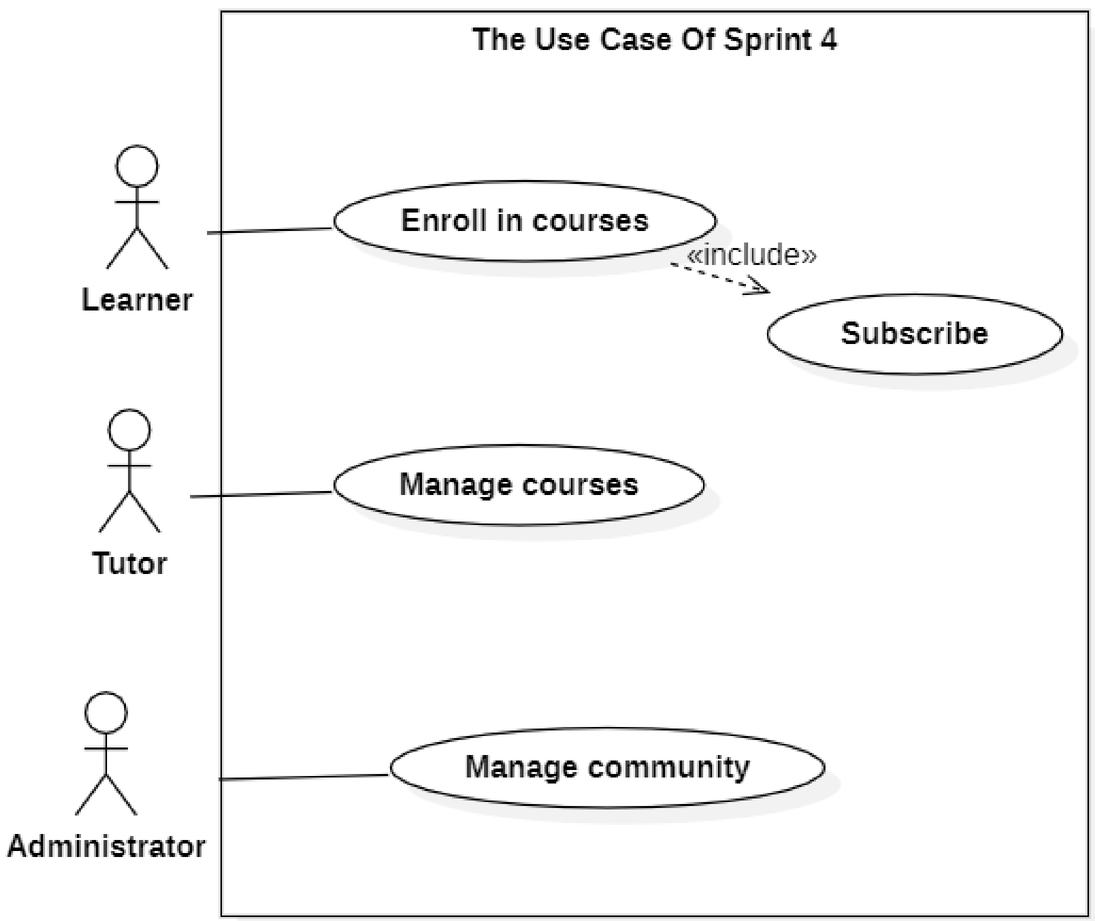

Figure VI.1: Use case diagram sprint 4

3.2.2 Refinement of Use Cases

In this section, we will start with detailing the "Enroll in courses" use case from sprint 4 as illustrated in Figure VI.2.

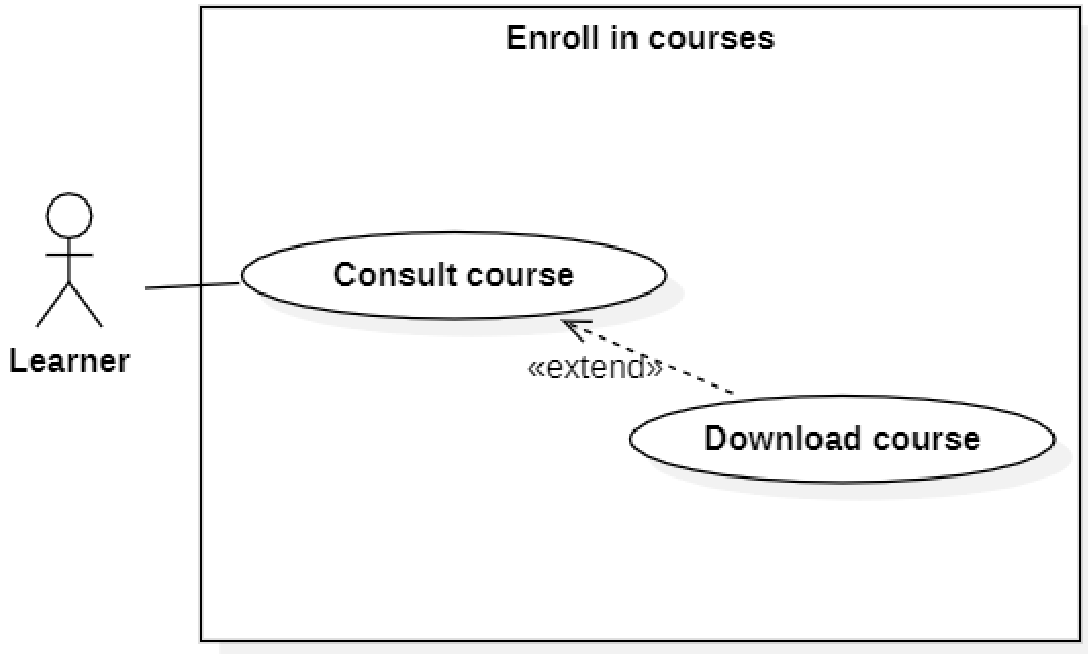

Figure VI.2: Refinement of «Enroll in courses» use case

The table VI.4 represents a description for the Enroll in courses use case.

IDENTIFICATION SUMMARY	
Title	Enroll in courses
Actor(s)	Learner
DESCRIPTION OF SEQUENCES	
Pre conditions	User needs to be authenticated in order to access the system.
Principal Scenario	
1.The user selects the "Courses" button located in the navigation bar. 2.The system presents the Courses page to the user. 3.The user utilizes the search bar to find courses of interest. 4.The user clicks on a course layout to view its details. 5.The system displays the course information and associated file. 6.The user downloads the course file if it meets their approval.	

Table VI.2: Textual description of the use case Book lessons from calendar

The figure VI.3 presents the sequence diagram for Manage courses.

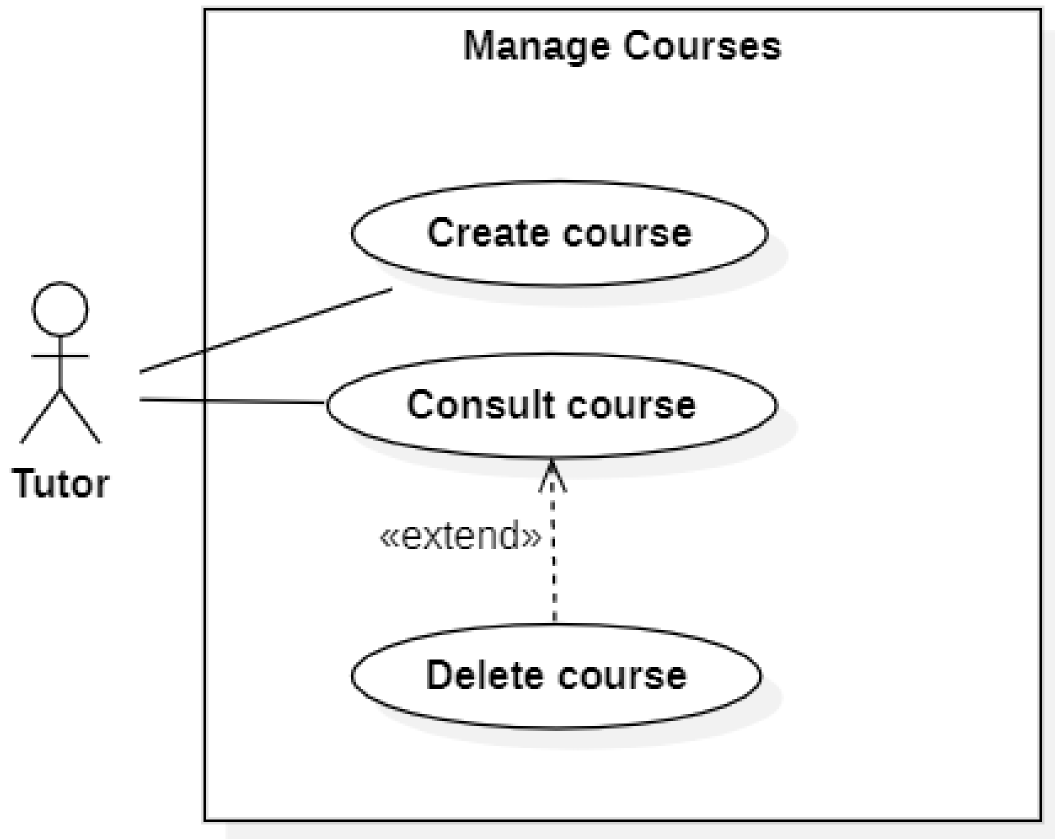

Figure VI.3: Refinement of «Manage courses» use case

Table VI.3 represents a description for the "Manage courses" use case.

IDENTIFICATION SUMMARY	
Title	Manage courses
Actor(s)	Tutor
DESCRIPTION OF SEQUENCES	
Pre conditions	User needs to be authenticated in order to access the system.
Principal Scenario	
1.The user selects the "Courses" button located in the navigation bar. 2.The system presents the Courses page to the user. 3.The user fills in the fields that describe the course and uploads the associated files. 4.The user clicks on the "Create Course" button. 5.System adds the course to the tutor's published courses.	

Table VI.3: Textual description of the use case the Manage courses

3.2.3 Refinement of Use Cases

In this section, we will start with detailing the "Community management" use case from sprint 4 as illustrated in Figure VI.4.

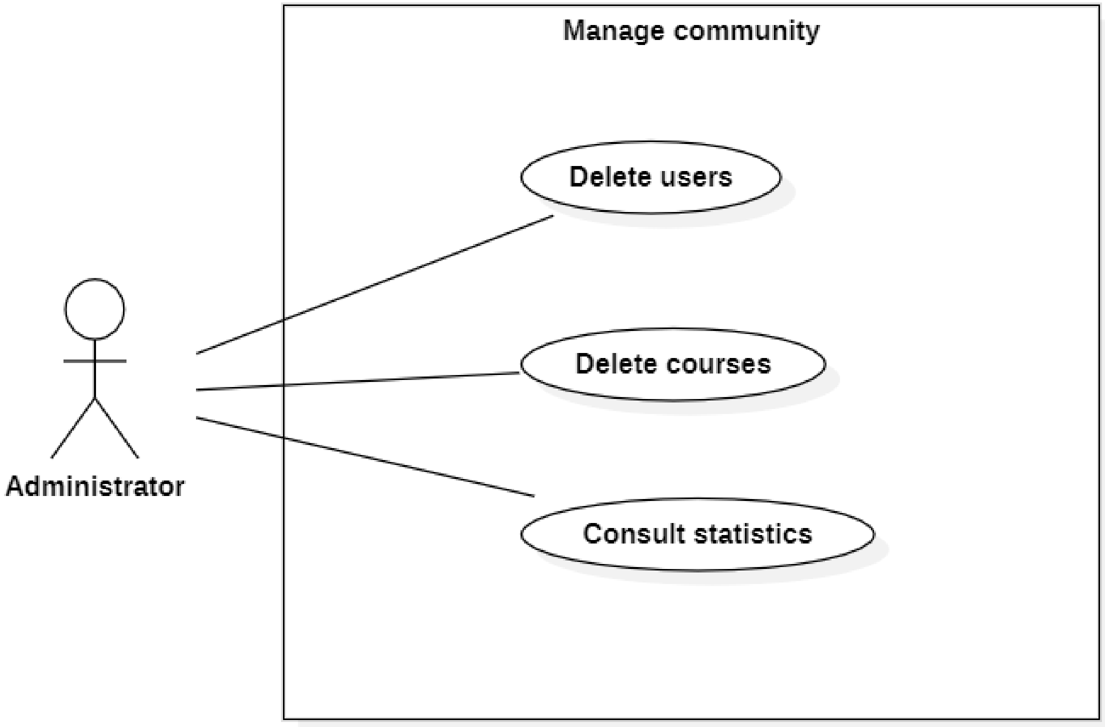

Figure VI.4: Refinement of «Community management» use case

The table VI.4 represents a description for the Community management use case.

IDENTIFICATION SUMMARY	
Title	Community management
Actor(s)	Administrator
DESCRIPTION OF SEQUENCES	
Pre conditions	User needs to be authenticated in order to access the system.
Principal Scenario	
1.NOTDONEYET. 2.NOTDONEYET. 3.NOTDONEYET. 4.NOTDONEYET. 5.NOTDONEYET. 6.NOTDONEYET.	

Table VI.4: Textual description of the use case Community management.

3.2.4 Sequence diagram

The sequence diagram in Figure VI.5 represents the steps to be followed by a learner in order to enroll in a course.

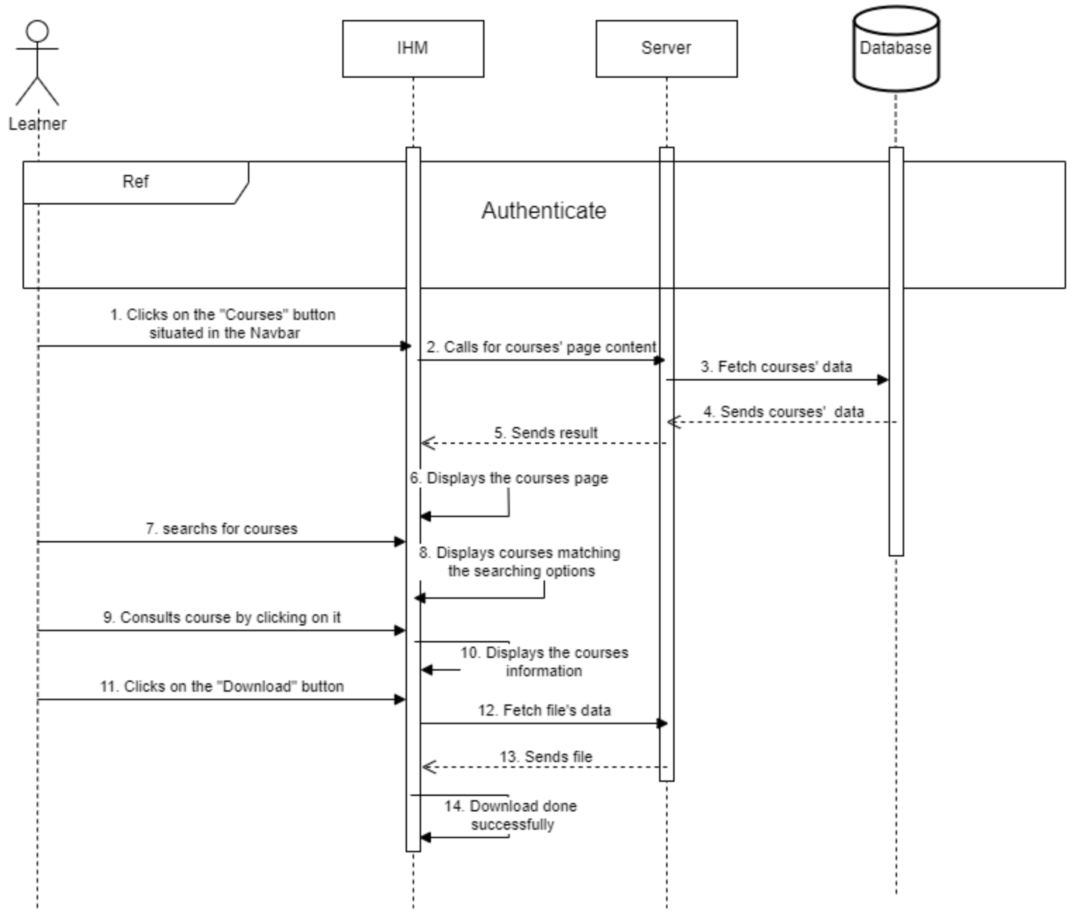

Figure VI.5: Sequence diagram of "Enroll in course"

The figure VI.6 presents the sequence diagram for create course

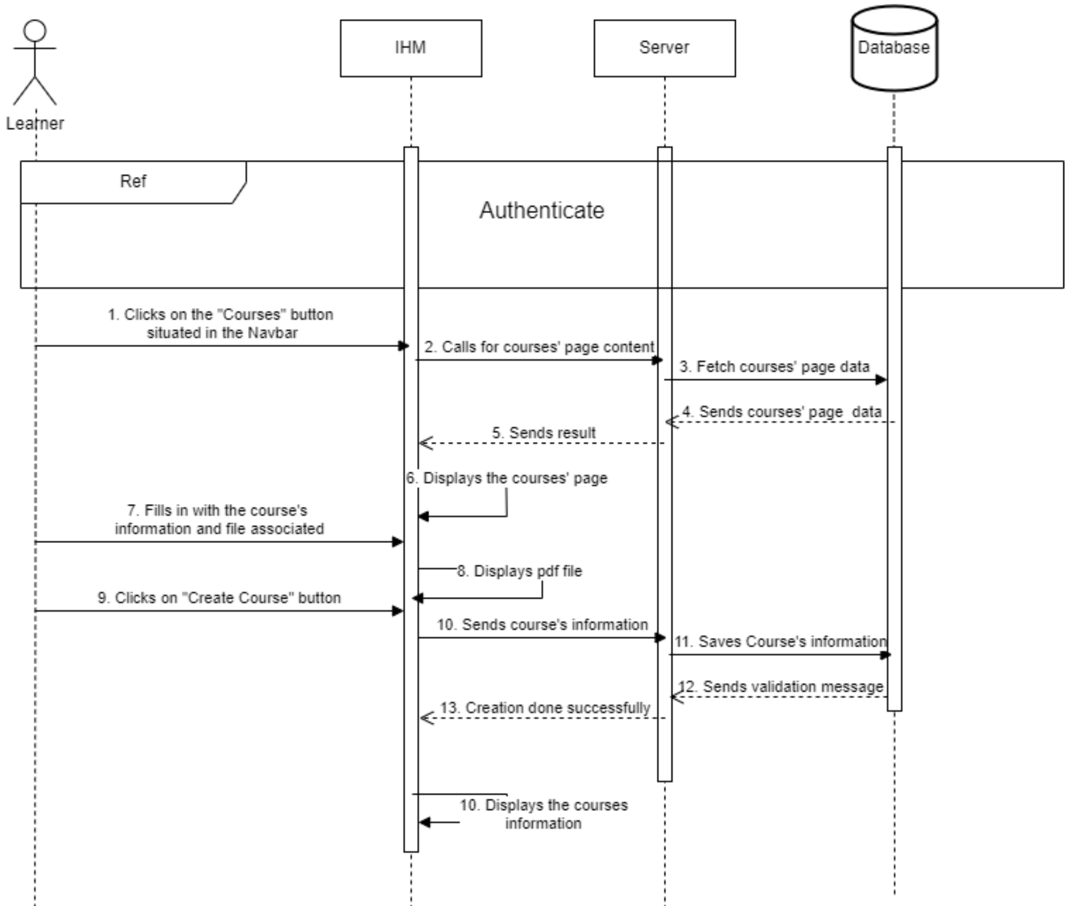

Figure VI.6: Sequence diagram of "Create course"

3.2.5 Class diagrams

The figure VI.7 shows the class diagram of sprint 4.

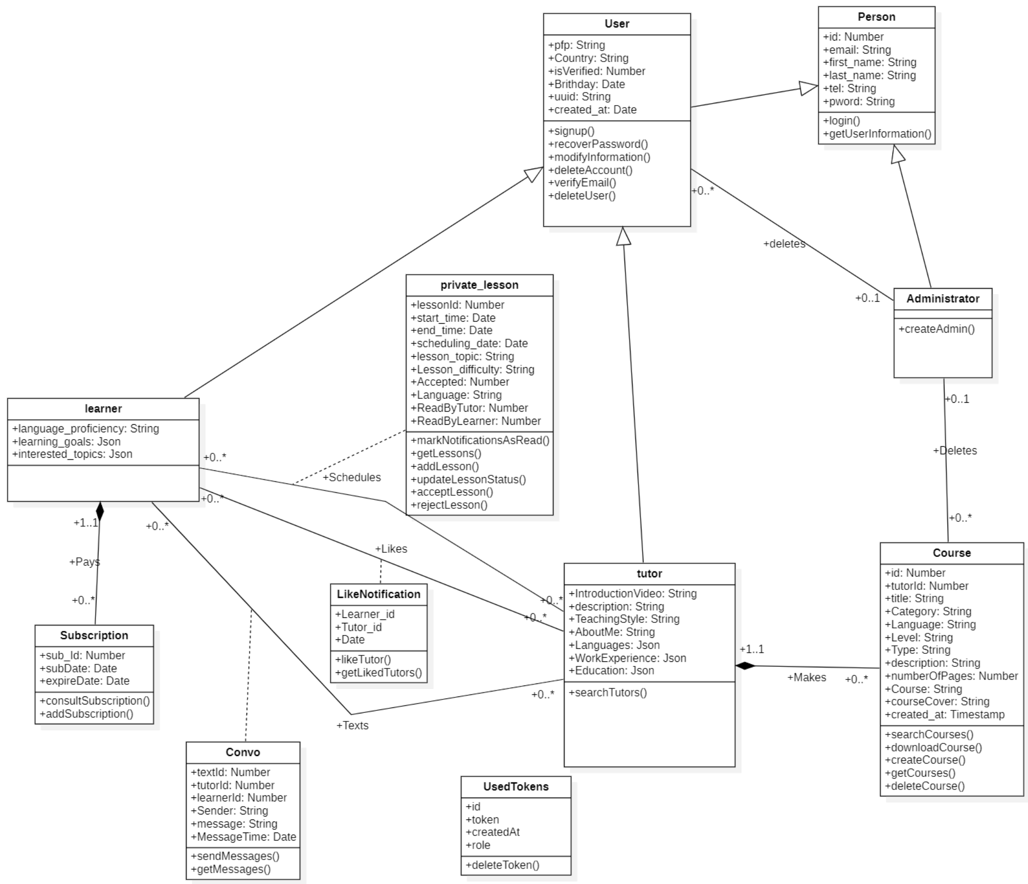

Figure VI.7: Class diagrams for sprint 4

4 Realisation

This section will present multiple screenshots to describe the interfaces of our tool and thus provide a better understanding of its functionality.

Upon clicking on the "Courses" situated in the navigation bar, our tutors will have access to their published courses along with the creating an account interface, as illustrated in the figure IV.6.

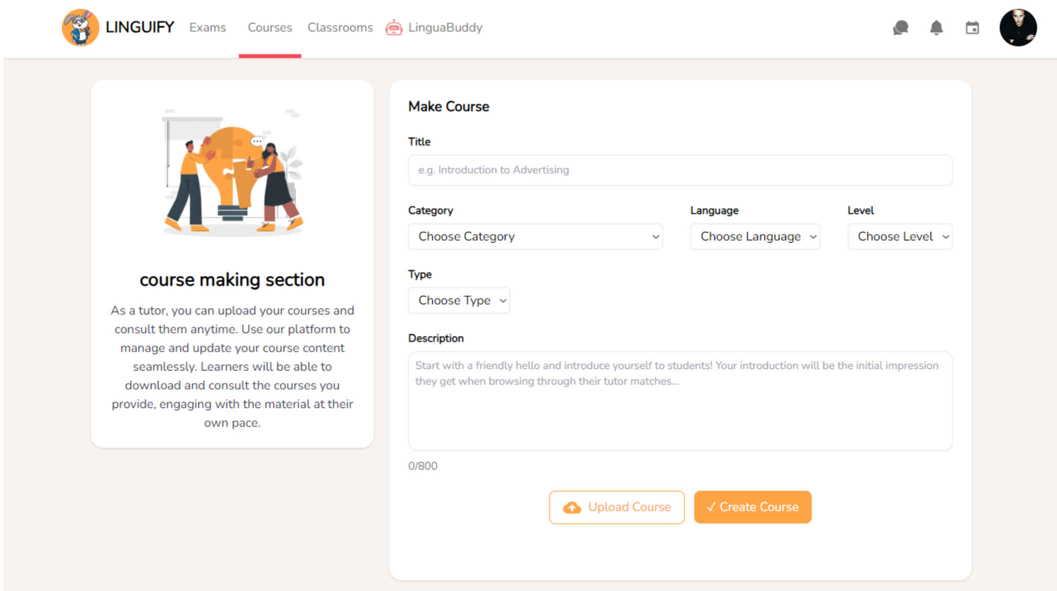

Figure VI.8: Tutor's courses' page - Interface

As you scroll down, you'll find the remainder of the courses page, where the tutor can view their published courses along with essential and useful information for each one, as illustrated in the figure VI.9.

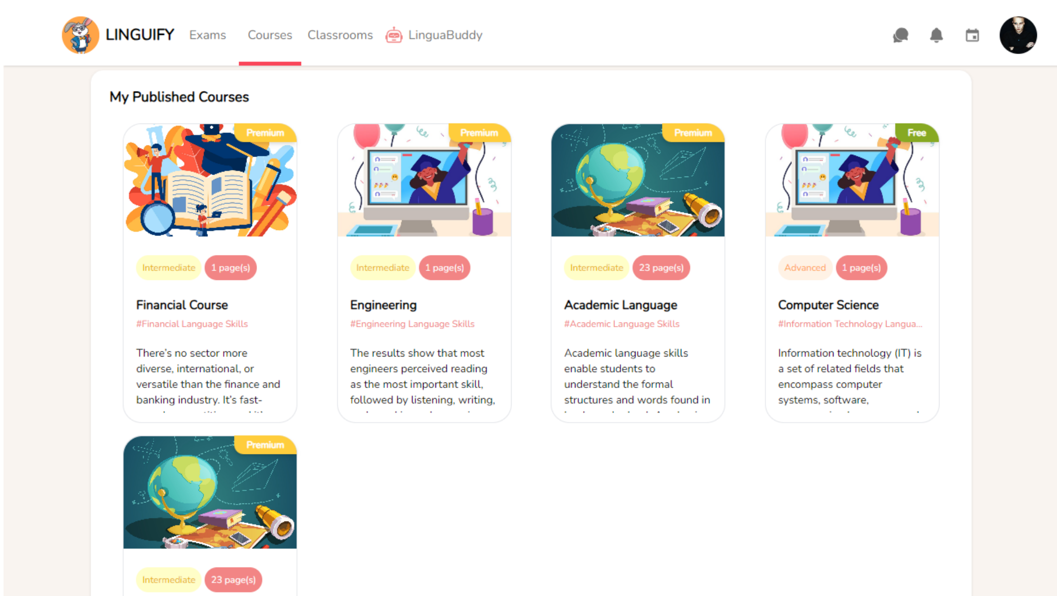

Figure VI.9: Tutor's published courses - Interface

When the tutor hovers over any of the courses, a delete icon appears, allowing them to delete the course if desired, as illustrated in the figure VI.10.

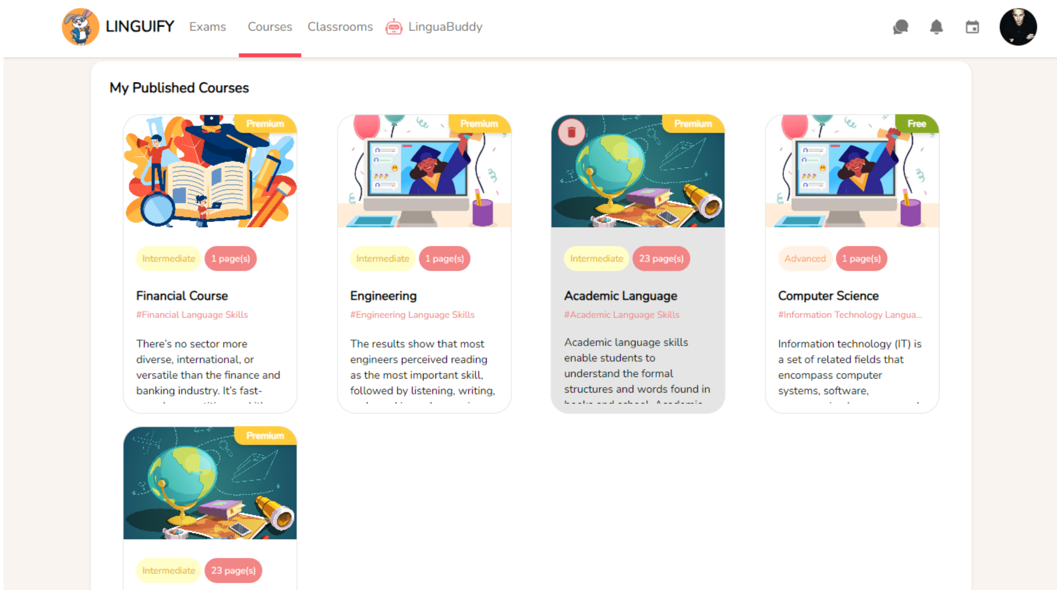

Figure VI.10: Hover on course to delete - Interface

Upon clicking the delete icon, a confirmation message pops up to verify the intent to delete the course, as illustrated in the figure VI.11.

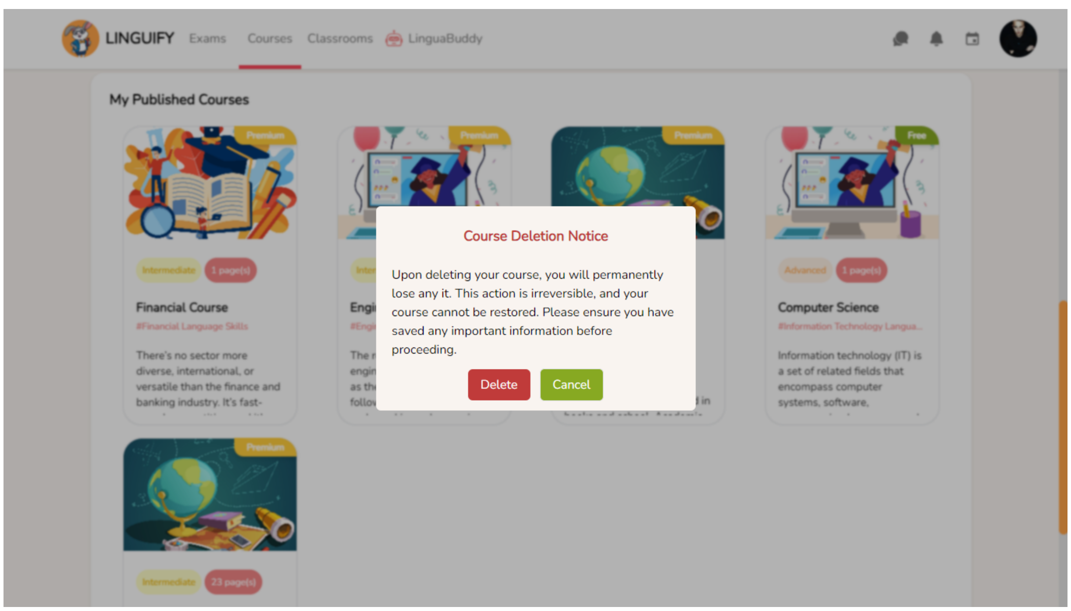

Figure VI.11: Delete message pop up - Interface

To create a course, the tutor needs to fill in the required information related to the course. They must then choose the course file by clicking the "Upload Course" button and finally clicking on the "Create Course" button, as illustrated in the figure VI.12.

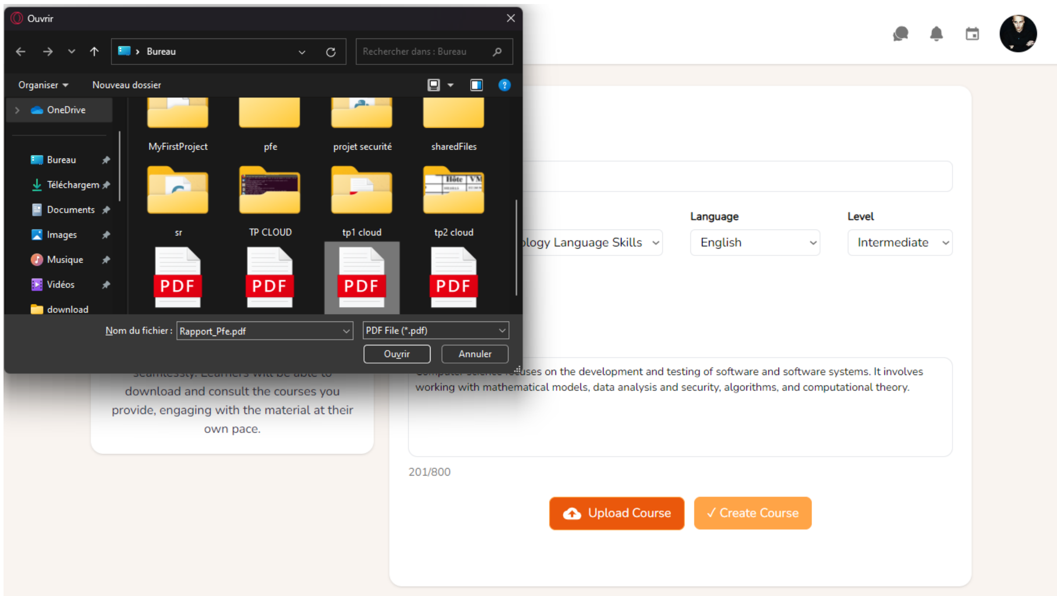

Figure VI.12: Upload course - Interface

On the learner's side, the courses page will display all available courses, featuring a search bar to help them find courses easily and a categorizing feature to enhance user-friendliness, as illustrated in the figure VI.13.

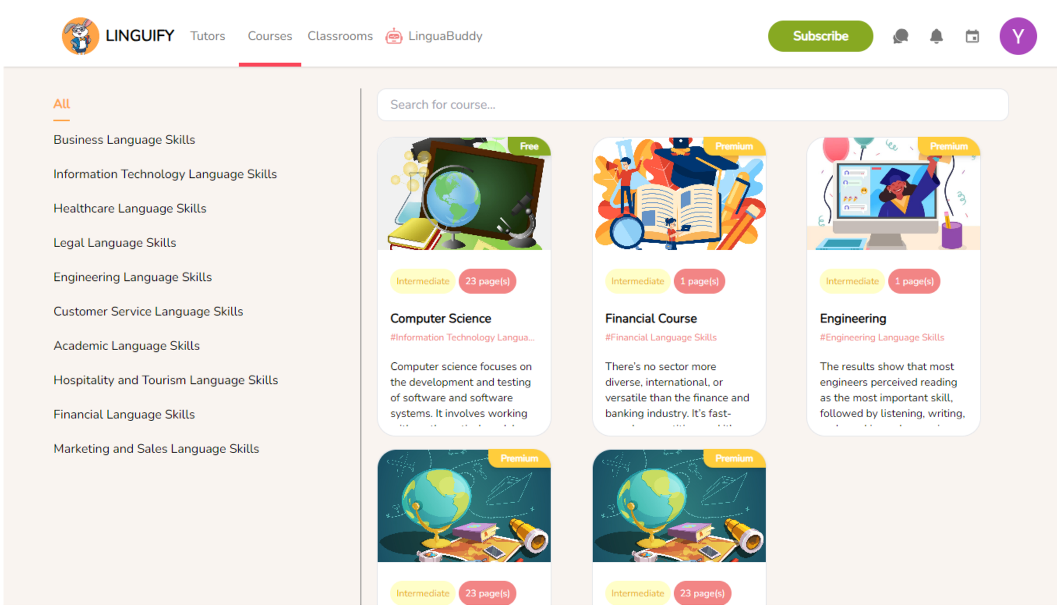

Figure VI.13: Learner's Courses' page - Interface

On the learner's side, the courses page will display all available courses, featuring a search bar to help them find courses easily and a categorizing feature to enhance user-friendliness, as illustrated in the figure VI.14.

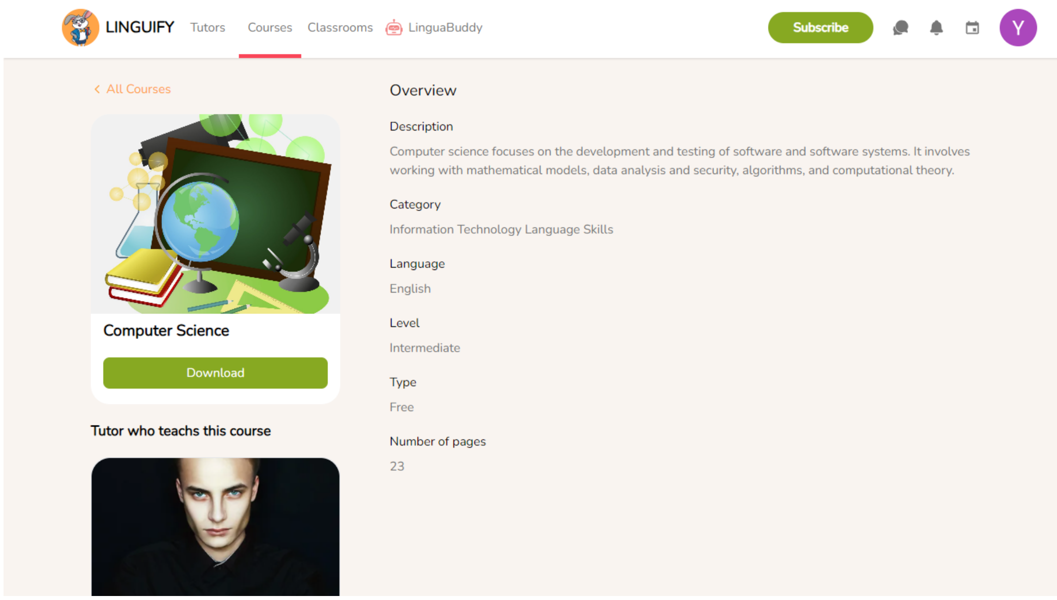

Figure VI.14: Learner's Courses' page p2 - Interface

This page contains all the information related to the course, including a download button. If you scroll down, you'll find the tutor's name and be able to access their profile from there, as illustrated in the figure VI.15.

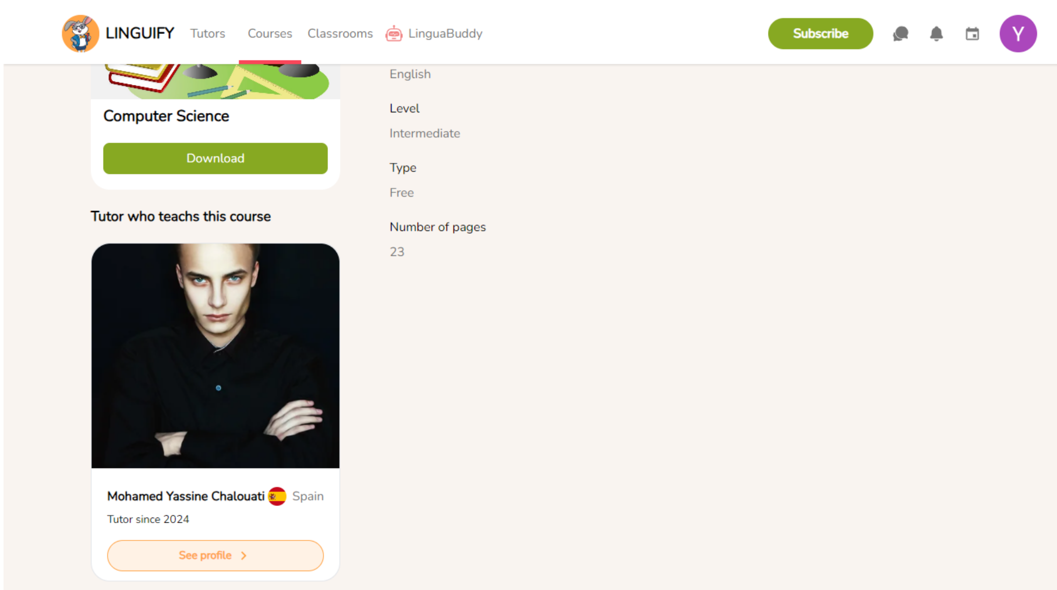

Figure VI.15: Learner's Courses' page - Interface

5 Sprint Review

Overall, the sprint was a success, and we achieved all our objectives. We developed comprehensive systems for both course and community management.

For course management, learners can now enroll in courses, download related files, and update their profiles, account information, and subscription details. Tutors can upload, consult, and delete course files.

For community management, administrators can manage user activities, report accounts and delete them when needed.

We are very satisfied with the results accomplished during this sprint.

6 Conclusion

The successful completion of this final sprint marks a significant milestone in our project. We developed a user-friendly course enrollment and management system, enabling learners to enroll effortlessly and tutors to manage their courses. Additionally, we implemented a comprehensive community management system, providing administrators with essential tools to oversee and manage user activities effectively.

We are excited about the significant impact these systems will have on our platform. The achievements of this sprint have laid a strong foundation for the future, and we are confident that these improvements will greatly enhance our platform.

General Conclusion and Perspectives

The final year project concluded in the creation of an e-learning web application. Throughout the project, we followed a rigorous methodology to design, develop, and test the application, ensuring that it reflected the needs of the users.

This report begins by providing context and a problem description for the project, as well as the host organization, Beecoders. We then did an analysis of the current situation to have a better understanding of the existing e-learning apps and identify any potential constraints. Following that, we presented our recommended solution and explained the work method and strategy we employed.

In addition, we provided a thorough preparatory research that included a needs assessment, a global use case and class diagram, architectural style, requirements description, and a product backlog. Finally, we discussed the first sprint, which is focused on account management.

At the end of this project, we have created a dependable, user-friendly e-learning application designed for achieving language proficiency across multiple fields with artificial intelligence features. We anticipate that our application will be helpful not just to individuals interested in learning new languages, but also to companies looking to facilitate the linguistic acclimatization of their international workers to technical terms in their native languages.

In terms of future plans, there is the prospect of transitioning to a cloud-based storage solution, particularly for multimedia assets such as pictures and videos. We can improve our platform's scalability and dependability by leveraging powerful cloud services like Amazon S3. This strategic transition to cloud-based file storage optimizes resource allocation while also boosting data security and accessibility, in line with our goal to offering a cutting-edge e-learning experience that matches our users' dynamic requirements and encourages long-term growth for our platform.

We are very pleased with the work we completed as part of this final-year project. We are confident that our online learning software will substantially simplify the process of effortlessly connecting learners and instructors in real time, offering important help to both parties. Throughout the project, we worked carefully, following best development standards and meeting deadlines.

We are thrilled about the possibilities for improving and expanding this project. Our goal is to continue developing this application in order to put the ideas and opportunities we've uncovered into action. We are convinced that we can make major improvements to our system, increasing its performance and effectiveness. We are considering the following areas of improvement :

- We intend on expanding our database, and target new potential users by making a section dedicated for kids, that can connect with tutors to help them learn and explore new languages at a young age, to do that We will use interactive language learning tools like games, quizzes, and exercises to keep children interested while learning new languages. Additionally, we intend to implement audio-visual content such as videos, songs, and stories to make language learning more engaging and immersive. To ensure safety, we aim to give parents the ability to monitor their children's platform activities, set usage restrictions, control access to specific features or content, and review and approve tutor recommendations, lesson plans, and learning materials before they are offered to their child.

- We intend to expand by creating a mobile app version with cross-platform frameworks like flutter for the front-end to assure compatibility with iOS and Android devices. For the back-end, we will emphasize an integration with our existing infrastructure, which is developed with Express.js and MySQL, to ensure data consistency and harmony across our online version and mobile platforms. We will also focus on creating an outstanding product in terms of user interface and experience for mobile devices, taking into account screen sizes and device capabilities, as well as following best practices in mobile app design and development.

- We intend to develop our own video calling system to make the process of adding features and tweaking options easier by introducing WebRTC on the client side, which allows for real-time communication between users' browsers. This involves establishing peer connections, capturing and transmitting audio/video streams, and handling network traversal through the STUN (Session Traversal Utilities for NAT) and TURN (Traversal Using Relays Around NAT) servers.

- We intend to use machine learning techniques to develop an artificial intelligence-powered solution to prohibit inappropriate behavior in conversations, such as abusive words and files. To accomplish this, we must first collect a diverse dataset of chat messages containing examples of bad manners, like offensive language, insults, harassment, or inappropriate content, and then annotate the dataset to label each message with the type of bad manners it demonstrates. This annotated data will serve as the training corpus for our AI model, choose suitable machine learning model and train it using the annotated dataset while optimizing its parameters to maximize performance and as a final part, integrate the trained model with the chat while ensuring a real time prevention and detection.

By pursuing these opportunities for improvement, we are committed to taking our solution to new levels and offering an even better experience for our users. We would want to thank everyone who helped make this project possible, especially our staff, mentors, and hosting organization. Their assistance and helpful advice have been critical to the success of this project.

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